Group Assignment Trimester 1 25/26

2025-09-16

Task

Description

You are a consultant for Pacific Insights Analytics (PIA), a boutique data analytics firm specializing in workforce performance. PIA has been hired by Huaxia Communications, a major Chinese company operating several large call centers across the country. Huaxia is facing rising operational challenges. Staff attrition is driving up recruitment and training costs and retaining top performers is becoming increasingly difficult. To address these issues, Huaxia has provided you with historical data from one of its call centers. This data comes from multiple administrative sources, including worker surveys, HR records, and call center productivity logs. Because the data was collected from different systems, it contains inconsistencies, formatting problems, and other quality issues. Your task is to clean, explore, and analyze the data in order to generate clear, evidence-based recommendations that can help Huaxia's management improve retention and boost productivity. Important: You should not run any regression models. The goal is to understand the data and identify patterns that can inform management decisions through EDA and hypothesis testing.

Declaration of AI Usage

In the preparation of this project, I made use of AI tools, specifically OpenAI's ChatGPT, as a support resource. The tool was employed to enhance efficiency in tasks such as structuring ideas, clarifying concepts, refining wording, and checking technical accuracy.

At all times, the use of AI served as a complement to my own knowledge, analysis, and critical thinking. The outputs generated were carefully reviewed, adapted, and integrated by us to ensure originality, accuracy, and academic integrity. The final work reflects my own understanding, reasoning, and skills, with AI functioning only as a leverage of these.

Setup

Packages

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(haven)
library(dplyr)
library(ggplot2)
library(ggcorrplot)
library(ISOweek)
library(stargazer)
```

```
##
## Please cite as:
##
##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
library(fmsb)
```

Loading the data

```r
data_perf <- read_dta("../data/performance_labeled.dta")
data_attitude <- read_dta("../data/attitude_labeled.dta")
data_endperiod <- read_dta("../data/endperiod_outcomes_labeled.dta")
data_sum_vol <- read_dta("../data/summary_volunteer_labeled.dta")
data_wage_new <- read_dta("../data/wage_new_labeled.dta")

# convert to tibbles
performance <- as_tibble(data_perf)
attitude <- as_tibble(data_attitude)
endperiod <- as_tibble(data_endperiod)
volunteer <- as_tibble(data_sum_vol)
wage <- as_tibble(data_wage_new)

# remove raw data
rm(data_perf)
rm(data_attitude)
rm(data_endperiod)
rm(data_sum_vol)
rm(data_wage_new)
```

Cleaning data

Handy functions for cleaning

Here come some handy functions to reduce duplicate code for cleaning the data:

```r
# convert to numeric, if its a number (not NA)
as_numeric <- function (x) {
  x <- as.character(x)
  # only convert if it looks numeric
  if (grepl("^[-+]?[0-9]*\\.?[0-9]+([eE][-+]?[0-9]+)?$", x)) {
    as.numeric(x)
  } else {
    NA_real_
  }
}
```

```r
# convert to integer, if its a number (not NA)
as_integer <- function (x) {
  # only convert if it looks numeric
  x <- as.character(x)
  if (grepl("^[-+]?[0-9]*\\.?[0-9]+([eE][-+]?[0-9]+)?$", x)) {
    as.integer(x)
  } else {
    NA_real_
  }
}
# Extract a numeric values correctly from a string value, considering multiple cases
extract_numeric <- function(x) {
  # remove leading/trailing spaces
  x <- trimws(x)

  # normalize dash types (en-dash, em-dash) to minus
  x <- gsub("-|-", "-", x)

  # allow scientific notation: replace commas in mantissa
  if (grepl("[eE]", x)) {
    # if comma before e → decimal comma
    x <- sub(",", ".", x)
    # remove thousand separators before e
    x <- gsub("\\.(?=\\d{3}[eE])", "", x, perl = TRUE)
    x <- gsub(",(?=\\d{3}[eE])", "", x, perl = TRUE)
    return(as.numeric(x))
  }

  # remove spaces
  x <- gsub(" ", "", x)

  # case 1: comma used as decimal
  if (grepl(",", x) & !grepl("\\.", x)) {
    x <- gsub(",", ".", x)
  }

  # case 2: both comma and dot appear
  # assume comma is thousands separator, remove it
  if (grepl(",.*\\.", x)) {
    x <- gsub(",", "", x)
  }

  as_numeric(x)
}

# find all non numeric characters in a string and return them
find_non_numeric_chars <- function(x) {
  non_numeric <- gsub("[-0-9]", "", x)  # remove digits, minus, dot, comma
  unique(non_numeric[non_numeric != ""])
}
# find all rows with non numeric characters in a string and return them
find_non_numeric_rows <- function(x) {
  non_numeric <- !grepl("^[-]?[0-9]+([0-9]+)?$",x) # remove digits, minus, dot, comma
```

```r
  non_numeric <- non_numeric[non_numeric != ""]
  x[non_numeric]
}
```

Clean volunteer

```r
glimpse(volunteer)
```

```
## Rows: 140
## Columns: 9
## $ personid  <chr> "33350", "40034", "31292", "21654", "36908", "13980", "44794~
## $ age       <dbl> 26, 19, 27, 29, 23, 1, 23, 25, 21, 23, 23, 30, 22, 22, 23, 2~
## $ tenure    <dbl> 22, 9, 24, 37, 15, 51, 3, 42, 3, 34, 24, 25, 3, 66, 9, 70, 4~
## $ children  <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", ~
## $ bedroom   <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", ~
## $ commute   <chr> "120", "300", "135", "80", "60", "  120", "40 min", "60", "1~
## $ gender    <chr> "m", "woman", "F", "f", "m", "woman", "male", "woman", "MALE~
## $ married   <chr> "SINGLE", "N", "NO", "N", "NO", "  single", " N", "NO", "sin~
## $ high_educ <chr> "Y", "no", "YES", "YES", "N", "NO", "n", "YES", "no", "Y", "~
```

```r
stargazer(as.data.frame(volunteer), type = "text", median = TRUE, header = FALSE)
```

```
##
## =====================================================
## Statistic  N   Mean  St. Dev.  Min  Median   Max
## -----------------------------------------------------
## age       140 23.536  4.352     1     23     35
## tenure    140 22.971  24.296  2.000 13.000 150.000
## -----------------------------------------------------
```

```r
# check character columns with numerical values to extract for non numeric chars to
# be aware for special cases
find_non_numeric_chars(volunteer$commute)
```

```
## [1] "  "       " min"     "  MIN"    " "        " minute"  " minutes" ". h"
## [8] " MIN"
```

```r
find_non_numeric_rows(volunteer$commute)
```

```
##  [1] "  120"       "40 min"      " 50 MIN"     " 170"        " 210"
##  [6] "40 minute"   "240 minutes" "1.2 h"       "  180"       " 90 MIN"
## [11] "40 minute"   "30 minutes"  "  180"       "120 MIN"     " 120"
## [16] "50 min"      "100 min"     " 120"        "60 minutes"  "1.2 h"
## [21] "180 minutes" "  150"       "2.3 h"
```

```r
# here we only need to take care of the decimal dots (will be added to the extraction function)
```

We can see a lot of columns with numerical values within strings and additional chars. Furthermore, categorical variables like married and gender contain different labels for the same categories.

```r
# convert to integer to remove leading zeros an than to str
volunteer$personid <- as.character(as.integer(volunteer$personid))
# Clean Numbers with extra characters
# Remove non-numeric characters, except dots and commas and convert to numeric
volunteer$commute <- sapply(volunteer$commute, extract_numeric)
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
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## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```
volunteer$commute <- sapply(volunteer$commute, as_numeric)

# convert "no" strings in the children column to 0 children
volunteer$children <- ifelse(volunteer$children %in% c("No", ""), "0", volunteer$children)
# convert to boolean by first converting to integer
volunteer$children <- as.logical(sapply(volunteer$children, as_integer))
# Convert Tenure to integer, as float dont make sense for months
volunteer$tenure <- sapply(volunteer$tenure, as_integer)

# convert bedroom to boolean by first converting 0s to False, 1 to true
volunteer$bedroom<- ifelse(
    volunteer$bedroom == "0",
    0,
    1
    )
volunteer$bedroom <- sapply(volunteer$bedroom, as_integer)
# normalize gender data by
# lower all charcters
volunteer$gender <- tolower(volunteer$gender)
# trim whitespaces
volunteer$gender <- trimws(volunteer$gender)
# cast all possible options for male and female
volunteer$gender <- ifelse(volunteer$gender %in% c("m", "male", "man"), "male", volunteer$gender)
volunteer$gender <- ifelse(volunteer$gender %in% c("woman", "female", "f", "w","fem"), "female", volunte

# using a similar approach to the gender we convert the married status now
volunteer$married <- tolower(volunteer$married)
# trim whitespaces
volunteer$married <- trimws(volunteer$married)
# cast all possible options for married/not married
volunteer$married <- ifelse(volunteer$married %in% c("married", "yes", "y", "true"), "TRUE", volunteer$m
volunteer$married <- ifelse(volunteer$married %in% c("no", "not married", "n", "na", "single","false"),

# using a similar approach to the married column we convert the high_educ status now
volunteer$high_educ <- tolower(volunteer$high_educ)
# trim whitespaces
volunteer$high_educ <- trimws(volunteer$high_educ)
# cast all possible options for high_educ/not high_educ
volunteer$high_educ <- ifelse(volunteer$high_educ %in% c("yes", "y"), "TRUE", volunteer$high_educ)
volunteer$high_educ <- ifelse(volunteer$high_educ %in% c("no", "n","false"), "FALSE", volunteer$high_edu

# convert to boolean
volunteer$high_educ <- as.logical(volunteer$high_educ)

# create numerical variable cols for gender and married
volunteer$gender_num <- sapply(ifelse(volunteer$gender=="male",0,1), as_integer)
volunteer$married_num <- sapply(ifelse(volunteer$married==FALSE,0,1), as_integer)
volunteer$high_educ_num <- sapply(ifelse(volunteer$high_educ==FALSE,0,1), as_integer)
```

```r
volunteer$children_num <-sapply(ifelse(volunteer$children==FALSE,0,1), as_integer)

# remove duplicates on same personid
volunteer <- volunteer %>%
  distinct(personid, .keep_all = TRUE)
```

```r
# Now we have much better insights using stargazer
stargazer(as.data.frame(volunteer), type = "text", median = TRUE, header = TRUE)
```

```
##
## ==================================================
## Statistic      N   Mean   St. Dev. Min Median Max
## --------------------------------------------------
## age           135 23.452   4.315    1    23    35
## tenure        135 23.385   24.608   2    13    150
## children      135  0.148   0.357    0    0      1
## bedroom       135  0.978   0.148    0    1      1
## commute       122 104.721  68.544   1    85    300
## high_educ     135  0.415   0.495    0    0      1
## gender_num    135  0.496   0.502    0    0      1
## married_num   135  0.200   0.401    0    0      1
## high_educ_num 135  0.415   0.495    0    0      1
## children_num  135  0.148   0.357    0    0      1
## --------------------------------------------------
```

Clean attitude

```r
glimpse(attitude)
```

```
## Rows: 2,379
## Columns: 5
## $ personid   <dbl> 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122,~
## $ year_week  <chr> "202249", "202250", "202251", "202252", "202253", "202302",~
## $ exhaustion <dbl> 9, 8, 8, 6, 12, 12, 12, 12, 10, 12, 11, 12, 12, 12, 14, 12,~
## $ negative   <dbl> 20, 21, 20, 17, 19, 18, 20, 16, 18, 24, 18, 19, 20, 19, 18,~
## $ positive   <dbl> 20, 25, 24, 22, 19, 19, 19, 22, 20, 23, 23, 22, 22, 24, 23,~
```

```r
stargazer(as.data.frame(attitude), type = "text", median = TRUE, header = TRUE)
```

```
##
## ===========================================================
## Statistic    N      Mean     St. Dev.   Min  Median   Max
## -----------------------------------------------------------
## personid   2,379 33,005.050 10,331.780 4,122 36,288 45,442
## exhaustion 2,379   8.612      7.794    0.000 7.000  36.000
## negative   2,379   16.651     6.848    8.000 16.000 40.000
## positive   2,379   24.215     6.668    8.000 24.000 40.000
## -----------------------------------------------------------
```

```r
# checking unique values to decide int vs dbl
unique(attitude$exhaustion)
```

```
##  [1]  9.00  8.00  6.00 12.00 10.00 11.00 14.00 13.00 18.00 15.00 16.00 19.00
## [13] 20.00 23.00 24.00 29.00  2.00  3.00  4.00  8.58  0.00  1.00  7.00 11.64
## [25]  6.32  5.00  6.40 17.00 25.00 27.00  7.55 22.00  9.55 21.00 31.00  9.56
```

```
## [37] 36.00 26.00 32.00 28.00 12.20 33.00 30.00 12.47  9.78  6.27  6.21 34.00
## [49] 35.00  7.04  6.86  8.55  5.99
```

```r
unique(attitude$positive)
```

```
##  [1] 20.00000 25.00000 24.00000 22.00000 19.00000 23.00000 21.00000 18.00000
##  [9] 17.00000  8.00000 16.00000 32.00000 34.00000 30.00000 29.00000 31.00000
## [17] 24.24000 28.00000 22.56000 27.00000 26.00000 28.13000 33.00000 35.00000
## [25]  9.00000 12.00000 14.00000 13.00000 15.00000 27.19000 11.00000 10.00000
## [33] 38.00000 26.28000 36.00000 23.79000 23.48000 37.00000 39.00000 40.00000
## [41] 21.91000 22.13000 23.46000 27.18000 26.88000 27.11000 26.04000 25.04065
## [49] 27.05000
```

```r
unique(attitude$negative)
```

```
##  [1] 20.00000 21.00000 17.00000 19.00000 18.00000 16.00000 24.00000 28.00000
##  [9] 22.00000 26.00000 25.00000 27.00000 10.00000 15.00000 14.00000 13.00000
## [17] 17.55000 11.00000 12.00000 18.53000  9.00000 17.87716  8.00000 14.85000
## [25] 15.47000 23.00000 37.00000 16.22000 30.00000 17.98000 18.19000 32.00000
## [33] 29.00000 40.00000 18.77000 31.00000 19.75000 17.57000 14.89000 15.04000
## [41] 34.00000 39.00000 36.00000 35.00000 15.55000 15.94000 16.75610 15.10000
```

We see that we do not have missing values, and the data types for all columns already fit. We only need to convert personid to a string.

```r
# convert to integer to remove leading zeros an than to str
attitude$personid <- as.character(as.integer(attitude$personid))
# remove duplicates on same personid and year_week
attitude <- attitude %>%
  distinct(personid, year_week, .keep_all = TRUE)
```

Clean endperiod

```r
glimpse(endperiod)
```

```
## Rows: 135
## Columns: 4
## $ personid       <dbl> 4122, 6278, 7720, 8834, 8854, 10098, 10356, 12426, 1297~
## $ promote_switch <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 0~
## $ quitjob        <dbl> 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0~
## $ costofcommute  <chr> "18", "12", "9", "0", "4", "12", "0", "0", "0", "6.00 "~
```

```r
stargazer(as.data.frame(endperiod), type = "text", median = TRUE, header = TRUE)
```

```
##
## =============================================================
## Statistic       N     Mean     St. Dev.   Min  Median  Max
## -------------------------------------------------------------
## personid       135 32,716.780 10,835.700 4,122 37,292 45,442
## promote_switch 135   0.163      0.371      0     0      1
## quitjob        135   0.296      0.458      0     0      1
## -------------------------------------------------------------
```

```r
# check character columns with numerical values to extract for non numeric chars to
# be aware for special cases
find_non_numeric_chars(endperiod$costofcommute)
```

```
##  [1] ". "      ".  / "  "."        "   ." "NA"       "CNY ."
```

```
##  [7] "  "       ","        ".  "     "¥."       " "
```

```r
find_non_numeric_rows(endperiod$costofcommute)
```

```
##  [1] "6.00 "           "-"                 "25.00  / "      "11.8181819915771"
##  [5] "   14.00"     "NA"                "CNY 3.00"         "  6"
##  [9] "2,00"             "17.7272720336914" "  0"              "4.54545450210571"
## [13] "2,00"             "0.00  "           "¥10.00"           " 10"
## [17] "0,00"
```

```r
# here we only need to take care of the decimal dots (will be added to the extraction function), any ot
```

No missing values, promote_switch and quitjob are logical values, though we also will create a numerical version (0-1) for them to compute statistics. costofcommute need to be converted to doubles.

```r
# convert to integer to remove leading zeros an than to str
endperiod$personid <- as.character(as.integer(endperiod$personid))
# convert to integers
endperiod$promote_switch_num <- sapply(endperiod$promote_switch, as_integer)
endperiod$quitjob_num <- sapply(endperiod$quitjob, as_integer)
# convert to logical (too have numerical and categorical variables)
endperiod$promote_switch <- as.logical(endperiod$promote_switch_num)
endperiod$quitjob <- as.logical(endperiod$quitjob_num)
# extract numbers from string. Both, dots and commas are used as decimal separators
endperiod$costofcommute<- sapply(endperiod$costofcommute, extract_numeric)
# remove duplicates on same personid
endperiod <- endperiod %>%
  distinct(personid, .keep_all = TRUE)
```

```r
glimpse(endperiod)
```

```
## Rows: 135
## Columns: 6
## $ personid           <chr> "4122", "6278", "7720", "8834", "8854", "10098", "1~
## $ promote_switch     <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FA~
## $ quitjob            <lgl> FALSE, TRUE, FALSE, FALSE, FALSE, TRUE, FALSE, FALS~
## $ costofcommute      <dbl> 18.00000, 12.00000, 9.00000, 0.00000, 4.00000, 12.0~
## $ promote_switch_num <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, ~
## $ quitjob_num        <int> 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, ~
```

```r
stargazer(as.data.frame(endperiod), type = "text", median = TRUE, header = TRUE)
```

```
##
## =========================================================
## Statistic           N   Mean  St. Dev.  Min  Median  Max
## ---------------------------------------------------------
## promote_switch     135 0.163  0.371     0      0      1
## quitjob            135 0.296  0.458     0      0      1
## costofcommute      127 7.308  7.195   0.000 6.000  55.000
## promote_switch_num 135 0.163  0.371     0      0      1
## quitjob_num        135 0.296  0.458     0      0      1
## ---------------------------------------------------------
```

Cost of commute has two missing values, which needs to be considered during analysis.

Clean performance

```r
glimpse(performance)
```

```
## Rows: 9,870
## Columns: 12
## $ personid          <dbl> 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122~
## $ year_week         <chr> "202201", "202202", "202203", "202204", "202205", "2~
## $ perform1          <chr> "-1.14160859584808", "0.414592355489731", "1.0139772~
## $ phonecall         <chr> "-1.13219404220581", "0.312170952558517", "1.0971519~
## $ phonecallraw      <chr> "223", "499", "649", "709", "403", "760", "268", "73~
## $ homethatweek      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ logphonecall      <chr> "5.40717172622681", "6.21260595321655", "6.475432872~
## $ logcallpersec     <chr> "-5.10147857666016", "-5.15978050231934", "-5.142978~
## $ logcalllength     <chr> "10.5086498260498", "11.372386932373", "11.618411064~
## $ logcall_dayworked <chr> "9.41003799438477", "9.58062744140625", "9.826651573~
## $ logdaysworked     <chr> "1.0986123085022", "1.7917594909668", "1.79175949096~
## $ date              <chr> "2022-01-03", "2022-01-10", "2022-01-17", "2022-01-2~
```

```r
stargazer(as.data.frame(performance), type = "text", median = TRUE, header = TRUE)
```

```
##
## =============================================================
## Statistic      N      Mean     St. Dev.   Min   Median   Max
## -------------------------------------------------------------
## personid     9,870 32,493.740 10,623.780 4,122 36,908 45,442
## homethatweek 9,870   0.196      0.397      0     0      1
## -------------------------------------------------------------
```

```r
unique(performance$homethatweek)
```

```
## [1] 0 1
```

```r
# check character columns with numerical values to extract for non numeric chars to
# be aware for special cases
find_non_numeric_chars(performance$perform1)
```

```
## [1] "."  "-." ".e"
```

```r
#find_non_numeric_rows(performance$perform1)
find_non_numeric_chars(performance$phonecall)
```

```
## [1] "."  "-."
```

```r
find_non_numeric_chars(performance$phonecallraw)
```

```
## [1] "NA" "  " " "
```

```r
find_non_numeric_chars(performance$homethatweek)
```

```
## character(0)
```

```r
find_non_numeric_chars(performance$logphonecall)
```

```
## [1] "."  "NA" ","
```

```r
find_non_numeric_chars(performance$logcallpersec)
```

```
## [1] "."  "NA" ","
```

```r
find_non_numeric_chars(performance$logcalllength)
```

```
## [1] "."  "NA" ","
```

```r
find_non_numeric_chars(performance$logcall_dayworked)
```

```
## [1] "."  "NA" ","
```

```r
find_non_numeric_chars(performance$logdaysworked)
```

```
## [1] "." ","
```

```r
# here we  need to take care of the decimal dots, dashes to negative(negative) and scientifics notion w
```

Personid needs to be converted to a string, as well as all all the call and performance measurements need to be converted to doubles. Homethatweek needs to be converted to a logical and numerical

```r
# convert to integer to remove leading zeros an than to str
performance$personid <- as.character(as.integer(performance$personid))
# trim year_week and remove all not nuemerical characters
performance$year_week <- trimws(performance$year_week)
performance$year_week<- gsub("[^0-9.]", "", performance$year_week)
# convert to float/doubles
performance$perform1 <- sapply(performance$perform1, as_numeric)
performance$phonecall <- sapply(performance$phonecall, as_numeric)
# convert to integer
performance$phonecallraw <- sapply(performance$phonecallraw, as_integer)
# convert to boolean
performance$homethatweek_num <- sapply(performance$homethatweek, as_integer)
performance$homethatweek <- sapply(performance$homethatweek, as_integer)
# convert to float/doubles
performance$logphonecall <- sapply(performance$logphonecall, as_numeric)
performance$logcallpersec <- sapply(performance$logcallpersec, as_numeric)
performance$logcalllength <- sapply(performance$logcalllength, as_numeric)
performance$logcall_dayworked <- sapply(performance$logcall_dayworked, as_numeric)
performance$logdaysworked <- sapply(performance$logdaysworked, as_numeric)
# convert to Date
performance$date <- as.Date(performance$date)

# remove duplicates on same personid, year_week
performance <- performance %>%
  distinct(personid, year_week, .keep_all = TRUE)
```

```r
glimpse(performance)
```

```
## Rows: 9,870
## Columns: 13
## $ personid          <chr> "4122", "4122", "4122", "4122", "4122", "4122", "412~
## $ year_week         <chr> "202201", "202202", "202203", "202204", "202205", "2~
## $ perform1          <dbl> -1.1416086, 0.4145924, 1.0139773, 1.4812315, -0.1150~
## $ phonecall         <dbl> -1.13219404, 0.31217095, 1.09715199, 1.41114438, -0.~
## $ phonecallraw      <dbl> 223, 499, 649, 709, 403, 760, 268, 73, 301, 629, 529~
## $ homethatweek      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ logphonecall      <dbl> 5.407172, 6.212606, 6.475433, 6.563856, 5.998937, 6.~
## $ logcallpersec     <dbl> -5.101479, -5.159781, -5.142978, -5.161007, -5.15217~
## $ logcalllength     <dbl> 10.508650, 11.372387, 11.618411, 11.724862, 11.15110~
## $ logcall_dayworked <dbl> 9.410038, 9.580627, 9.826652, 9.778953, 9.359349, 9.~
```

```
## $ logdaysworked     <dbl> 1.098612, 1.791759, 1.791759, 1.945910, 1.791759, 1.~
## $ date              <date> 2022-01-03, 2022-01-10, 2022-01-17, 2022-01-24, 202~
## $ homethatweek_num  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
```

```
stargazer(as.data.frame(performance), type = "text", median = TRUE, header = TRUE)
```

```
##
## ==============================================================
## Statistic           N     Mean   St. Dev.  Min   Median  Max
## --------------------------------------------------------------
## perform1          9,827 -0.015   0.988   -3.031 0.049   4.163
## phonecall         9,716 -0.006   0.963   -3.113 0.070   5.828
## phonecallraw      9,561 440.258 142.558    1      445   1,264
## homethatweek      9,870  0.196   0.397     0       0       1
## logphonecall      9,574  6.009   0.481   0.000  6.098   7.142
## logcallpersec     9,576 -5.170   0.160   -5.951 -5.169 -1.099
## logcalllength     9,576 11.175   0.521   2.485  11.273 12.116
## logcall_dayworked 9,576  9.474   0.457   2.485  9.552  10.359
## logdaysworked     9,855  1.685   0.282   0.000  1.792   1.946
## homethatweek_num  9,870  0.196   0.397     0       0       1
## --------------------------------------------------------------
```

Almost all measurements have missing data.

Clean wage

```
glimpse(wage)
```

```
## Rows: 3,007
## Columns: 5
## $ basewage    <chr> "1650", "1650", "1650", "1650", "1650", "1650", "1800", "18~
## $ grosswage   <chr> "3650.76000976562", "4775.52001953125", "4358", "4801", "40~
## $ personid    <chr> "4122", "4122", "4122", "4122", "4122", "4122", "4122", "41~
## $ wage_month <chr> "202201", "202202", "202203", "202204", "202205", "202206",~
## $ bonustotal <chr> "2000.76000976562", "3125.52001953125", "2708", "3151", "23~
```

```
stargazer(as.data.frame(wage), type = "text", median = TRUE, header = TRUE)
```

```
##
## ========================================
## Statistic N Mean St. Dev. Min Median Max
## ========================================
```

```
# check character columns with numerical values to extract for non numeric chars to
# be aware for special cases
find_non_numeric_chars(wage$basewage)
```

```
## [1] ","  " "  "  " " ." ",."
```

```
find_non_numeric_rows(wage$basewage)
```

```
##  [1] "1600,00"   " 1600"    " 1700"    " 1600"    " 1700"    "  1600"
##  [7] "  1600"   "1800,00"   "2 000.00" "1700,00"  "  1700"   "  1700"
## [13] " 1700"    " 1700"    "  1750"   "  2100"   "2,300.00" "1,500.00"
## [19] " 1650"    "  2300"   " 1700"    " 1450"    " 1600"    "1400,00"
## [25] "  1800"   "1400,00"   "1400,00"  "1,550.00" " 1750"    "1,550.00"
## [31] " 1500"    "1 400.00" "  1550"   "  1800"   " 1400"    "  1400"
```

```
## [37] "1550,00"  " 1500"    "  1750"   " 1850"    "1 400.00" "  1600"
## [43] "  1600"   "1650,00"  "1,500.00" "  1750"   " 1500"    "1800,00"
## [49] "1 500.00" " 1500"    "1500,00"  "1,500.00"
```

```
# only trimming needed
```

We have no numerical values yet. basewage, can be converted to integer, gross_wage and bonustotal can be converted to double

```
# convert to integer to remove leading zeros an than to str
wage$personid <- as.character(as.integer(wage$personid))
wage$basewage <- as.integer(wage$basewage)
```

```
## Warning: NAs introduced by coercion
```

```
wage$grosswage <- sapply(wage$grosswage, as_numeric)
wage$wage_month <- trimws(wage$wage_month)
wage$wage_month<- gsub("[^0-9.]", "", wage$wage_month)
wage <- rename(wage, year_month=wage_month)
wage$bonustotal <- sapply(wage$bonustotal, as_numeric)
# remove duplicates on same personid, yearmonth
wage <- wage %>%
  distinct(personid, year_month, .keep_all = TRUE)
```

```
glimpse(wage)
```

```
## Rows: 3,007
## Columns: 5
## $ basewage   <int> 1650, 1650, 1650, 1650, 1650, 1650, 1800, 1800, 1800, 1800,~
## $ grosswage  <dbl> 3650.76, 4775.52, 4358.00, 4801.00, 4045.12, 5497.76, 3123.~
## $ personid   <chr> "4122", "4122", "4122", "4122", "4122", "4122", "4122", "41~
## $ year_month <chr> "202201", "202202", "202203", "202204", "202205", "202206",~
## $ bonustotal <dbl> 2000.76, 3125.52, 2708.00, 3151.00, 2395.12, 3847.76, 1323.~
```

```
stargazer(as.data.frame(wage), type = "text", median = TRUE, header = TRUE)
```

```
##
## ================================================================
## Statistic    N     Mean    St. Dev.   Min    Median      Max
## ----------------------------------------------------------------
## basewage   2,987 1,666.271  221.174   650     1,600     2,850
## grosswage  2,978 3,135.544 1,066.355 48.850 2,979.950 14,553.000
## bonustotal 2,993 1,502.263  926.276  0.000  1,325.170 12,853.000
## ----------------------------------------------------------------
```

All columns again contain missing data. Grosswage is not the sum of basewage+bonustotal all the time: Cases:

- NA in basewage and grosswage or basewage and bonus total or bonustotal and grosswage: remove row
- NA in basewage but grosswage and bonustotal are given: basewage = grosswage - bonustotal
- NA in bonustotal but grosswage and basewage are given: bonustotal = grosswage - basewage
- Grosswage != bonuswage+basewage: recalculate
- wage yearmonth sometimes misses the month

```
# case 2
wage <- wage %>% filter(!is.na(basewage) & !is.na(grosswage) | !is.na(bonustotal) & !is.na(grosswage) |
```

```r
# case 3
wage <- wage %>% mutate(basewage = ifelse(is.na(basewage), grosswage-bonustotal, basewage))

# case 4
wage <- wage %>% mutate(bonustotal = ifelse(is.na(bonustotal), grosswage-basewage, bonustotal))

# case 5
wage <- wage %>% mutate(grosswage = ifelse(grosswage != bonustotal+basewage, bonustotal+basewage, grossw

# case 6
wage <- wage %>% filter(nchar(year_month)==6)
```

Merging data together

Merging

```r
# First we merge all volunteer related data together on personid (unique for a person)
volunteer_endperiod <- merge(volunteer, endperiod, by="personid", all=TRUE)
# Second we merge all weekly based timeseries data together per personid
attitude_performance <- merge(attitude, performance, by=c("personid","year_week"), all=TRUE)
# As wage only has a yearmonth column, we cannot directly merge it with an other table containing times
# The goal now is to create one big dataframe with all volunteer information, their performance, attitu
attitude_performance <- attitude_performance %>%
  mutate(
    week_date = ISOweek2date(
      paste0(substr(year_week,1,4), "-W", substr(year_week,5,6), "-1")
    ),
    year_month = format(week_date, "%Y%m")  # gives e.g. "202401", "202412"
  )
# merge on year_month
attitude_performance_wage <- merge(attitude_performance, wage, by=c("personid","year_month"), all.x=TRUE
# merge on personid
volunteer_endperiod_attitude_performance_wage <- merge(attitude_performance_wage, volunteer_endperiod, b

# lets try to get a dataframe with ALL data and no NAs at all
distinct_all <- na.omit(volunteer_endperiod_attitude_performance_wage)
```

Analyzing merged dataframes

```r
stargazer(as.data.frame(volunteer_endperiod), type = "text", median = TRUE, header = TRUE)
```

```
##
## ==============================================================
## Statistic           N   Mean   St. Dev.  Min  Median  Max
## --------------------------------------------------------------
## age                135 23.452   4.315     1     23     35
## tenure             135 23.385   24.608    2     13    150
## children           135  0.148   0.357     0     0      1
## bedroom            135  0.978   0.148     0     1      1
## commute            122 104.721  68.544    1     85    300
## high_educ          135  0.415   0.495     0     0      1
## gender_num         135  0.496   0.502     0     0      1
```

```
## married_num        135  0.200   0.401     0     0      1
## high_educ_num      135  0.415   0.495     0     0      1
## children_num       135  0.148   0.357     0     0      1
## promote_switch     135  0.163   0.371     0     0      1
## quitjob            135  0.296   0.458     0     0      1
## costofcommute      127  7.308   7.195   0.000 6.000   55.000
## promote_switch_num 135  0.163   0.371     0     0      1
## quitjob_num        135  0.296   0.458     0     0      1
## -------------------------------------------------------------
```

```r
# looks good
stargazer(as.data.frame(attitude_performance), type = "text", median = TRUE, header = TRUE)
```

```
##
## ===============================================================
## Statistic           N    Mean   St. Dev.  Min   Median  Max
## ---------------------------------------------------------------
## exhaustion        2,379  8.612   7.794   0.000  7.000  36.000
## negative          2,379 16.651   6.848   8.000  16.000 40.000
## positive          2,379 24.215   6.668   8.000  24.000 40.000
## perform1          9,827 -0.015   0.988   -3.031 0.049  4.163
## phonecall         9,716 -0.006   0.963   -3.113 0.070  5.828
## phonecallraw      9,561 440.258 142.558    1     445   1,264
## homethatweek      9,870  0.196   0.397     0      0      1
## logphonecall      9,574  6.009   0.481   0.000  6.098  7.142
## logcallpersec     9,576 -5.170   0.160   -5.951 -5.169 -1.099
## logcalllength     9,576 11.175   0.521   2.485  11.273 12.116
## logcall_dayworked 9,576  9.474   0.457   2.485  9.552  10.359
## logdaysworked     9,855  1.685   0.282   0.000  1.792  1.946
## homethatweek_num  9,870  0.196   0.397     0      0      1
## ---------------------------------------------------------------
```

```r
# when using this dataframe we have to be cautions as attitude has way less columns than attitude n(att
stargazer(as.data.frame(attitude_performance_wage), type = "text", median = TRUE, header = TRUE)
```

```
##
## ============================================================================
## Statistic           N      Mean    St. Dev.    Min      Median      Max
## ----------------------------------------------------------------------------
## exhaustion        2,379    8.612    7.794     0.000     7.000     36.000
## negative          2,379   16.651    6.848     8.000     16.000    40.000
## positive          2,379   24.215    6.668     8.000     24.000    40.000
## perform1          9,827   -0.015    0.988    -3.031     0.049     4.163
## phonecall         9,716   -0.006    0.963    -3.113     0.070     5.828
## phonecallraw      9,561   440.258  142.558      1        445      1,264
## homethatweek      9,870    0.196    0.397       0         0         1
## logphonecall      9,574    6.009    0.481     0.000     6.098     7.142
## logcallpersec     9,576   -5.170    0.160    -5.951    -5.169    -1.099
## logcalllength     9,576   11.175    0.521     2.485     11.273    12.116
## logcall_dayworked 9,576    9.474    0.457     2.485     9.552     10.359
## logdaysworked     9,855    1.685    0.282     0.000     1.792     1.946
## homethatweek_num  9,870    0.196    0.397       0         0         1
## basewage          10,023 1,601.750 176.618   527.350  1,600.000 2,450.000
## grosswage         9,919  3,097.672 990.788  1,140.280 2,877.000 14,553.000
## bonustotal        10,023 1,494.926 912.498    0.000   1,284.590 12,853.000
```

```
## ---------------------------------------------------------------------------
```

```r
stargazer(as.data.frame(volunteer_endperiod_attitude_performance_wage), type = "text", median = TRUE, he
```

```
##
## ============================================================================
## Statistic             N      Mean    St. Dev.    Min      Median      Max
## ----------------------------------------------------------------------------
## exhaustion          2,379    8.612     7.794     0.000     7.000     36.000
## negative            2,379   16.651     6.848     8.000    16.000     40.000
## positive            2,379   24.215     6.668     8.000    24.000     40.000
## perform1            9,827   -0.015     0.988    -3.031     0.049     4.163
## phonecall           9,716   -0.006     0.963    -3.113     0.070     5.828
## phonecallraw        9,561   440.258  142.558      1         445      1,264
## homethatweek        9,870    0.196     0.397      0          0         1
## logphonecall        9,574    6.009     0.481     0.000     6.098     7.142
## logcallpersec       9,576   -5.170     0.160    -5.951    -5.169    -1.099
## logcalllength       9,576   11.175     0.521     2.485    11.273    12.116
## logcall_dayworked   9,576    9.474     0.457     2.485     9.552    10.359
## logdaysworked       9,855    1.685     0.282     0.000     1.792     1.946
## homethatweek_num    9,870    0.196     0.397      0          0         1
## basewage           10,023 1,601.750 176.618   527.350  1,600.000 2,450.000
## grosswage           9,919 3,097.672 990.788  1,140.280 2,877.000 14,553.000
## bonustotal         10,023 1,494.926 912.498    0.000   1,284.590 12,853.000
## age                10,079   23.485     4.191      1         23        35
## tenure             10,079   23.893    24.773      2         18        150
## children           10,079    0.141     0.348      0          0         1
## bedroom            10,079    0.974     0.160      0          1         1
## commute             9,118   103.519   67.620      1         80        300
## high_educ          10,079    0.400     0.490      0          0         1
## gender_num         10,079    0.498     0.500      0          0         1
## married_num        10,079    0.181     0.385      0          0         1
## high_educ_num      10,079    0.400     0.490      0          0         1
## children_num       10,079    0.141     0.348      0          0         1
## promote_switch     10,079    0.182     0.386      0          0         1
## quitjob            10,079    0.238     0.426      0          0         1
## costofcommute       9,488    7.213     7.231     0.000     6.000     55.000
## promote_switch_num 10,079    0.182     0.386      0          0         1
## quitjob_num        10,079    0.238     0.426      0          0         1
## ----------------------------------------------------------------------------
```

```r
stargazer(as.data.frame(distinct_all), type = "text", median = TRUE, header = TRUE)
```

```
##
## ============================================================================
## Statistic             N     Mean    St. Dev.     Min      Median      Max
## ----------------------------------------------------------------------------
## exhaustion          1,785   8.983     8.105     0.000     8.000     36.000
## negative            1,785  16.699     7.088     8.000    16.000     40.000
## positive            1,785  23.718     6.730     8.000    24.000     40.000
## perform1            1,785   0.070     0.866    -2.906     0.096     2.772
## phonecall           1,785   0.030     0.814    -3.098     0.046     2.266
## phonecallraw        1,785  439.319   126.850      2        442       835
## homethatweek        1,785   0.508     0.500      0          1         1
## logphonecall        1,785   6.032     0.367     0.693     6.091     6.727
```

```
## logcallpersec      1,785  -5.158     0.129    -5.541    -5.157     -4.075
## logcalllength      1,785  11.191     0.375     5.561    11.242     11.952
## logcall_dayworked  1,785   9.465     0.354     3.951     9.506     10.104
## logdaysworked      1,785   1.726     0.182     0.693     1.792      1.946
## homethatweek_num   1,785   0.508     0.500       0         1          1
## basewage           1,785 1,683.305  158.587    1,300     1,650      2,150
## grosswage          1,785 3,328.000 1,046.364 1,740.000 3,095.000 14,553.000
## bonustotal         1,785 1,644.695  969.980   40.000   1,408.500 12,853.000
## age                1,785  23.519     3.254      18        23         34
## tenure             1,785  23.685    26.683       2        13        150
## children           1,785   0.076     0.264       0         0          1
## bedroom            1,785   0.941     0.235       0         1          1
## commute            1,785  104.871   67.073      20        80        300
## high_educ          1,785   0.345     0.475       0         0          1
## gender_num         1,785   0.459     0.498       0         0          1
## married_num        1,785   0.111     0.314       0         0          1
## high_educ_num      1,785   0.345     0.475       0         0          1
## children_num       1,785   0.076     0.264       0         0          1
## promote_switch     1,785   0.182     0.386       0         0          1
## quitjob            1,785   0.000     0.000       0         0          0
## costofcommute      1,785   6.749     6.160     0.000     5.000     30.000
## promote_switch_num 1,785   0.182     0.386       0         0          1
## quitjob_num        1,785   0.000     0.000       0         0          0
## ----------------------------------------------------------------------------
```

When joining everything together we can see that we do not have all data for each person and its work weeks, what was to be expected. When removing all NAs we can see that the numbers slightly change and for examples all quitters are gone. Therefore we have to use the volunteer_endperiod_attitude_performance_wage with caution and eliminate NAs for each calculation individually. Though this dataframe is handy to use for the further work.

volunteer_enperiod: Cross sectional data of all employee related information.

attitude_performance: Panel data on how an employee performed in a certain week of the year and how he/she felt.

attitude_performance_wage: Panel data on how an employee performed in a certain week of the year, how he/she felt in that week and their wage during that month. Caution: contains two different time series.

volunteer_endperiod_attitude_performance_wage: Like attitude_performance_wage, but with volunteer_endperod data.

General Analysis

Population: All employees of Huaxia
Our sample: Depending on the table we use and what we collect the unit of interest might differ:
Attitude, performance (time series): weekly observations of a person
volunteer (cross sectional): the person itself
wage (time series): the wage of a person in a month
endperiod (cross sectional): behavior of a person(?)
volunteer_endperiod_attitude_performance_wage(panel data): repeated observations of different employees across multiple time periods, combining attitudes performance, volunteer, endperiod, and wages

Get first basic insights of the tables before eliminating null values and merging data together. For a better overview we will look at each table alone, not on the composition table.

Statistics

Performance

```
stargazer(as.data.frame(performance), type = "text", median = TRUE, header = TRUE)
```

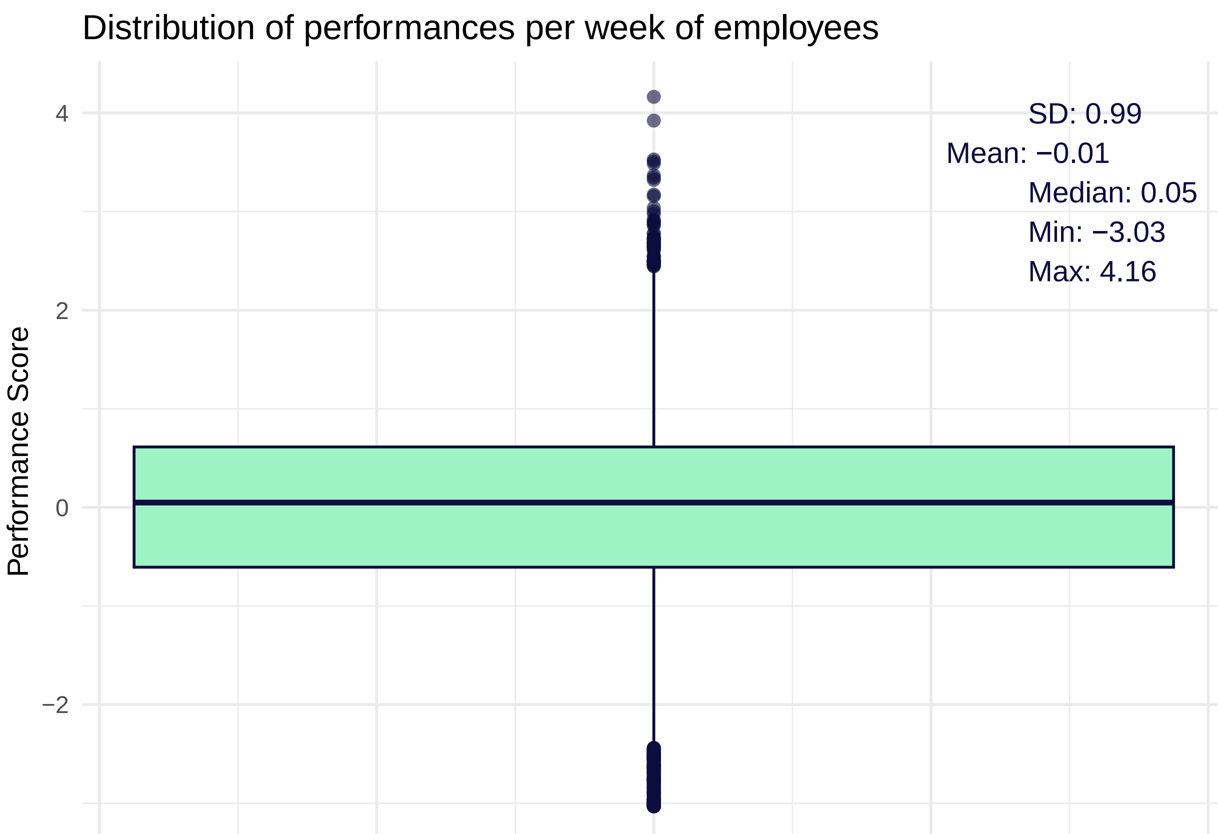
```
##
## ==============================================================
## Statistic           N    Mean   St. Dev.  Min   Median  Max
## --------------------------------------------------------------
## perform1          9,827 -0.015   0.988   -3.031 0.049   4.163
## phonecall         9,716 -0.006   0.963   -3.113 0.070   5.828
## phonecallraw      9,561 440.258 142.558    1     445    1,264
## homethatweek      9,870  0.196   0.397     0      0       1
## logphonecall      9,574  6.009   0.481   0.000  6.098   7.142
## logcallpersec     9,576 -5.170   0.160   -5.951 -5.169 -1.099
## logcalllength     9,576 11.175   0.521   2.485  11.273 12.116
## logcall_dayworked 9,576  9.474   0.457   2.485  9.552  10.359
## logdaysworked     9,855  1.685   0.282   0.000  1.792   1.946
## homethatweek_num  9,870  0.196   0.397     0      0       1
## --------------------------------------------------------------
```

```
# perform plot
stats <- performance %>% summarise(
  min   = min(perform1, na.rm = TRUE),
  max   = max(perform1, na.rm = TRUE),
  mean = mean(perform1, na.rm = TRUE),
  median = median(perform1, na.rm = TRUE),
  sd    = sd(perform1, na.rm = TRUE)
)

p<-ggplot(performance, aes(y = perform1)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Performance Score", title = "Distribution of performances per week of employees") +
  theme_minimal() +
  theme(axis.title.x = element_blank(),
      axis.text.x  = element_blank(),
      axis.ticks.x = element_blank())

# add annotations
p + annotate("text", x = 0.27, y = 4,
          label = paste0("SD: ", round(stats$sd,2)),
          hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 0.27, y = 3.6,
          label = paste0("Mean: ", round(stats$mean,2)),
          text = 0, color = "#0d0c3f")+
  annotate("text", x = 0.27, y = 3.2,
          label = paste0("Median: ", round(stats$median,2)),
          hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 0.27, y = 2.8,
```

```
          label = paste0("Min: ", round(stats$min,2)),
          hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 0.27, y = 2.4,
          label = paste0("Max: ", round(stats$max,2)),
          hjust = 0, color = "#0d0c3f")
```

```
## Warning in annotate("text", x = 0.27, y = 3.6, label = paste0("Mean: ", :
## Ignoring unknown parameters: `text`
```

```
## Warning: Removed 43 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
ggsave("performance_distribution.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 43 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

A lot of missing values, 296 at max, which will fall out from the data set when working with it. Working from home rate is at 19,64% on average, probably as a result of a Work-from-home-policy.

Volunteer

```
stargazer(as.data.frame(volunteer), type = "text", median = TRUE, header = TRUE)
```

```
##
## ===================================================
## Statistic       N   Mean   St. Dev. Min Median Max
## ---------------------------------------------------
## age            135 23.452   4.315    1    23    35
```

```
## tenure        135 23.385   24.608   2    13   150
## children      135  0.148   0.357    0    0     1
## bedroom       135  0.978   0.148    0    1     1
## commute       122 104.721  68.544   1    85   300
## high_educ     135  0.415   0.495    0    0     1
## gender_num    135  0.496   0.502    0    0     1
## married_num   135  0.200   0.401    0    0     1
## high_educ_num 135  0.415   0.495    0    0     1
## children_num  135  0.148   0.357    0    0     1
## ------------------------------------------------
```

No missing values. The average worker is 23,5 years old with a mean commute time of 100 minutes per week while there are greater differences, and is for almost 23 months within the company. 15% have children, 20% are married, more than 40% have a higher education. Both, men and women work at the company.

Endperiod

```
stargazer(as.data.frame(endperiod), type = "text", median = TRUE, header = TRUE)
```

```
##
## ==========================================================
## Statistic         N   Mean  St. Dev.  Min  Median  Max
## ----------------------------------------------------------
## promote_switch    135 0.163  0.371     0     0      1
## quitjob           135 0.296  0.458     0     0      1
## costofcommute     127 7.308  7.195   0.000 6.000  55.000
## promote_switch_num 135 0.163  0.371     0     0      1
## quitjob_num       135 0.296  0.458     0     0      1
## ----------------------------------------------------------
```

On average, an employee had cost of commute of 7.5 but this number may heavily differ to some employees, 16% got a promotion, while almost 30% quit their job at some point.

Wage

```
stargazer(as.data.frame(wage), type = "text", median = TRUE, header = TRUE)
```

```
##
## ====================================================================
## Statistic    N     Mean     St. Dev.     Min      Median      Max
## --------------------------------------------------------------------
## basewage   2,997 1,665.835  222.317    527.350   1,600.000 2,850.000
## grosswage  2,968 3,170.622 1,023.961    53.100   2,986.500 14,553.000
## bonustotal 2,997 1,502.992  929.753  -1,596.900 1,325.170 12,853.000
## --------------------------------------------------------------------
```

```
stats <- wage %>% summarise(
  min  = min(grosswage, na.rm = TRUE),
  max  = max(grosswage, na.rm = TRUE),
  mean = mean(grosswage, na.rm = TRUE),
  median = median(grosswage, na.rm = TRUE),
  sd   = sd(grosswage, na.rm = TRUE)
)

p <- ggplot(wage, aes(y = grosswage)) +
```

```r
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "€ per month", title = "Distribution of Gross Wages") +
  theme_minimal()

# add annotations
p + annotate("text", x = 0.27, y = 14000,
           label = paste0("SD: ", round(stats$sd,0), "€"),
           hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 0.27, y = 13000,
           label = paste0("Mean: ", round(stats$mean,0), "€"),
           hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 0.27, y = 12000,
           label = paste0("Median: ", round(stats$median,0), "€"),
           hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 0.27, y = 11000,
           label = paste0("Min: ", round(stats$min,0), "€"),
           hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 0.27, y = 10000,
           label = paste0("Max: ", round(stats$max,0), "€"),
           hjust = 0, color = "#0d0c3f")
```

```
## Warning: Removed 29 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```


```
ggsave("gross_wage_distribution.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 29 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

On average an employee had a base wage of around 1650€ which can slightly differ. A heavily uneven distribution can be seen for the bonuses for employees, while some refer no bonus at all others receive almost 13.000€ as bonus, on average it is 900€ per employee.

Attitude

```
stargazer(as.data.frame(attitude), type = "text", median = TRUE, header = TRUE)
```

```
##
## ====================================================
## Statistic    N    Mean  St. Dev.  Min  Median  Max
## ----------------------------------------------------
## exhaustion 2,379 8.612   7.794   0.000 7.000  36.000
## negative   2,379 16.651  6.848   8.000 16.000 40.000
## positive   2,379 24.215  6.668   8.000 24.000 40.000
## ----------------------------------------------------
```

Exhaustion scores are mostly low and clustered around 7–9 but some people reach much higher (up to 36), negative scores center near 16 with moderate variation across the group, and positive scores center near 24 with a similar moderate spread, meaning most participants stay within roughly one standard deviation (about ±7 points) of these averages but a few reach the extreme minimum or maximum values

All together

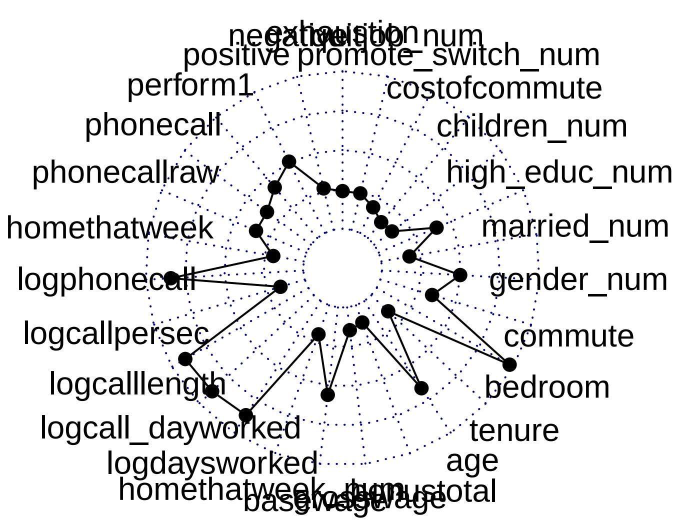
```r
stargazer(as.data.frame(volunteer_endperiod_attitude_performance_wage), type = "text", median = TRUE, ho
```

```
##
## ==============================================================================
## Statistic             N      Mean    St. Dev.    Min      Median      Max
## ------------------------------------------------------------------------------
## exhaustion          2,379    8.612    7.794     0.000     7.000      36.000
## negative            2,379   16.651    6.848     8.000    16.000      40.000
## positive            2,379   24.215    6.668     8.000    24.000      40.000
## perform1            9,827   -0.015    0.988    -3.031     0.049      4.163
## phonecall           9,716   -0.006    0.963    -3.113     0.070      5.828
## phonecallraw        9,561  440.258  142.558       1        445       1,264
## homethatweek        9,870    0.196    0.397       0          0         1
## logphonecall        9,574    6.009    0.481     0.000     6.098      7.142
## logcallpersec       9,576   -5.170    0.160    -5.951    -5.169     -1.099
## logcalllength       9,576   11.175    0.521     2.485    11.273     12.116
## logcall_dayworked   9,576    9.474    0.457     2.485     9.552     10.359
## logdaysworked       9,855    1.685    0.282     0.000     1.792      1.946
## homethatweek_num    9,870    0.196    0.397       0          0         1
## basewage            10,023 1,601.750 176.618   527.350  1,600.000 2,450.000
## grosswage           9,919  3,097.672 990.788 1,140.280 2,877.000 14,553.000
## bonustotal          10,023 1,494.926 912.498   0.000   1,284.590 12,853.000
## age                 10,079  23.485    4.191       1         23         35
## tenure              10,079  23.893    24.773      2         18        150
## children            10,079   0.141    0.348       0          0         1
## bedroom             10,079   0.974    0.160       0          1         1
## commute             9,118   103.519   67.620      1         80        300
## high_educ           10,079   0.400    0.490       0          0         1
## gender_num          10,079   0.498    0.500       0          0         1
## married_num         10,079   0.181    0.385       0          0         1
## high_educ_num       10,079   0.400    0.490       0          0         1
## children_num        10,079   0.141    0.348       0          0         1
## promote_switch      10,079   0.182    0.386       0          0         1
## quitjob             10,079   0.238    0.426       0          0         1
## costofcommute       9,488    7.213    7.231     0.000     6.000      55.000
## promote_switch_num  10,079   0.182    0.386       0          0         1
## quitjob_num         10,079   0.238    0.426       0          0         1
## ------------------------------------------------------------------------------
```

```r
df <- volunteer_endperiod_attitude_performance_wage
df <- dplyr::select(df, where(is.numeric))

# create a summary df as preparation for the radar plot rendering
create_summary <- function(df, drop_all_na = TRUE, keep_group_index = TRUE) {
  num <- df %>%
    select(where(is.numeric)) %>%
    mutate(across(everything(), ~ ifelse(is.finite(.x), .x, NA_real_)))

  if (drop_all_na) {
    keep <- vapply(num, function(x) any(!is.na(x)), logical(1))
    num <- num[, keep, drop = FALSE]
  }
  stopifnot(ncol(num) > 0)
```

```r
  summary <- num %>%
    summarise(
      across(everything(),
         list(mean = ~mean(.x, na.rm = TRUE),
              min  = ~min(.x,  na.rm = TRUE),
              max  = ~max(.x,  na.rm = TRUE)),
         .names = "{.col}_{.fn}"))
  summary
}



# Expects a dataframe with *_min, *_max, *_mean columns
render_radar_plot <- function (df, pad = 1e-6) {
  # separate min, max, mean
  max_vals <- df %>% select(ends_with("_max")) %>% rename_with(~ sub("_max$", "", .x))
  min_vals <- df %>% select(ends_with("_min")) %>% rename_with(~ sub("_min$", "", .x))
  mean_vals <- df %>% select(ends_with("_mean")) %>% rename_with(~ sub("_mean$", "", .x))

  # avoid zero-range axes (min == max) that break fmsb scaling
  rng <- as.numeric(max_vals[1, ] - min_vals[1, ])
  z   <- which(is.na(rng) | rng == 0)
  if (length(z)) {
    max_vals[1, z] <- as.numeric(max_vals[1, z]) + pad
    min_vals[1, z] <- as.numeric(min_vals[1, z]) - pad
  }

  web <- rbind(max_vals, min_vals, mean_vals)
  web[] <- lapply(web, as.numeric)

  radarchart(web, na.itp = TRUE)
}

radar_by_group <- function(df, group_col = "group_index", stat = "mean") {
  num <- df %>%
    mutate(across(where(is.numeric), ~ ifelse(is.finite(.x), .x, NA_real_)))

  # 1) aggregate per group
  fun <- switch(stat,
                mean = ~mean(.x, na.rm = TRUE),
                median = ~median(.x, na.rm = TRUE),
                stop("stat must be 'mean' or 'median'"))
  grp <- num %>%
    group_by(.data[[group_col]]) %>%
    summarise(across(where(is.numeric), fun), .groups = "drop")

  # 2) build fmsb input: first two rows = per-variable max/min over all groups
  vals <- grp %>% select(-all_of(group_col))
  rownames(vals) <- grp[[group_col]]
  max_vals <- summarise_all(vals, max, na.rm = TRUE)
  min_vals <- summarise_all(vals, min, na.rm = TRUE)

  # pad zero-range axes
```

```
  rng <- as.numeric(max_vals[1, ] - min_vals[1, ])
  z <- which(is.na(rng) | rng == 0)
  if (length(z)) {
    max_vals[1, z] <- as.numeric(max_vals[1, z]) + 1e-6
    min_vals[1, z] <- as.numeric(min_vals[1, z]) - 1e-6
  }

  web <- rbind(max_vals, min_vals, vals)

  # 3) colors and legend
  n <- nrow(web) - 2
  pal <- brewer.pal(max(3, min(8, n)), "Set2")[seq_len(n)]
  fill <- adjustcolor(pal, alpha.f = 0.25)

  radarchart(
    web,
    pcol   = pal,
    pfcol  = fill,
    plwd   = 2,
    plty   = 1,
    cglcol = "grey80",
    cglty  = 1,
    cglwd  = 0.8,
    axislabcol = "grey30",
    vlcex  = 0.8
  )

  legend("topright",
         legend = rownames(web)[-c(1,2)],
         bty = "n", pch = 15, pt.cex = 1.3,
         col = fill, text.col = "grey20")
}

summary <- create_summary(df)
render_radar_plot(summary)
```

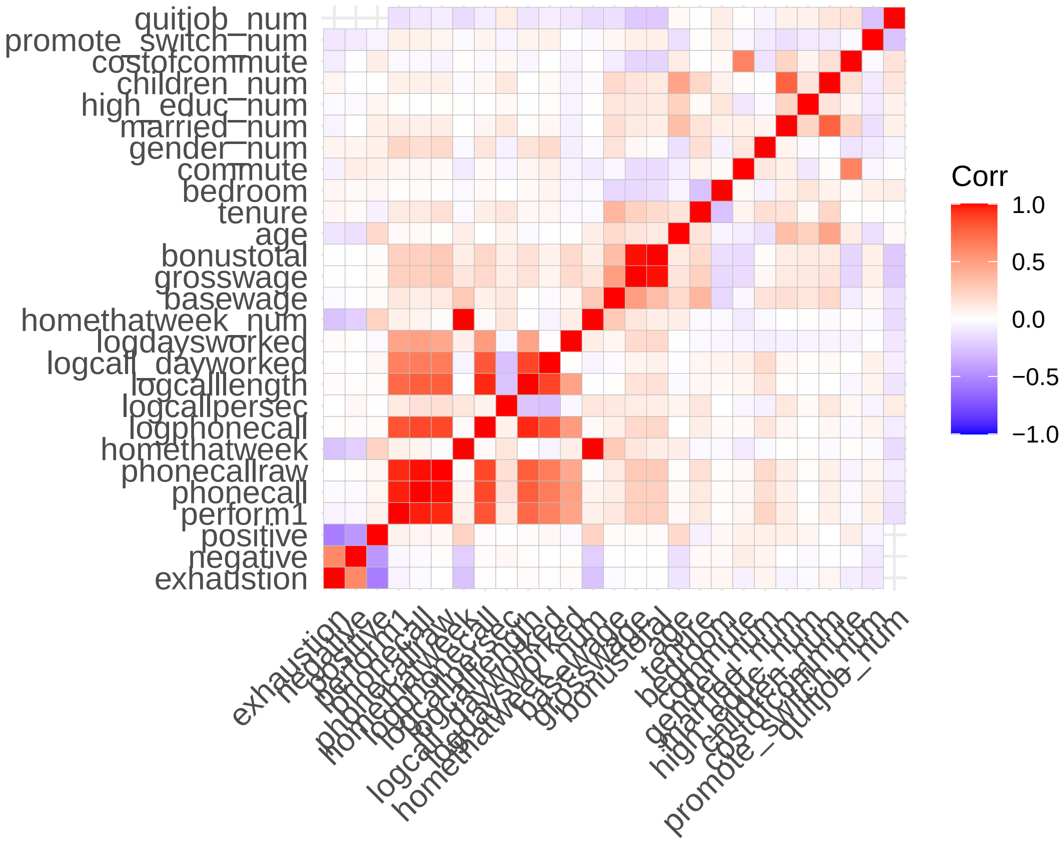


Basic correlation analysis

```r
nums <- dplyr::select(volunteer_endperiod_attitude_performance_wage, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```r
ggcorrplot(cm, lab = FALSE)
```



Here we can see that the performance metric "perform1" has a correlation to phonecall, phonecallraw, logphonecall, logcalllength, which gives us a hint on how this metrics was calculated. Furthermore we can see, a negative correlation between "negative" and "positive" from the attitude dataset, which makes sense as these are opposites. Because of the timeseries data of the performance data set, the light correlation can be explained as the attitude may change from day to day, or not (what we can not infer from that). There is also a slight correlation between exhaustion and negative attitude, which seems logical. What we cannot see are significant correlations between quit_job and any variable and other than the cluster itself between perfrom1 and any other variable. Quit_job has moderate negative correlations to bonuses/grosswage and promotions.

We cannot see any clear indicators on how to fight churn rate and/or what motivates people to perform better.

Detailed exploration analysis

Function for radar plots

```
# A funcion to render web plots for out data with some modificatino parameters
# The function first selects the desired column and than summarises the data based on what groups to co
radar_by_group <- function(df, include_group_indices = c(),group_col = "group_index", stat = "mean", ti
  # open device to safe as png
  if(save==TRUE){
    png(paste0(title,".png"), width = 2400, height = 2000, res = 300)
  }
  # check for parameters
  stopifnot(group_col %in% names(df))
  # select, based on parameters, columns
 final_selection = c(group_col, "gender_num", "married_num", "bedroom", "children_num", "high_educ_num"
  if (show_full_attitude==TRUE) {
    selection = c("negative","positive")
    final_selection <- c(selection, final_selection)
  }
  if (show_full_performance==TRUE) {
    selection = c("logdaysworked","logcall_dayworked","logcalllength","logcallpersec","logphonecall","p
    final_selection <- c(selection, final_selection)
  }
  if (show_full_wage==TRUE) {
      selection = c("bonustotal","basewage")
      final_selection <- c(selection, final_selection)
  }
  if (show_commute==TRUE) {
      selection = c("commute","costofcommute")
      final_selection <- c(selection, final_selection)
  }
  # chek if group col is in any of the selected field, if so, remove it
  if (group_col %in% sub("_num$", "", final_selection)){
    final_selection <- final_selection[final_selection!=paste0(group_col,"_num")]
  }
  print(final_selection)
  df <- df%>% select(final_selection)
  # only used numeric columns (+ group column) for computation and only real numbers
  num <- df %>%
    select(all_of(group_col), where(is.numeric)) %>%
    mutate(across(where(is.numeric), ~ ifelse(is.finite(.x), .x, NA_real_)))

  # remove "_num" to shorten names of var
  num <- num %>% rename_with(~ sub("_num$", "", .x))
  # select the min max values based on the 90%/10% quantile of the overall data
  max_vals <- summarise(num, across(where(is.numeric), ~ quantile(.x, 0.9, na.rm = TRUE)))
  min_vals <- summarise(num, across(where(is.numeric), ~ quantile(.x, 0.1, na.rm = TRUE)))
  # aggregate per group (mean or median)
  agg_fun <- switch(stat,
                    mean = ~mean(.x, na.rm = TRUE),
                    median = ~median(.x, na.rm = TRUE),
                    stop("stat must be 'mean' or 'median'"))
  vals <- num %>%
    group_by(.data[[group_col]]) %>%
```

```r
  summarise(across(where(is.numeric), agg_fun), .groups = "drop")

# get the correct label
if (length(include_group_indices)>0){
  print(vals)
  print(group_col)
  print(include_group_indices)
  labels <- vals %>% filter(.data[[group_col]] %in% include_group_indices)
  print(labels)
  labels <- as.character(labels[[group_col]])
  print(labels)
}
# again check for not numeric values and drop them if needed
keep <- vapply(vals, function(x) any(!is.na(x)), logical(1))
vals <- vals[, keep, drop = FALSE]
# check for enough values to display
if (ncol(vals) < 3) stop("Need  3 valid numeric variables after cleaning.")
# fill remaining NA cells with column means to avoid NA polygons
cm <- vapply(vals, function(x) mean(x, na.rm = TRUE), numeric(1))
for (j in seq_along(vals)) vals[[j]][is.na(vals[[j]])] <- cm[j]
# per-variable max/min, pad zero ranges
rng <- as.numeric(max_vals[1, ] - min_vals[1, ])
z <- which(is.na(rng) | rng == 0)
pad <- 1e-6
rng <- as.numeric(max_vals[1, ] - min_vals[1, ])
z <- which(is.na(rng) | rng == 0)
if (length(z)) {
  mx <- as.numeric(max_vals[1, ])
  mn <- as.numeric(min_vals[1, ])
  mx[z] <- mx[z] + pad
  mn[z] <- mn[z] - pad
  max_vals[1, ] <- as.list(mx)
  min_vals[1, ] <- as.list(mn)
}
# ensure base data.frames to simplify row assignment
max_vals <- as.data.frame(max_vals)
min_vals <- as.data.frame(min_vals)
# sub select values from desired groups
if (length(include_group_indices)>0) {
  vals <- vals %>% filter(.data[[group_col]] %in% include_group_indices)
}
# remove group index to not be displayed as category in the web chart
vals <- vals %>% select(-all_of(group_col))
if (group_col %in% colnames(max_vals)) {
  max_vals <- max_vals %>% select(-all_of(group_col))
}
if (group_col %in% colnames(min_vals)) {
  min_vals <- min_vals %>% select(-all_of(group_col))
}
# cap values if they exceed the bounds
vals <- vals %>%
  mutate(across(
    everything(),
```

```
      ~ pmax(
          pmin(.x, max_vals[[cur_column()]][1]),   # upper bound for this var
          min_vals[[cur_column()]][1]              # lower bound for this var
        )
    ))
  # put everything together
  web <- rbind(max_vals, min_vals, vals)
  web[] <- lapply(web, function(x) as.numeric(x))  # ensure numeric matrix

  # # make it look nice
  n <- nrow(web) - 2
  cols <- grDevices::hcl(h = seq(15, 375, length.out = n + 1)[1:n],
                         c = 80, l = 60)
  fills <- grDevices::adjustcolor(cols, alpha.f = 0.25)
  cols <- c("#2CB1A6",  # teal-mint
          "#1B1E3F",  # deep navy
          "#94A3B8",  # cool gray
          "#A7F3D0",  # light mint
          "#F4A261") # muted coral
  fills <- adjustcolor(cols, alpha.f = 0.25)
  # plot it
  radarchart(web,
             pcol   = cols[seq_len(nrow(web)-2)],
             pfcol  = fills[seq_len(nrow(web)-2)],
             plwd = 2, plty = 1,
             cglcol = "grey80", cglwd = 0.8, axislabcol = "grey30",
             vlcex = 1.2, na.itp = TRUE)
  title(main = title)
  # add a little legend
  if(!is.null(group_col)){
    legend("topright", legend = labels,
         bty = "n", pch = 15, pt.cex = 1.3, col = cols[seq_len(nrow(web)-2)], text.col = "grey20")
  }
  # close device
  if(save==TRUE){
    dev.off()
  }
}
```

Quitter vs no-quitters

Lets compare quitter vs non-quitter to better understand both groups:

```
df <- volunteer_endperiod_attitude_performance_wage
quitter_df <- df %>% filter(quitjob == TRUE)
non_quitter_df <- df %>% filter(quitjob == FALSE)

# general impression
stargazer(as.data.frame(quitter_df), type = "text", median = TRUE, header = TRUE)
```

```
##
## ========================================================================
## Statistic             N      Mean     St. Dev.    Min      Median      Max
## ------------------------------------------------------------------------
```

```
## perform1           2,383  -0.239    1.020    -3.031    -0.209     3.367
## phonecall          2,316  -0.182    1.036    -3.113    -0.145     5.828
## phonecallraw       2,269  419.852  150.717      1        419       1,264
## homethatweek       2,401   0.088    0.283      0          0          1
## logphonecall       2,267   5.939    0.563    0.000     6.038      7.142
## logcallpersec      2,264  -5.141    0.192    -5.721    -5.148    -1.099
## logcalllength      2,266  11.075    0.608    2.485     11.198    12.002
## logcall_dayworked  2,266   9.414    0.517    2.485     9.506     10.210
## logdaysworked      2,398   1.636    0.327    0.000     1.609      1.946
## homethatweek_num   2,401   0.088    0.283      0          0          1
## basewage           2,383 1,560.699 185.905   527.350  1,550.000 2,450.000
## grosswage          2,360 2,687.155 793.984 1,140.280 2,523.000 10,830.070
## bonustotal         2,383 1,125.618 725.063   0.000    971.690  9,330.070
## age                2,401  23.676    3.088      19        23         32
## tenure             2,401  23.439    28.649     2         10        150
## children           2,401   0.221    0.415      0          0          1
## bedroom            2,401   1.000    0.000      1          1          1
## commute            2,118  104.710   76.718     1         120        300
## high_educ          2,401   0.461    0.499      0          0          1
## gender_num         2,401   0.450    0.498      0          0          1
## married_num        2,401   0.239    0.427      0          0          1
## high_educ_num      2,401   0.461    0.499      0          0          1
## children_num       2,401   0.221    0.415      0          0          1
## promote_switch     2,401   0.000    0.000      0          0          0
## quitjob            2,401   1.000    0.000      1          1          1
## costofcommute      2,332   9.136    10.395   0.000     6.000     55.000
## promote_switch_num 2,401   0.000    0.000      0          0          0
## quitjob_num        2,401   1.000    0.000      1          1          1
## -----------------------------------------------------------------------
```

```r
stargazer(as.data.frame(non_quitter_df), type = "text", median = TRUE, header = TRUE)
```

```
##
## ========================================================================
## Statistic            N      Mean    St. Dev.     Min     Median      Max
## ------------------------------------------------------------------------
## exhaustion         2,379   8.612     7.794     0.000     7.000     36.000
## negative           2,379  16.651     6.848     8.000    16.000     40.000
## positive           2,379  24.215     6.668     8.000    24.000     40.000
## perform1           7,444   0.057     0.966    -3.031     0.119     4.163
## phonecall          7,400   0.049     0.932    -3.113     0.127     3.758
## phonecallraw       7,292  446.607   139.324      1        452      1,141
## homethatweek       7,469   0.231     0.422       0         0          1
## logphonecall       7,307   6.031     0.451     0.000     6.114     7.040
## logcallpersec      7,312  -5.179     0.147    -5.951    -5.175    -3.296
## logcalllength      7,310  11.205     0.487     3.989    11.296     12.116
## logcall_dayworked  7,310   9.492     0.435     2.639     9.567     10.359
## logdaysworked      7,457   1.701     0.263     0.000     1.792     1.946
## homethatweek_num   7,469   0.231     0.422       0         0          1
## basewage           7,640 1,614.554  171.635   650.000  1,600.000 2,300.000
## grosswage          7,559 3,225.839 1,011.125 1,264.430 3,010.000 14,553.000
## bonustotal         7,640 1,610.117  934.160   0.000   1,383.000 12,853.000
## age                7,678  23.426     4.479       1        23         35
## tenure             7,678  24.035    23.430       2        19        150
## children           7,678   0.116     0.320       0         0          1
```

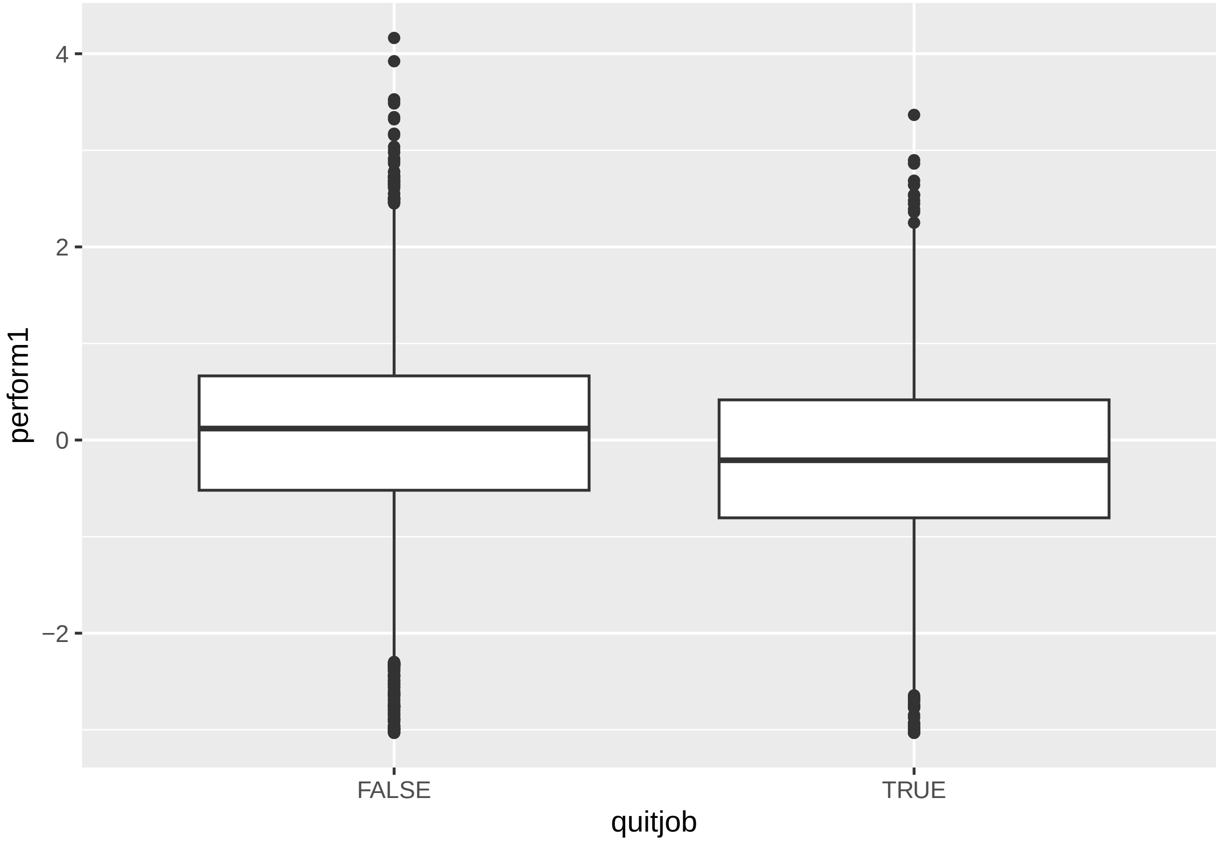
```
## bedroom            7,678   0.966     0.182       0          1          1
## commute            7,000  103.158    64.618       2         80        300
## high_educ          7,678   0.380     0.486       0          0          1
## gender_num         7,678   0.513     0.500       0          1          1
## married_num        7,678   0.163     0.369       0          0          1
## high_educ_num      7,678   0.380     0.486       0          0          1
## children_num       7,678   0.116     0.320       0          0          1
## promote_switch     7,678   0.239     0.427       0          0          1
## quitjob            7,678   0.000     0.000       0          0          0
## costofcommute      7,156   6.587     5.703     0.000      6.000     30.000
## promote_switch_num 7,678   0.239     0.427       0          0          1
## quitjob_num        7,678   0.000     0.000       0          0          0
## ---------------------------------------------------------------------------
```

```r
stargazer(as.data.frame(df), type = "text", median = TRUE, header = TRUE)
```

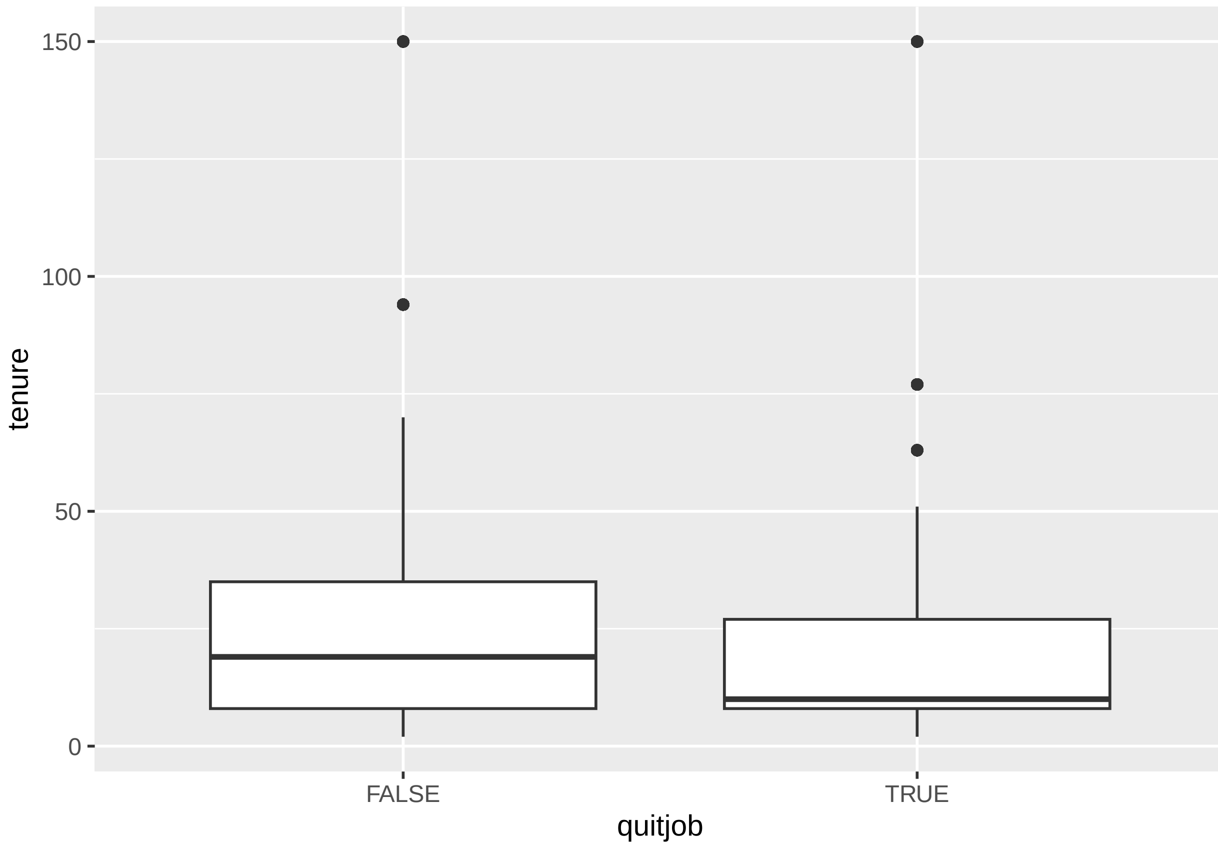
```
##
## ===========================================================================
## Statistic            N      Mean    St. Dev.    Min      Median      Max
## ---------------------------------------------------------------------------
## exhaustion         2,379    8.612     7.794     0.000     7.000     36.000
## negative           2,379   16.651     6.848     8.000    16.000     40.000
## positive           2,379   24.215     6.668     8.000    24.000     40.000
## perform1           9,827   -0.015     0.988    -3.031     0.049      4.163
## phonecall          9,716   -0.006     0.963    -3.113     0.070      5.828
## phonecallraw       9,561   440.258  142.558      1         445       1,264
## homethatweek       9,870    0.196     0.397       0          0          1
## logphonecall       9,574    6.009     0.481     0.000     6.098      7.142
## logcallpersec      9,576   -5.170     0.160    -5.951    -5.169     -1.099
## logcalllength      9,576   11.175     0.521     2.485    11.273     12.116
## logcall_dayworked  9,576    9.474     0.457     2.485     9.552     10.359
## logdaysworked      9,855    1.685     0.282     0.000     1.792      1.946
## homethatweek_num   9,870    0.196     0.397       0          0          1
## basewage           10,023 1,601.750 176.618   527.350  1,600.000 2,450.000
## grosswage          9,919  3,097.672 990.788  1,140.280 2,877.000 14,553.000
## bonustotal         10,023 1,494.926 912.498    0.000   1,284.590 12,853.000
## age                10,079  23.485     4.191       1         23         35
## tenure             10,079  23.893    24.773       2         18        150
## children           10,079   0.141     0.348       0          0          1
## bedroom            10,079   0.974     0.160       0          1          1
## commute            9,118   103.519   67.620       1         80        300
## high_educ          10,079   0.400     0.490       0          0          1
## gender_num         10,079   0.498     0.500       0          0          1
## married_num        10,079   0.181     0.385       0          0          1
## high_educ_num      10,079   0.400     0.490       0          0          1
## children_num       10,079   0.141     0.348       0          0          1
## promote_switch     10,079   0.182     0.386       0          0          1
## quitjob            10,079   0.238     0.426       0          0          1
## costofcommute      9,488    7.213     7.231     0.000     6.000     55.000
## promote_switch_num 10,079   0.182     0.386       0          0          1
## quitjob_num        10,079   0.238     0.426       0          0          1
## ---------------------------------------------------------------------------
```

```r
# plot per numeric variable
ggplot(df, aes(x=quitjob, y=perform1)) + geom_boxplot()
```

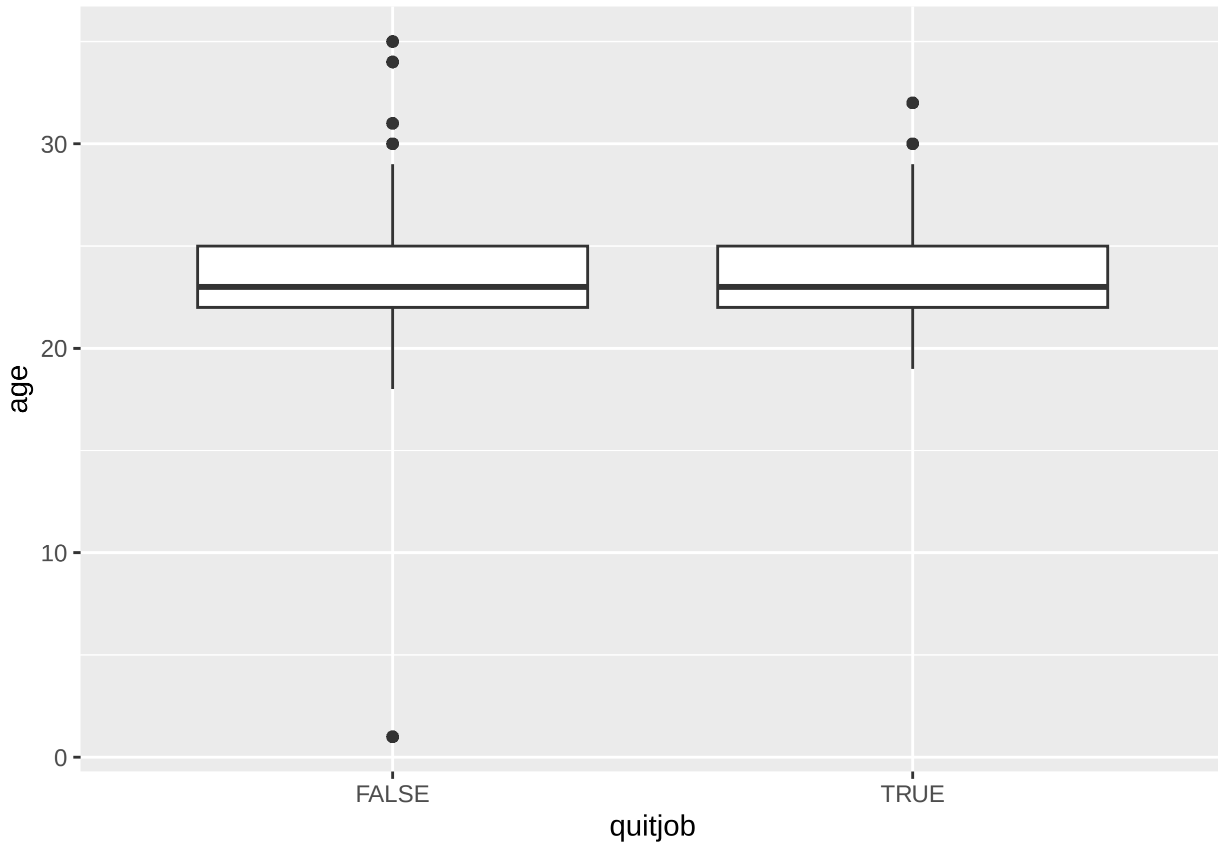
```
## Warning: Removed 252 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```r
ggplot(df, aes(x=quitjob, y=tenure)) + geom_boxplot()
```

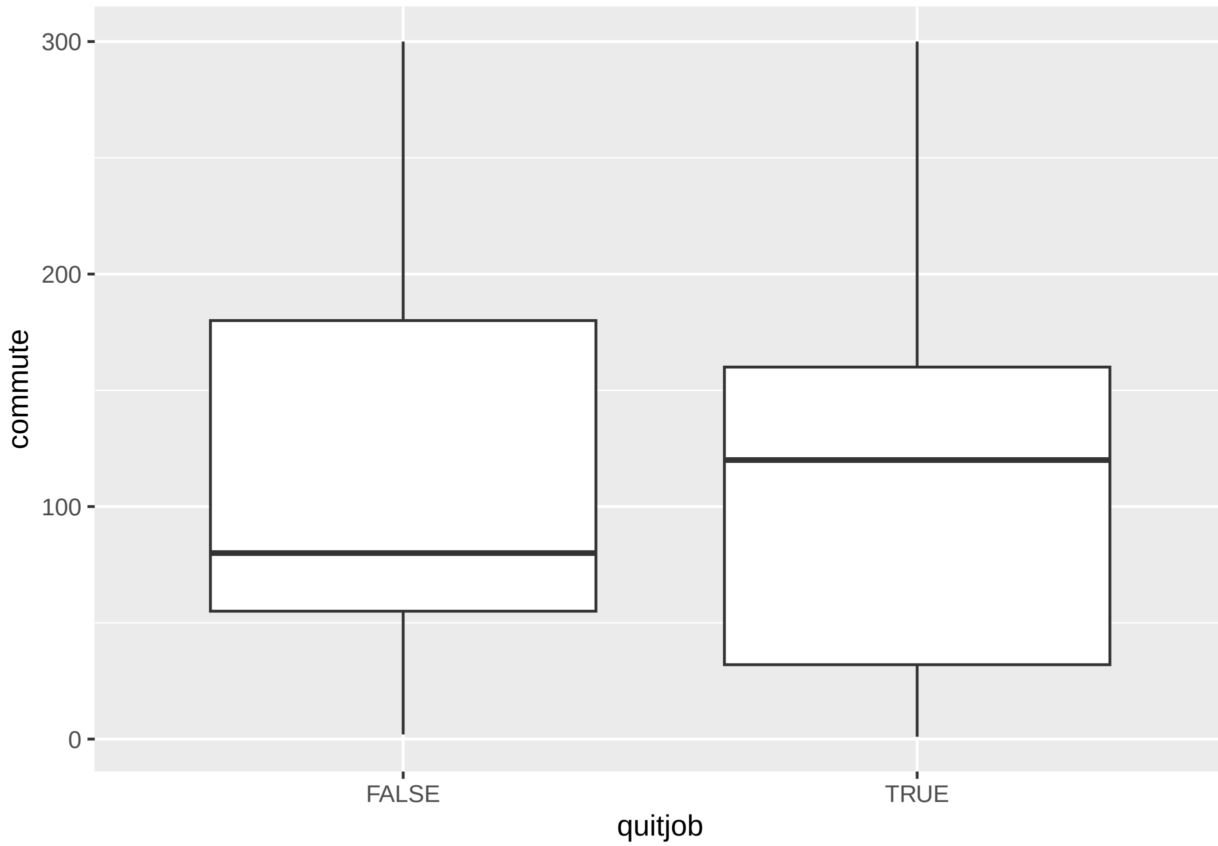



```
ggplot(df, aes(x=quitjob, y=age)) + geom_boxplot()
```



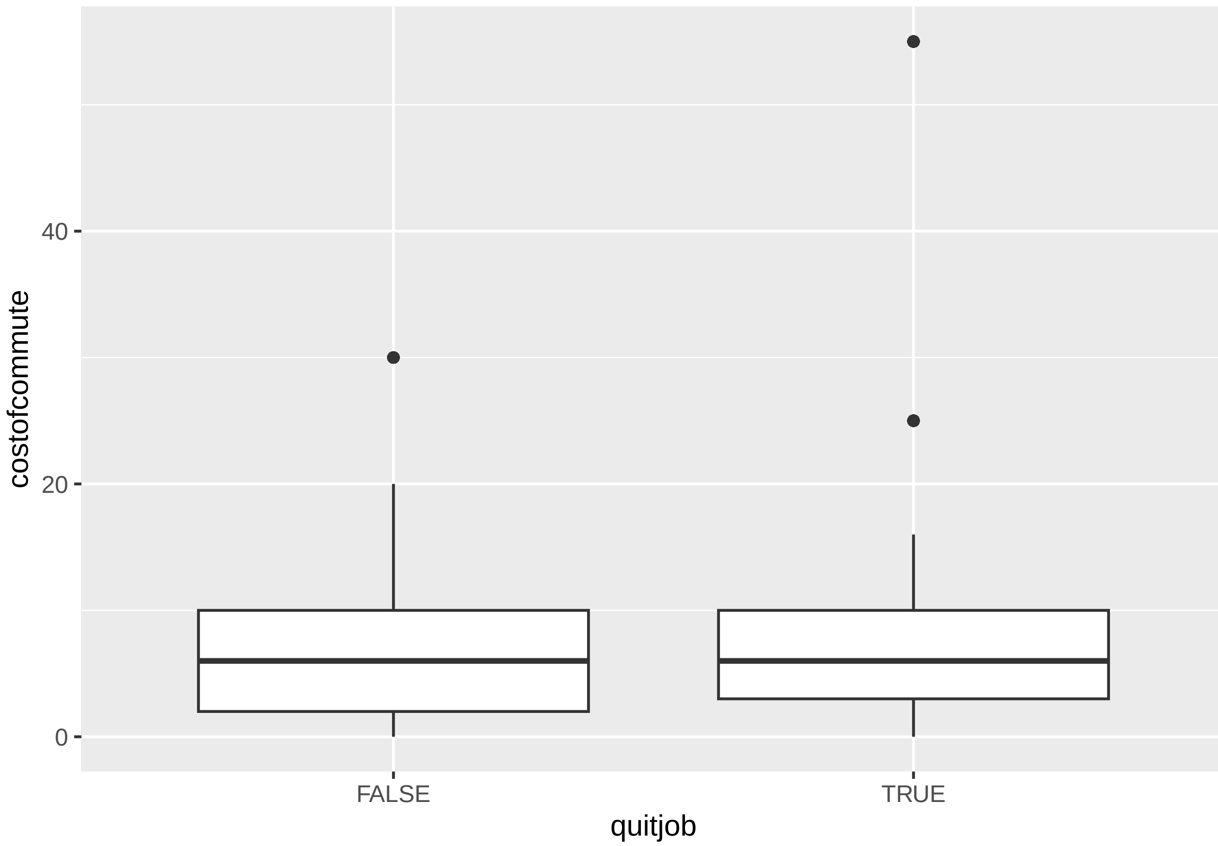

```
ggplot(df, aes(x=quitjob, y=commute)) + geom_boxplot()
```

```
## Warning: Removed 961 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



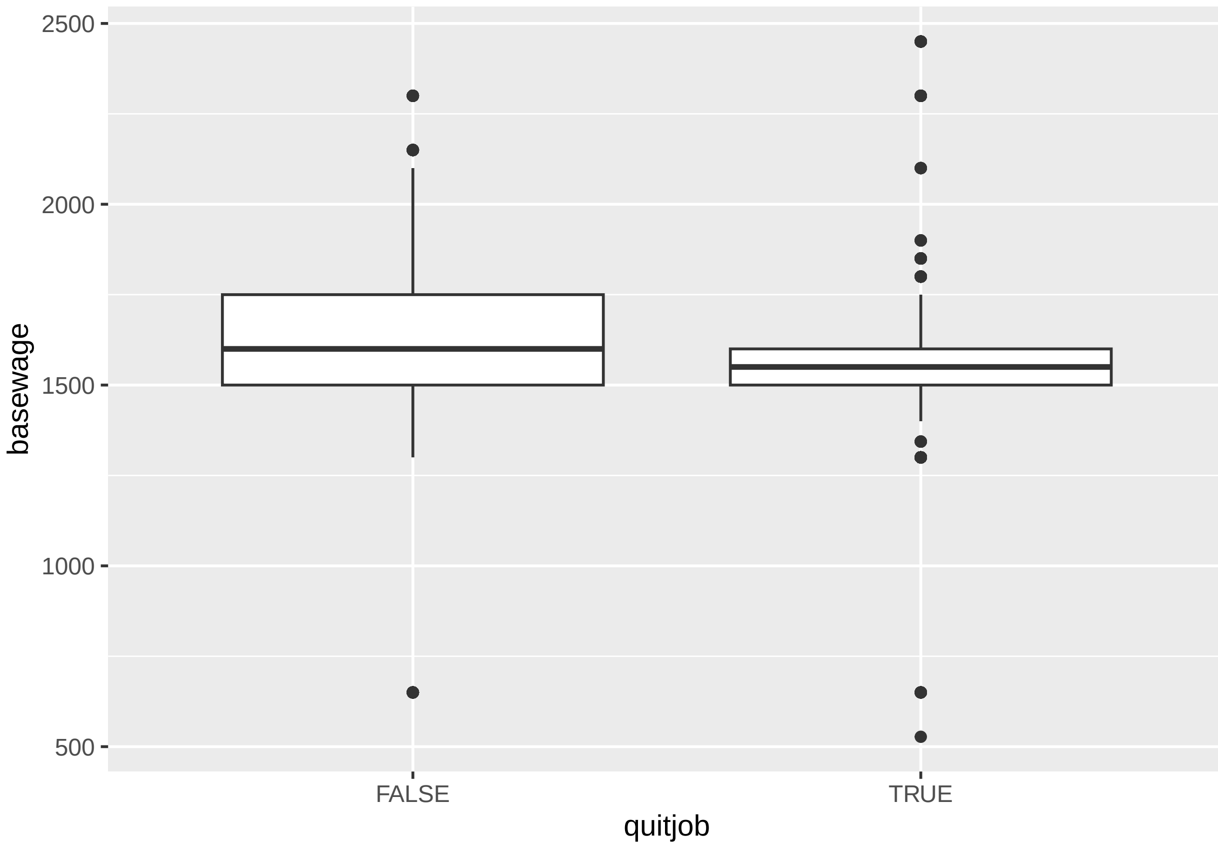

```
ggplot(df, aes(x=quitjob, y=costofcommute)) + geom_boxplot()
```

```
## Warning: Removed 591 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



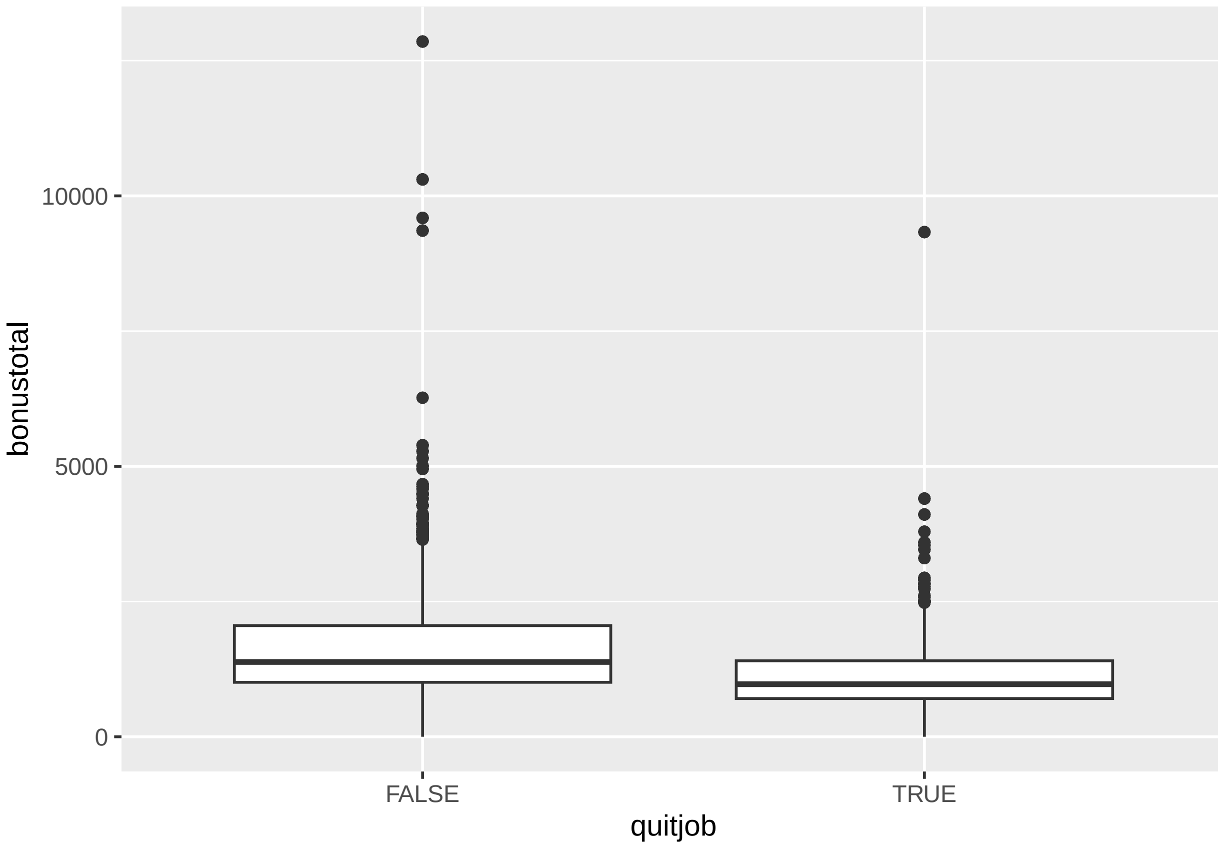

```r
ggplot(df, aes(x=quitjob, y=basewage)) + geom_boxplot()
```

```
## Warning: Removed 56 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



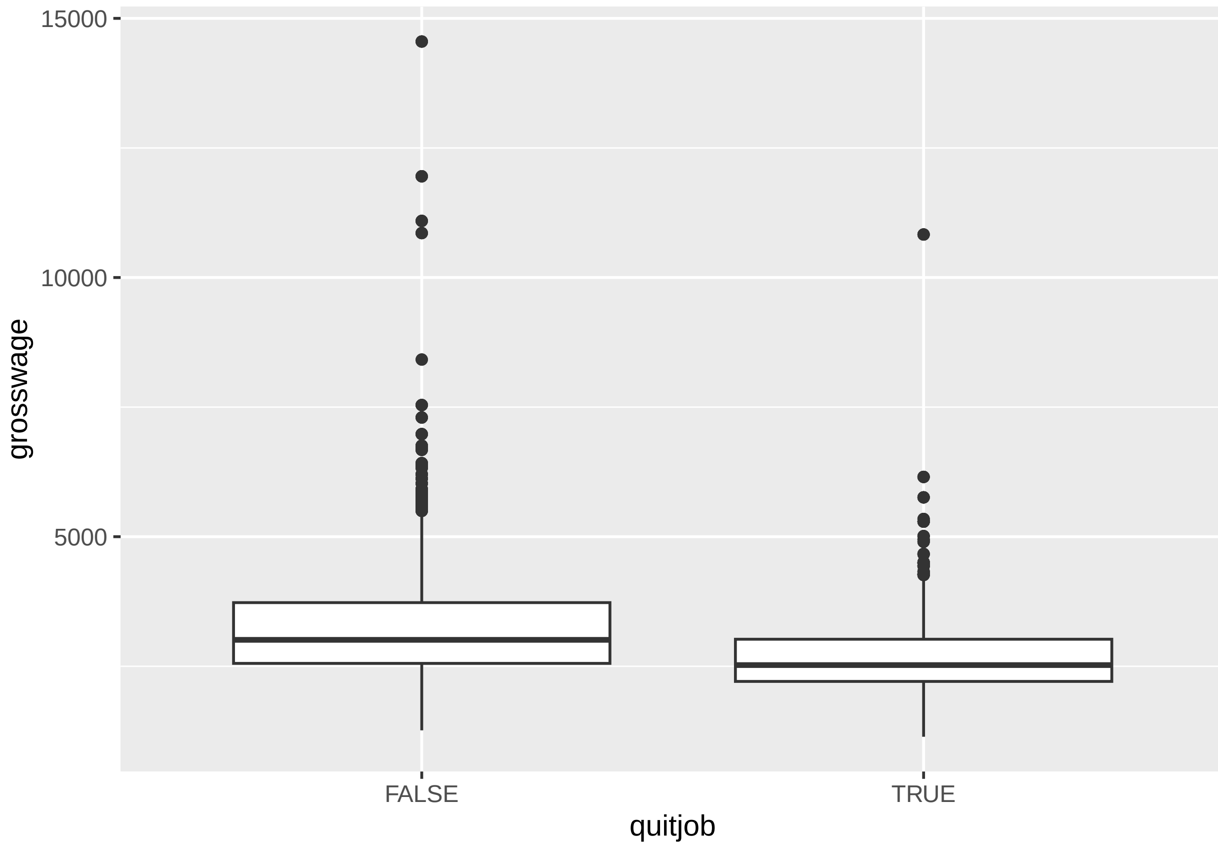

```
ggplot(df, aes(x=quitjob, y=bonustotal)) + geom_boxplot()
```

```
## Warning: Removed 56 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

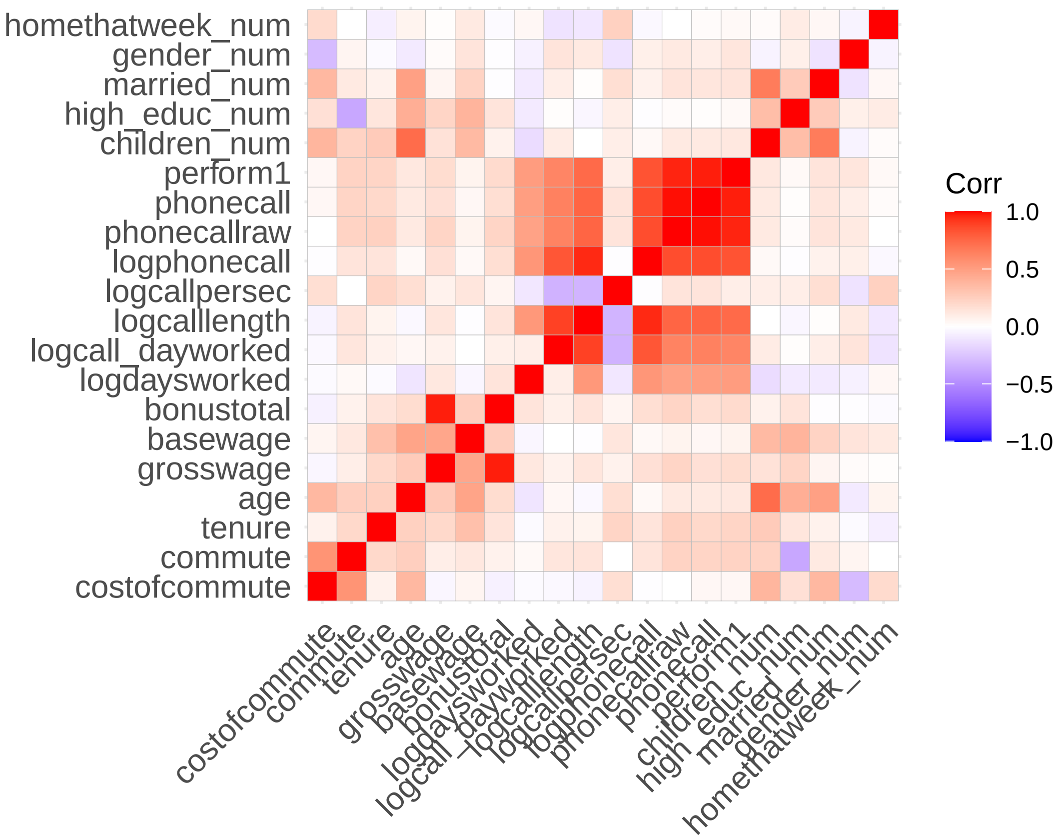



```r
ggplot(df, aes(x=quitjob, y=grosswage)) + geom_boxplot()
```

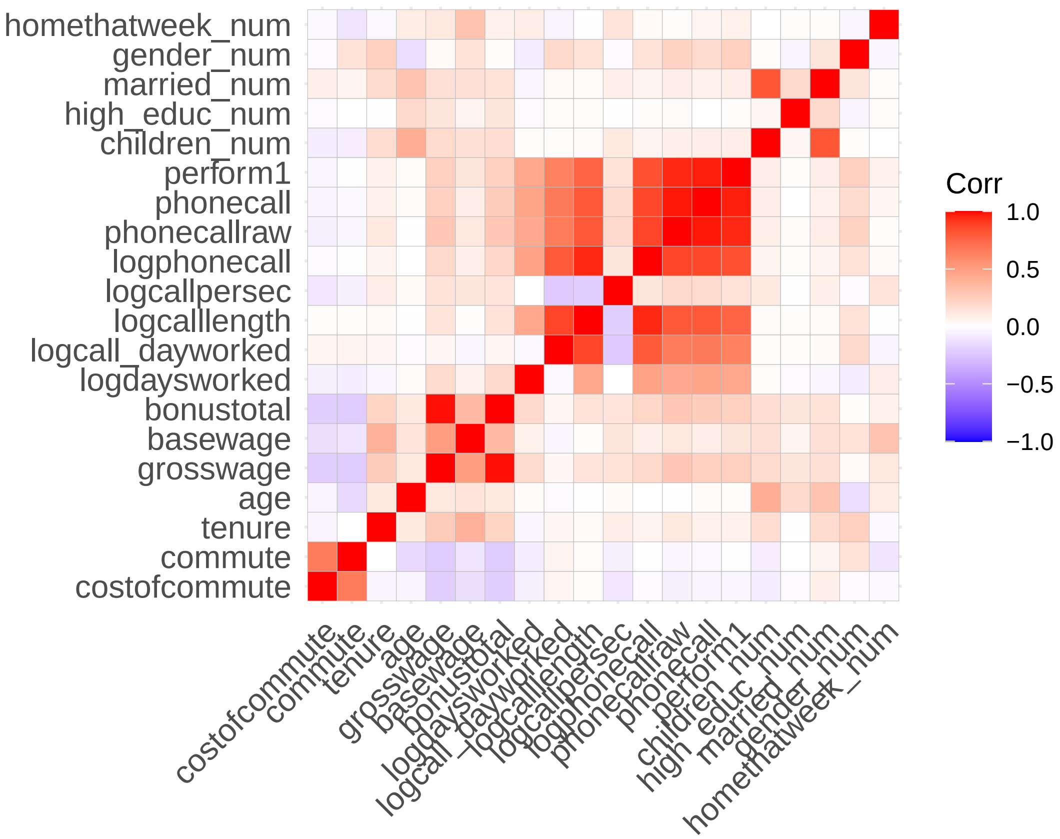
```
## Warning: Removed 160 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
# Look at correlations
# subselect variable available in both groups (e.g. attitude data not available for quitters)
non_quitter_df <- non_quitter_df %>% select(costofcommute,commute,tenure,age,grosswage,basewage,bonusto

quitter_df <- quitter_df %>% select(costofcommute,commute,tenure,age,grosswage,basewage,bonustotal,logda

nums <- dplyr::select(quitter_df, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
ggcorrplot(cm, lab = FALSE)
```



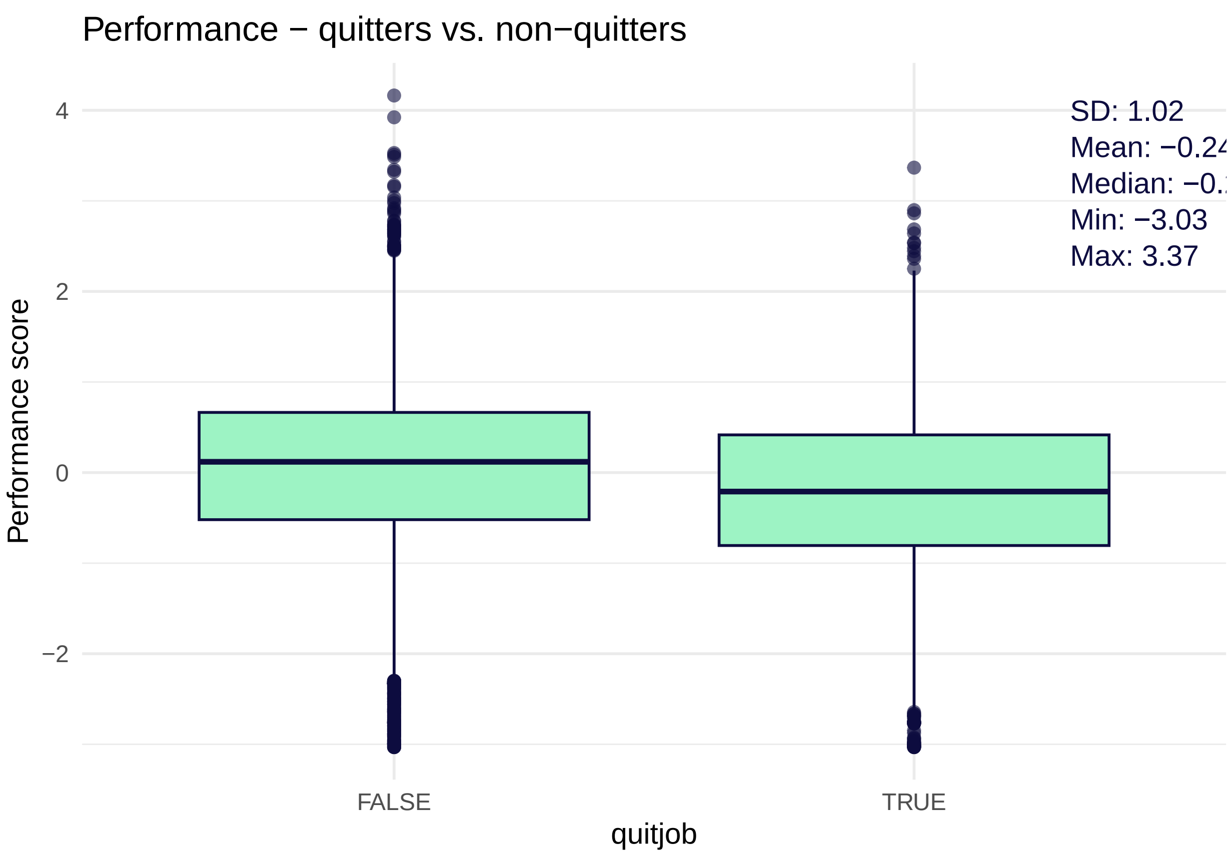

```r
nums <- dplyr::select(non_quitter_df, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
ggcorrplot(cm, lab = FALSE)
```




```r
df <- volunteer_endperiod_attitude_performance_wage
quitter_df <- df %>% filter(quitjob == TRUE)
non_quitter_df <- df %>% filter(quitjob == FALSE)
# Make some nice looking box plots for perform, commute
stats <- quitter_df %>% summarise(
  min  = min(perform1, na.rm = TRUE),
  max  = max(perform1, na.rm = TRUE),
  mean = mean(perform1, na.rm = TRUE),
  median = median(perform1, na.rm = TRUE),
  sd   = sd(perform1, na.rm = TRUE)
)

p <- ggplot(df, aes(y = perform1, x=quitjob)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Performance score", title = "Performance - quitters vs. non-quitters") +
  theme_minimal()

# add annotations
p + annotate("text", x = 2.3, y = 4,
           label = paste0("SD: ", round(stats$sd,2)),
           hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.3, y = 3.6,
```

```
          label = paste0("Mean: ", round(stats$mean,2)),
          hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.3, y = 3.2,
          label = paste0("Median: ", round(stats$median,2)),
          hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.3, y = 2.8,
          label = paste0("Min: ", round(stats$min,2)),
          hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.3, y = 2.4,
          label = paste0("Max: ", round(stats$max,2)),
          hjust = 0, color = "#0d0c3f")
```

```
## Warning: Removed 252 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



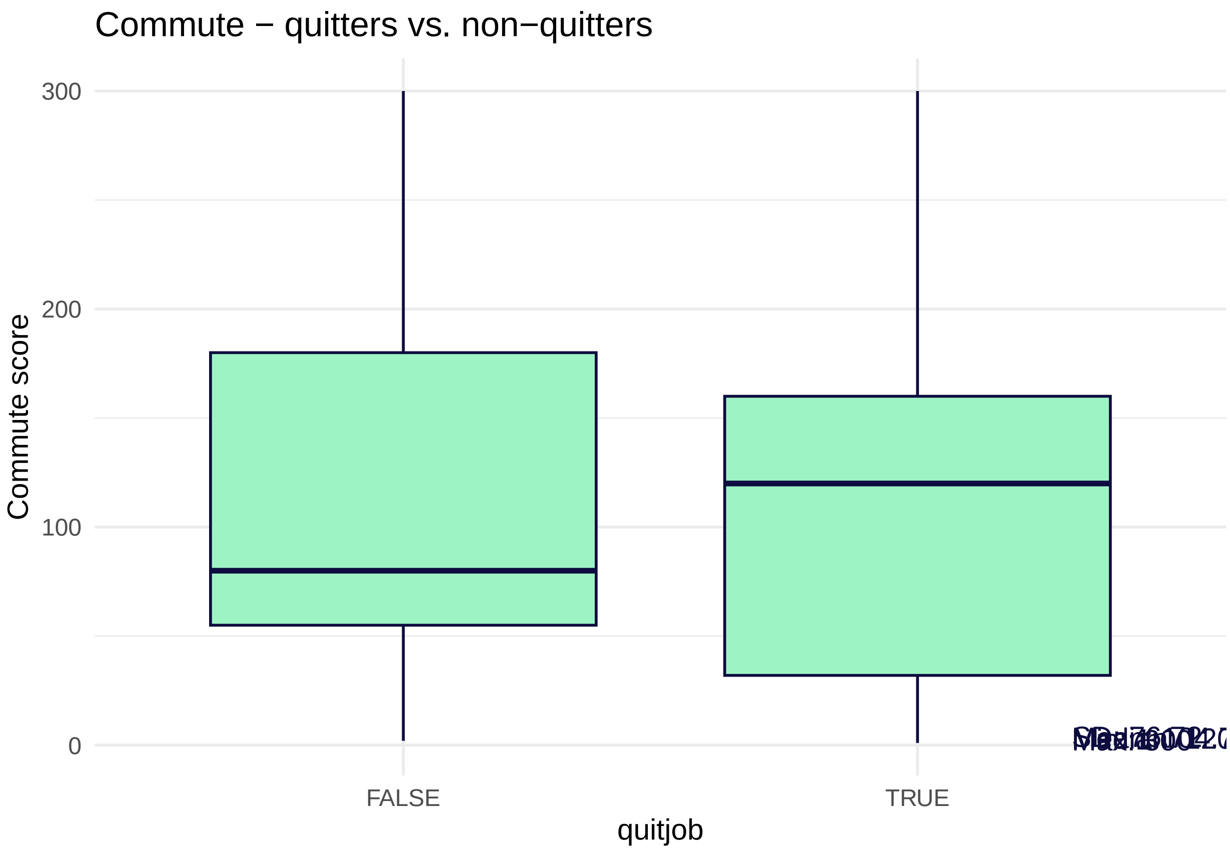

```
ggsave("performance_quitter_non_quitters.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 252 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
stats <- quitter_df %>% summarise(
  min   = min(commute, na.rm = TRUE),
  max   = max(commute, na.rm = TRUE),
  mean = mean(commute, na.rm = TRUE),
  median = median(commute, na.rm = TRUE),
  sd    = sd(commute, na.rm = TRUE)
)
```

```r
p <- ggplot(df, aes(y = commute, x=quitjob)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Commute score", title = "Commute - quitters vs. non-quitters") +
  theme_minimal()

# add annotations
p + annotate("text", x = 2.3, y = 4,
           label = paste0("SD: ", round(stats$sd,2)),
           hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.3, y = 3.6,
           label = paste0("Mean: ", round(stats$mean,2)),
           hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.3, y = 3.2,
           label = paste0("Median: ", round(stats$median,2)),
           hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.3, y = 2.8,
           label = paste0("Min: ", round(stats$min,2)),
           hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.3, y = 2.4,
           label = paste0("Max: ", round(stats$max,2)),
           hjust = 0, color = "#0d0c3f")
```

```
## Warning: Removed 961 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



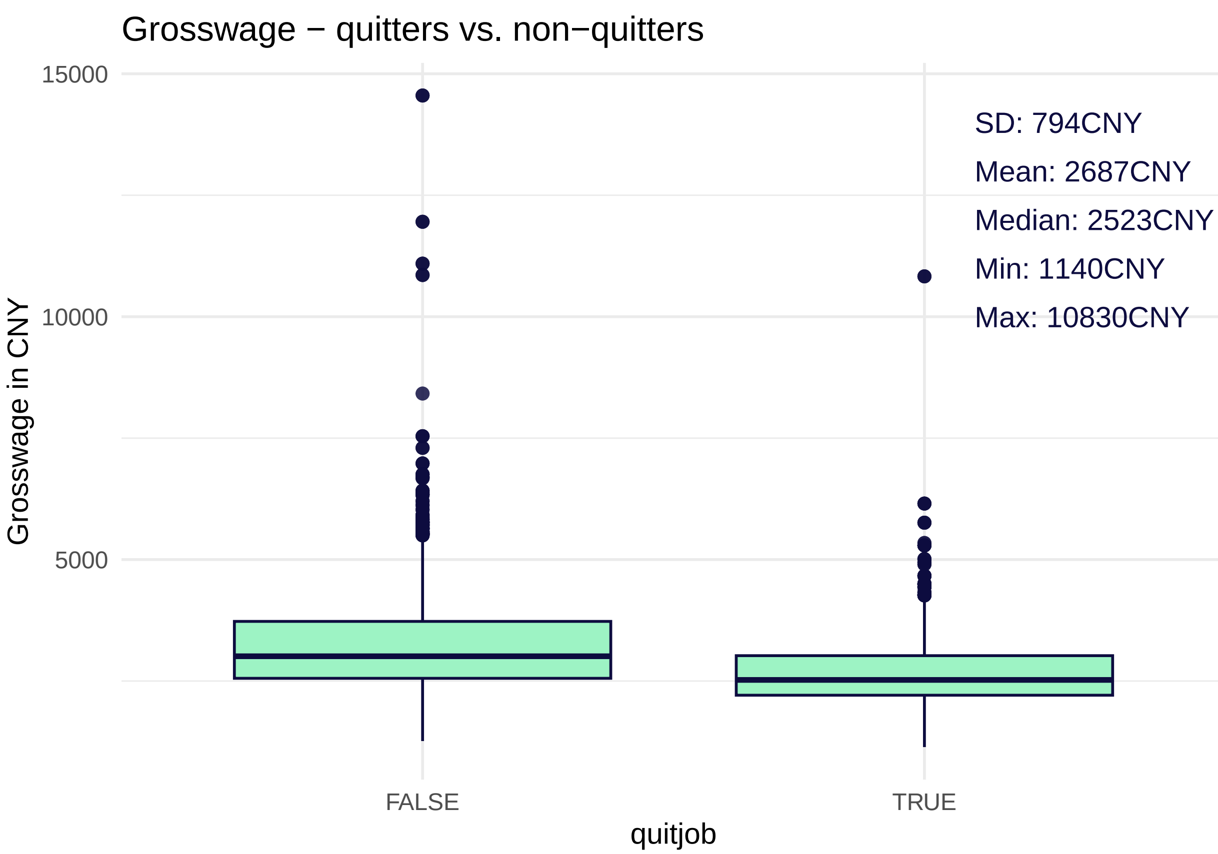

```
ggsave("commute_quitter_non_quitters.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 961 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
stats <- quitter_df %>% summarise(
  min   = min(grosswage, na.rm = TRUE),
  max   = max(grosswage, na.rm = TRUE),
  mean = mean(grosswage, na.rm = TRUE),
  median = median(grosswage, na.rm = TRUE),
  sd    = sd(grosswage, na.rm = TRUE)
)

p <- ggplot(df, aes(y = grosswage, x=quitjob)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Grosswage in CNY", title = "Grosswage - quitters vs. non-quitters") +
  theme_minimal()

# add annotations
p + annotate("text", x = 2.1, y = 14000,
           label = paste0("SD: ", round(stats$sd,0), "CNY"),
           hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.1, y = 13000,
```

```
           label = paste0("Mean: ", round(stats$mean,0), "CNY"),
           hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.1, y = 12000,
           label = paste0("Median: ", round(stats$median,0), "CNY"),
           hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.1, y = 11000,
           label = paste0("Min: ", round(stats$min,0), "CNY"),
           hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.1, y = 10000,
           label = paste0("Max: ", round(stats$max,0), "CNY"),
           hjust = 0, color = "#0d0c3f")
```

```
## Warning: Removed 160 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
ggsave("grosswage_quitter_non_quitters.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 160 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
# radar plot to compare quitter vs non quitter
radar_by_group(df, group_col="quitjob", include_group_indices=c(TRUE, FALSE), title="Quitter vs non-qui
```

```
##  [1] "commute"            "costofcommute"      "negative"
##  [4] "positive"           "quitjob"            "gender_num"
##  [7] "married_num"        "bedroom"            "children_num"
## [10] "high_educ_num"      "promote_switch_num" "homethatweek_num"
## [13] "perform1"           "grosswage"          "exhaustion"
```

```
## [16] "tenure"

## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
##   # Was:
##   data %>% select(final_selection)
##
##   # Now:
##   data %>% select(all_of(final_selection))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## # A tibble: 2 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1 FALSE      103.          6.59     16.7     24.2  0.513   0.163   0.966
## 2 TRUE       105.          9.14    NaN      NaN    0.450   0.239   1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "quitjob"
## [1]  TRUE FALSE
## # A tibble: 2 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1 FALSE      103.          6.59     16.7     24.2  0.513   0.163   0.966
## 2 TRUE       105.          9.14    NaN      NaN    0.450   0.239   1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "FALSE" "TRUE"

## pdf
##   2
```

Generally speaking quitters perform less, work for the company shorter than average, have an average age, have slightly higher costs of commute while commuting about the same in minutes per week, earn less and work not from home in comparison to workers who stay. When begin older, quitters have a higher tendency to have children, being married and have higher education. People who were older and left at some point generally performed better than younger quitters. People who stay, tend to have lower cost of commute and lower commute times (maybe because they commit to working for the company and move closer to it).

We can already get a clue from this comparison of the differences between quitters and non quitters about the group constellation and personas.

Groups

Lets have a look on all possible combinations of categorical values and which are represented within the volunteer dataset with a threshold of at least 100 people-work-performances per group. We choose this sample size to increase our confidence. Afterwards we will do the same general analysis we did for the whole population for each group.

Important note: Because all quitters do not have attitude data, positive, negative and exhaustion appear to be "average" on the radar plots, but are actually NULL which is caused by the radar plotting function.

Available groups

```
# make it easier to process
df <- volunteer_endperiod_attitude_performance_wage
# define categorical variables
cat_cols <- c("gender","married","high_educ","bedroom","children","promote_switch","quitjob", "homethatw

# factors
df <- df %>% mutate(across(all_of(cat_cols), ~ factor(.x)))

# build grid of categories
lvls <- lapply(df[cat_cols], levels)
grid <- tidyr::expand_grid(!!!lvls)

# create summary for each category/group of people working at the company
summary_df <- df %>%
  group_by(across(all_of(cat_cols))) %>%
  summarise(
    # add number per category to filter later
    n = n(),
    across(where(is.numeric),
           list(mean = ~mean(.x, na.rm = TRUE),
                sd   = ~sd(.x,   na.rm = TRUE),
                se = ~sd(.x, na.rm = TRUE) / sqrt(sum(!is.na(.x))),
                med  = ~median(.x, na.rm = TRUE),
                min  = ~min(.x, na.rm = TRUE),
                max  = ~max(.x, na.rm = TRUE)),
           .names = "{.col}_{.fn}"
           ),
    .groups = "drop"
) %>%
  mutate(group_index = row_number()) %>% # create an index for each group for bettler handling
  arrange(desc(n))
```

```
## Warning: There were 400 warnings in `summarise()`.
## The first warning was:
## i In argument: `across(...)`.
## i In group 7: `gender = female`, `married = FALSE`, `high_educ = FALSE`,
##   `bedroom = 1`, `children = FALSE`, `promote_switch = FALSE`, `quitjob =
##   TRUE`, `homethatweek = 0`.
## Caused by warning in `min()`:
## ! no non-missing arguments to min; returning Inf
## i Run `dplyr::last_dplyr_warnings()` to see the 399 remaining warnings.
```

```
# add group_index to the big d
group_characteristics <- summary_df %>% select(group_index, gender, married, bedroom, children, high_edu
volunteer_endperiod_attitude_performance_wage <- merge(volunteer_endperiod_attitude_performance_wage, gr
# filter by number of rows per group
summary_df <- summary_df %>% filter(n>=100)
```

As a result we receive 23 distinct groups out of 8x8=64 possible options where each groups represents a work week of an employee:

Top 6 groups by number

```
top <- summary_df %>% slice(1:6)

total_n <- summary_df %>% summarise(total_n = sum(n))
top_n <- top %>% summarise(top_n = sum(n))

top_n / total_n # about one quarter of the groups contain over 50% of the total observations
```

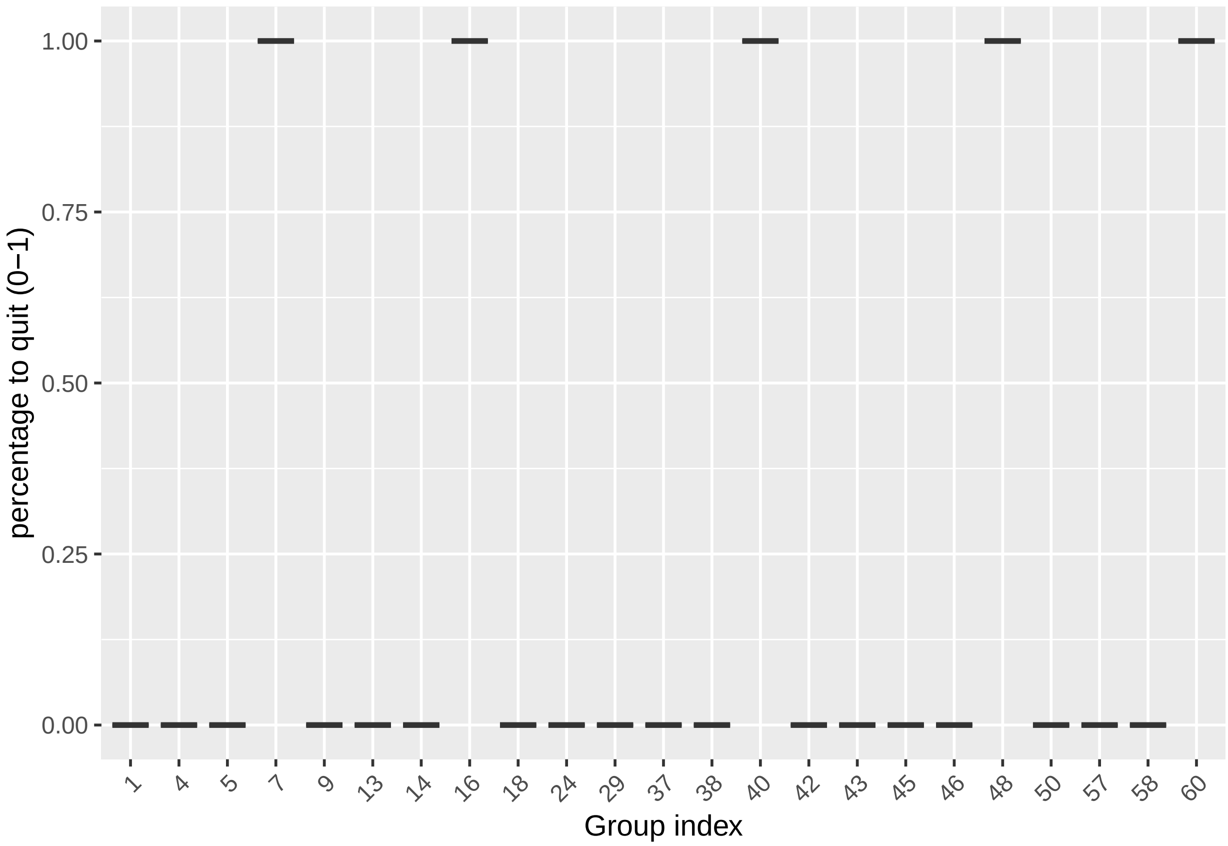
```
##       top_n
## 1 0.5245345
```

```
top %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatweek)
```

```
## # A tibble: 6 x 8
##   gender married bedroom children high_educ promote_switch quitjob homethatweek
##   <fct>  <fct>   <fct>   <fct>    <fct>     <fct>          <fct>   <fct>
## 1 female FALSE   1       FALSE    FALSE     FALSE          FALSE   0
## 2 male   FALSE   1       FALSE    FALSE     FALSE          FALSE   0
## 3 male   FALSE   1       FALSE    TRUE      FALSE          FALSE   0
## 4 male   FALSE   1       FALSE    FALSE     FALSE          TRUE    0
## 5 male   FALSE   1       FALSE    FALSE     TRUE           FALSE   0
## 6 female FALSE   1       FALSE    TRUE      FALSE          FALSE   0
```

```
#
```

```
# First we look at the quitters and try to see which are more likely to quit
ggplot(summary_df, aes(x = factor(group_index), y = quitjob_num_mean)) +
  geom_boxplot() +
  labs(x = "Group index", y = "percentage to quit (0-1)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
df <- volunteer_endperiod_attitude_performance_wage
radar_by_group(df, group_col="group_index", include_group_indices=c(7,16,40,48,60), title="Quitter grou
```

```
##  [1] "commute"            "costofcommute"      "negative"
##  [4] "positive"           "group_index"        "gender_num"
##  [7] "married_num"        "bedroom"            "children_num"
## [10] "high_educ_num"      "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num"   "perform1"           "grosswage"
## [16] "exhaustion"         "tenure"
## # A tibble: 62 x 17
##    group_index commute costofcommute negative positive gender married bedroom
##          <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
##  1           1   103.           6.45     12.4     29.2      1       0       0
##  2           2   180           18        21.8     20.5      1       0       0
##  3           3   120            9        18.8     23        1       0       0
##  4           4   119.           6.71     22.0     22.4      1       0       1
##  5           5    92.0          5.12     14.5     26.6      1       0       1
##  6           6   131.           7.58     18.2     24.8      1       0       1
##  7           7   118.           4.11    NaN      NaN        1       0       1
##  8           8    20            0       NaN      NaN        1       0       1
##  9           9   112.           3.88     13.8     16.2      1       0       1
## 10          10    28            0       NaN      NaN        1       0       1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
```

```
## [1]  7 16 40 48 60
## # A tibble: 5 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##         <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1           7   118.           4.11      NaN      NaN      1       0       1
## 2          16    52.1          3.57      NaN      NaN      1       0       1
## 3          40   120.           8.28      NaN      NaN      0       0       1
## 4          48    33.5         11.7       NaN      NaN      0       0       1
## 5          60   101.          20.5       NaN      NaN      0       1       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "7"  "16" "40" "48" "60"
```

```
## pdf
##   2
```

Here we can clearly identify groups 7,16,40,48 and 60 where all people quit. Within the rest of the groups nobody quit.

Analyze quitter groups

```r
df <- volunteer_endperiod_attitude_performance_wage
```

Group 7 357 observations of 6 volunteer

Persona observations:
female
not married
number of bedrooms: 1
no children
no high education
no promotion
worked for 7.6 months at the company when left (< avg)
20.5 years old (< avg)

Other average weekly observations:
not working from home
performed -0.308 (< avg)
basewage per month: 1453 (< avg)
bonus per month: 831 (« avg)
grosswage per month: 2265 (< avg)
cost of commute per month: 4 (< avg)
commute minutes per week: 103 (<avg)
logdaysworked: 1.6 (~= avg)

This groups shows a very similar correlation matrix in comparison to the overall sample, no specialities.

Interpretation
Looks like a group of female juniors (wage, performance, tenure, age), living not close by, probably the first or second job after high school(?) or even a job for a gap year. Probably left because of pursuing with school or another job for a pay increase.

Recommendation

This group is amoung the youngest and has a worse performance than the average. Offering working from home/remote working capabilities can be a recommended, as these group could for example continue working during university or traveling. When performance increases a raise would be appropiate but now as first action.

```r
# select group data
group7 <- df %>% filter(group_index==7)
stargazer(as.data.frame(group7), type = "text", median = TRUE, header = TRUE)
```

```
##
## ========================================================================
## Statistic           N    Mean    St. Dev.    Min      Median      Max
## ------------------------------------------------------------------------
## bedroom            357   1.000    0.000       1          1          1
## children           357   0.000    0.000       0          0          0
## high_educ          357   0.000    0.000       0          0          0
## promote_switch     357   0.000    0.000       0          0          0
## quitjob            357   1.000    0.000       1          1          1
## homethatweek       357   0.000    0.000       0          0          0
## perform1           345  -0.308    0.854    -3.031     -0.204      2.230
## phonecall          320  -0.220    0.983    -3.054     -0.157      5.828
## phonecallraw       315  411.311  137.944     22         419       1,264
## logphonecall       315   5.941    0.443     3.091      6.038      7.142
## logcallpersec      307  -5.217    0.114    -5.560     -5.198     -4.924
## logcalllength      308  11.142    0.444     8.473     11.222     11.900
## logcall_dayworked  308   9.493    0.369     7.087      9.566     10.049
## logdaysworked      357   1.603    0.325     0.000      1.609      1.946
## homethatweek_num   357   0.000    0.000       0          0          0
## basewage           357 1,451.197 160.260   527.350  1,500.000 1,750.000
## grosswage          352 2,281.242 477.662  1,140.280 2,222.000 4,487.660
## bonustotal         357  832.198  454.516    0.000    795.000  2,937.660
## age                357  20.501    1.315      19         21         22
## tenure             357   7.622    2.661       2          9         10
## commute            357  118.263   97.337     20         120        300
## gender_num         357   1.000    0.000       1          1          1
## married_num        357   0.000    0.000       0          0          0
## high_educ_num      357   0.000    0.000       0          0          0
## children_num       357   0.000    0.000       0          0          0
## costofcommute      357   4.113    4.491     0.000      4.545     10.000
## promote_switch_num 357   0.000    0.000       0          0          0
## quitjob_num        357   1.000    0.000       1          1          1
## group_index        357   7.000    0.000       7          7          7
## ------------------------------------------------------------------------
```

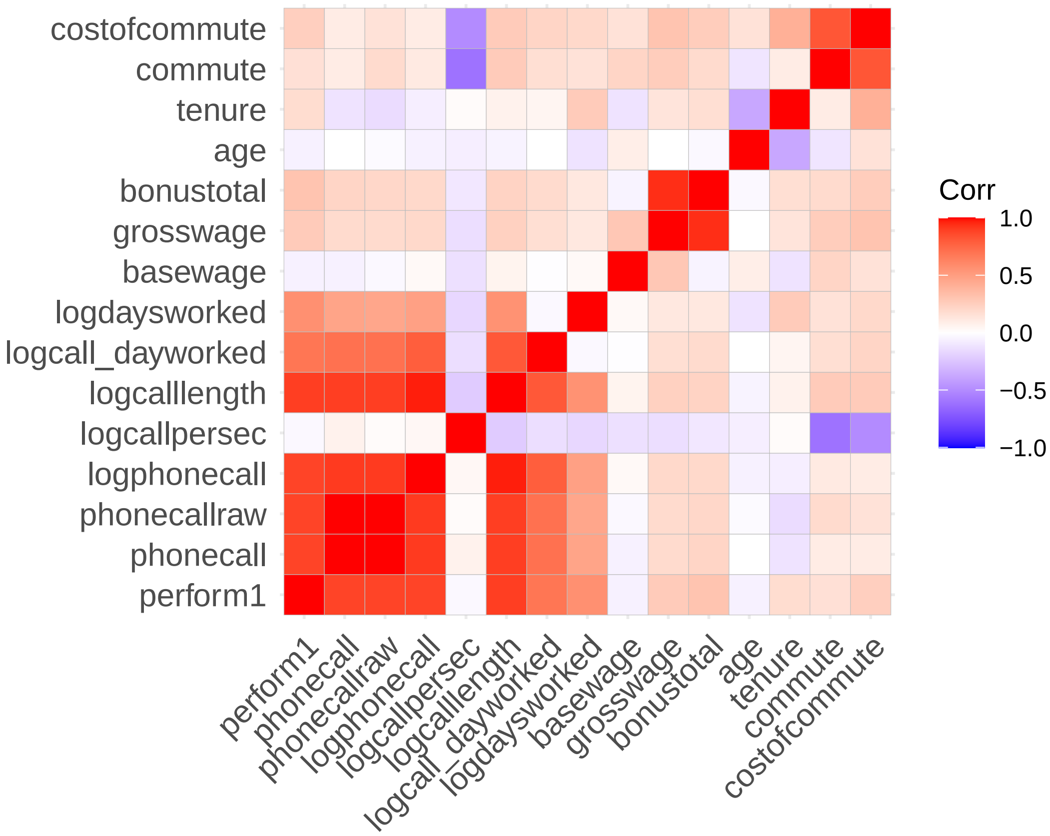
```r
# group definition
#group7 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatweek
# distinct persons
unique(group7 %>%select(personid))
```

```
##     personid
## 1      40034
## 2      38862
## 4      44256
## 5      40346
## 26     42104
## 110    39478
```

```r
nums <- dplyr::select(group7, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```r
ggcorrplot(cm, lab = FALSE)
```




```r
# radar plot
radar_by_group(df, group_col="group_index", include_group_indices=c(7), title="Group 7 characteristics"
```

```
##  [1] "commute"            "costofcommute"      "negative"
##  [4] "positive"           "group_index"        "gender_num"
##  [7] "married_num"        "bedroom"            "children_num"
## [10] "high_educ_num"      "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num"   "perform1"           "grosswage"
## [16] "exhaustion"         "tenure"
## # A tibble: 62 x 17
##    group_index commute costofcommute negative positive gender married bedroom
##          <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
##  1           1   103.           6.45     12.4     29.2      1       0       0
##  2           2   180           18        21.8     20.5      1       0       0
##  3           3   120            9        18.8     23        1       0       0
##  4           4   119.           6.71     22.0     22.4      1       0       1
##  5           5    92.0          5.12     14.5     26.6      1       0       1
##  6           6   131.           7.58     18.2     24.8      1       0       1
##  7           7   118.           4.11    NaN      NaN        1       0       1
##  8           8    20            0       NaN      NaN        1       0       1
```

```
## 9            9   112.           3.88     13.8     16.2      1       0       1
## 10          10    28            0       NaN      NaN       1       0       1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
## [1] 7
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##         <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1           7    118.          4.11      NaN      NaN      1       0       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "7"

## pdf
##   2
```

Group 16 338 observations for 6 volunteers

Persona observations:
female
not married
number of bedrooms: 1
no children
high education
no promotion
worked for 23.5 months at the company when left (= avg)
23.7 years old (>= avg)

Other average weekly observations:
not working from home
performed: 0 (= avg)
basewage per month: 1598 (< avg)
bonus per month: 1362 (< avg)
grosswage per month: 2940 (<=avg)
cost of commute per month: 3.9 (< avg)
commute min per week: 52 (« avg)
logdaysworked: 1.65 (~= avg)

For this group the strong positive correlation between tenure and age stands out. For the full sample it is slightly negative. Commute and age have a much stronger negative corrleation

Interpretation
Looks like a group of female juniors (wage, performance, tenure, age), who already worked for almost 2 years, living somewhat central, probably the first or second job college. Decent performance, not a top performer but average. Probably left because they did not receive a promotion and went for a higher paying job as their salary was still below average.

Recommendation
As this group has an average performance while still being young, and already has a higher education degree (so they probably wont leave for education) our client should prioritize this group and try to give employees in this group a raise and a promotion to keep them within the company. Additionally offer working from home as a benifit, though is not the greatest leverage for this group.

```r
# select group data
group16 <- df %>% filter(group_index==16)
stargazer(as.data.frame(group16), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====================================================================
## Statistic           N    Mean    St. Dev.    Min     Median     Max
## ---------------------------------------------------------------------
## bedroom            338   1.000    0.000       1        1         1
## children           338   0.000    0.000       0        0         0
## high_educ          338   1.000    0.000       1        1         1
## promote_switch     338   0.000    0.000       0        0         0
## quitjob            338   1.000    0.000       1        1         1
## homethatweek       338   0.000    0.000       0        0         0
## perform1           338  -0.0001   1.079    -3.031    0.049     2.863
## phonecall          320  -0.038    1.045    -3.054    0.048     2.463
## phonecallraw       307  454.397  154.865     51       457       910
## logphonecall       306   6.046    0.419     3.932    6.124     6.813
## logcallpersec      310  -5.159    0.141    -5.495   -5.151    -4.624
## logcalllength      310  11.183    0.562     5.489    11.316    11.991
## logcall_dayworked  310   9.483    0.485     3.697    9.552     10.100
## logdaysworked      337   1.645    0.395     0.000    1.792     1.946
## homethatweek_num   338   0.000    0.000       0        0         0
## basewage           338 1,598.521 93.866    1,450    1,600     1,750
## grosswage          325 2,967.382 823.790  1,655.000 2,713.000 6,153.380
## bonustotal         338 1,362.637 809.880    5.000   1,144.520 4,403.380
## age                338  23.683    1.309      22       23        26
## tenure             338  23.485    6.243      13       24        30
## commute            227  52.106    41.709      2       60        120
## gender_num         338   1.000    0.000       1        1         1
## married_num        338   0.000    0.000       0        0         0
## high_educ_num      338   1.000    0.000       1        1         1
## children_num       338   0.000    0.000       0        0         0
## costofcommute      338   3.565    2.123       0        3         6
## promote_switch_num 338   0.000    0.000       0        0         0
## quitjob_num        338   1.000    0.000       1        1         1
## group_index        338  16.000    0.000      16       16        16
## ---------------------------------------------------------------------
```

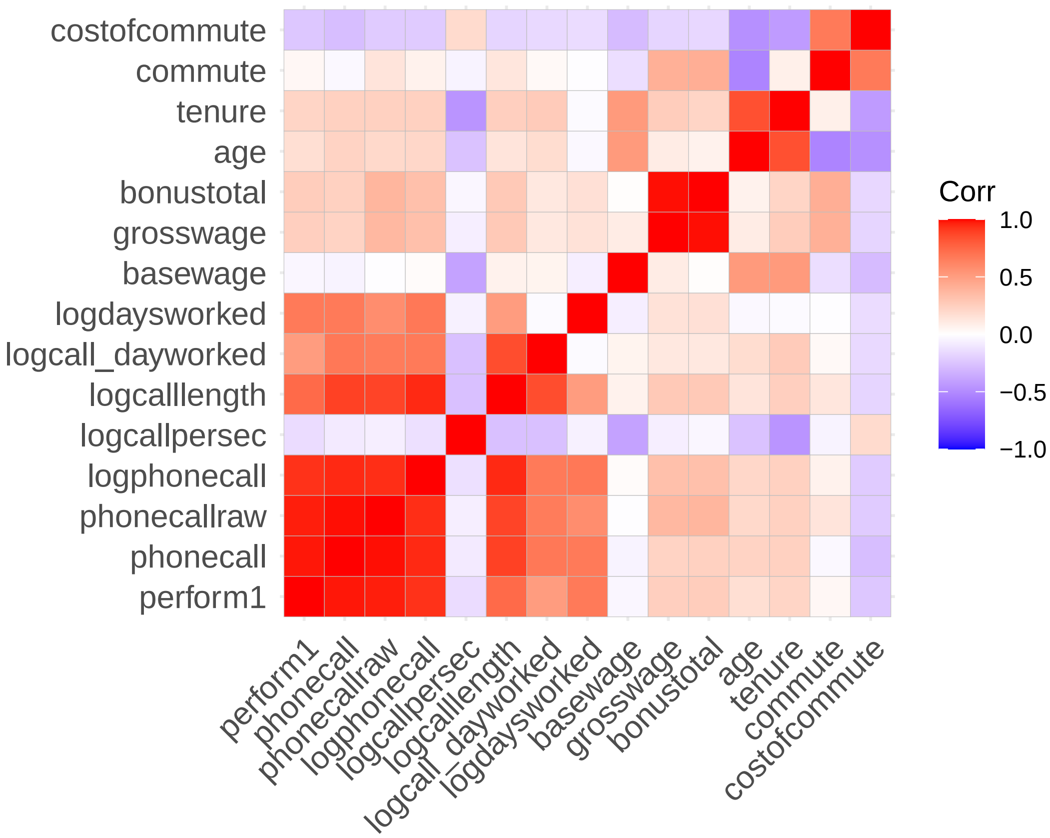
```r
# group definition
#group16 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatwee
# distinct persons
unique(group16 %>%select(personid))
```

```
##    personid
## 1     31150
## 2     37276
## 3     26634
## 5     36494
## 41    29230
## 85    26934
```

```r
nums <- dplyr::select(group16, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```r
ggcorrplot(cm, lab = FALSE)
```




```r
# radar plot
radar_by_group(df, group_col="group_index", include_group_indices=c(16), title="Group 16 characteristic
```

```
##  [1] "commute"            "costofcommute"      "negative"
##  [4] "positive"           "group_index"        "gender_num"
##  [7] "married_num"        "bedroom"            "children_num"
## [10] "high_educ_num"      "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num"   "perform1"           "grosswage"
## [16] "exhaustion"         "tenure"
## # A tibble: 62 x 17
##    group_index commute costofcommute negative positive gender married bedroom
##          <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
##  1           1   103.           6.45     12.4     29.2      1       0       0
##  2           2   180           18        21.8     20.5      1       0       0
##  3           3   120            9        18.8     23        1       0       0
##  4           4   119.           6.71     22.0     22.4      1       0       1
##  5           5    92.0          5.12     14.5     26.6      1       0       1
##  6           6   131.           7.58     18.2     24.8      1       0       1
##  7           7   118.           4.11    NaN      NaN        1       0       1
##  8           8    20            0       NaN      NaN        1       0       1
##  9           9   112.           3.88     13.8     16.2      1       0       1
## 10          10    28            0       NaN      NaN        1       0       1
## # i 52 more rows
```

```
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
## [1] 16
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##         <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1          16    52.1          3.57      NaN      NaN      1       0       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "16"
```

```
## pdf
##   2
```

Group 40 643 observations for 12 volunteers

Persona observations:
male
not married
number of bedrooms: 1
no children
no high education
no promotion
worked for 26.8 months at the company when left (= avg)
22.8 years old (<= avg)

Other average weekly observations:
not working from home
performed: -0.46 (« avg)
basewage per month: 1492 (< avg)
bonus per month: 1149 (< avg)
grosswage per month: 2604 (<=avg)
cost of commute per month: 8.3 (> avg)
commute min per week: 119 (« avg)
logdaysworked: 1.7 (~= avg)

This correlation matrix is again very similar to the full sample matrix, no outstanding differences.

Interpretation
Young male group, with no high education and family, which has never worked from home. Heavily underperformed, while working as many days as the average. Lives somewhat far away (over avg) with avg cost for commuting. Earns about 400 less than avereag despite the bad performance. Group7 had similar circumstances, despite commute, but a much better performance with lower salary. But the group is twice as big as this group.

Recommendation

As the performance is worse than average, trying to keep this group of workers should not be prioritized.

```r
# select group data
group40 <- df %>% filter(group_index==40)
stargazer(as.data.frame(group40), type = "text", median = TRUE, header = TRUE)
```

```
##
## ============================================================================
```

```
## Statistic           N     Mean    St. Dev.    Min      Median      Max
## -----------------------------------------------------------------------
## bedroom            643   1.000    0.000       1          1          1
## children           643   0.000    0.000       0          0          0
## high_educ          643   0.000    0.000       0          0          0
## promote_switch     643   0.000    0.000       0          0          0
## quitjob            643   1.000    0.000       1          1          1
## homethatweek       643   0.000    0.000       0          0          0
## perform1           642  -0.462    0.899    -3.024     -0.446     2.540
## phonecall          638  -0.392    0.900    -3.106     -0.377     2.554
## phonecallraw       631  391.097  135.837      1         385        883
## logphonecall       631   5.880    0.533     0.000      5.953      6.783
## logcallpersec      629  -5.157    0.174    -5.609     -5.162     -3.178
## logcalllength      630  11.038    0.572     3.178     11.131     11.824
## logcall_dayworked  630   9.327    0.503     2.485      9.406     10.021
## logdaysworked      642   1.697    0.265     0.000      1.792      1.946
## homethatweek_num   643   0.000    0.000       0          0          0
## basewage           638 1,491.016 141.827   650.000  1,500.000 1,900.000
## grosswage          633 2,643.891 737.108 1,360.000 2,485.170 5,011.380
## bonustotal         638 1,149.647 684.029    0.000     966.000  3,461.380
## age                643  22.879    2.293      19         23         27
## tenure             643  26.818   45.869       3         10        150
## commute            612  119.649  51.741      30        120        200
## gender_num         643   0.000    0.000       0          0          0
## married_num        643   0.000    0.000       0          0          0
## high_educ_num      643   0.000    0.000       0          0          0
## children_num       643   0.000    0.000       0          0          0
## costofcommute      643   8.283    2.400       4         10         12
## promote_switch_num 643   0.000    0.000       0          0          0
## quitjob_num        643   1.000    0.000       1          1          1
## group_index        643  40.000    0.000      40         40         40
## -----------------------------------------------------------------------
```

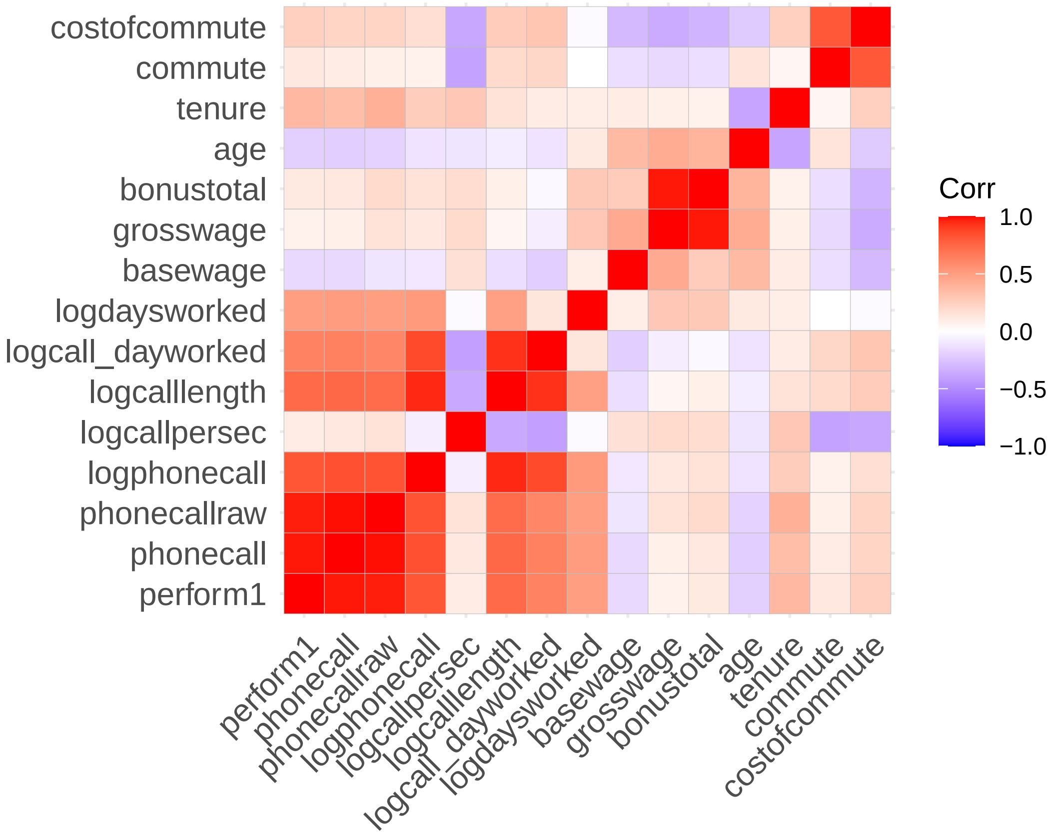
```r
# group definition
#group40 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatwee
# distinct persons
unique(group40 %>%select(personid))
```

```
##     personid
## 1      44794
## 3      43288
## 4      38552
## 7      38842
## 19     41332
## 55     40008
## 128    43524
## 136    35822
## 177    34890
## 184    41320
## 275    39096
## 405    42096
```

```r
nums <- dplyr::select(group40, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```
ggcorrplot(cm, lab = FALSE)
```




```
# radar plot
radar_by_group(df, group_col="group_index", include_group_indices=c(40), title="Group 40 characteristics
```

```
##  [1] "commute"           "costofcommute"      "negative"
##  [4] "positive"          "group_index"        "gender_num"
##  [7] "married_num"       "bedroom"            "children_num"
## [10] "high_educ_num"     "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num"  "perform1"           "grosswage"
## [16] "exhaustion"        "tenure"
## # A tibble: 62 x 17
##    group_index commute costofcommute negative positive gender married bedroom
##          <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
##  1           1   103.           6.45     12.4     29.2      1       0       0
##  2           2   180           18        21.8     20.5      1       0       0
##  3           3   120            9        18.8     23        1       0       0
##  4           4   119.           6.71     22.0     22.4      1       0       1
##  5           5    92.0          5.12     14.5     26.6      1       0       1
##  6           6   131.           7.58     18.2     24.8      1       0       1
##  7           7   118.           4.11    NaN      NaN        1       0       1
##  8           8    20            0       NaN      NaN        1       0       1
##  9           9   112.           3.88     13.8     16.2      1       0       1
## 10          10    28            0       NaN      NaN        1       0       1
## # i 52 more rows
```

```
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
## [1] 40
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##         <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1          40    120.          8.28      NaN      NaN      0       0       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "40"

## pdf
##   2
```

Group 48 218 observations for 4 volunteers

Persona observations:
male
not married
number of bedrooms: 1
no children
high education
no promotion
worked for 15.4 months at the company when left (= avg)
21.8 years old (<= avg)

Other average weekly observations:
not working from home
performed: -0.612 (« avg)
basewage per month: 1566 (< avg)
bonus per month: 1087 (< avg)
grosswage per month: 2627 (<=avg)
cost of commute per month: 9.5 (> avg)
commute min per week: 33.4 (« avg)
logdaysworked: 1.7 (~= avg)

We can see a strong correlation between costofcommute, tenure and age and a rather negative one between costofcommute, commute, tenure, age and grosswage/bonustotal. But small sample size (less confidence).

Interpretation
Young male group, with no education and no family, which has never worked from home. Heavily underperformed, even more than males with high education who quit, while working as many days as the average. Lives closer to work than the average with slightly higher than avg cost for commuting. Earns about 400 less than average despite the bad performance. Unclear why they left, probably also because they wanted more money/a promotion.

Recommendation

Because of the small sample size confidence is low. Tough, as the performance is bad, trying to keep this group of workers should not be prioritized. The company has to ask itself if it is willing to pay more money for this performance,

```r
# select group data
group48 <- df %>% filter(group_index==48)
```

```r
stargazer(as.data.frame(group48), type = "text", median = TRUE, header = TRUE)
```

```
##
## ========================================================================
## Statistic           N    Mean    St. Dev.    Min     Median      Max
## ------------------------------------------------------------------------
## bedroom            218   1.000    0.000       1         1         1
## children           218   0.000    0.000       0         0         0
## high_educ          218   1.000    0.000       1         1         1
## promote_switch     218   0.000    0.000       0         0         0
## quitjob            218   1.000    0.000       1         1         1
## homethatweek       218   0.000    0.000       0         0         0
## perform1           218  -0.627    0.690    -2.601    -0.614     1.202
## phonecall          216  -0.446    0.750    -2.618    -0.401     2.029
## phonecallraw       218  365.362  104.638     65      373.5       695
## logphonecall       216   5.845    0.379     4.174     5.923     6.544
## logcallpersec      217  -5.210    0.159    -5.721    -5.193    -4.581
## logcalllength      218  11.054    0.336     9.594    11.143    11.468
## logcall_dayworked  218   9.379    0.298     8.157     9.422     9.880
## logdaysworked      218   1.675    0.208     0.693     1.792     1.946
## homethatweek_num   218   0.000    0.000       0         0         0
## basewage           218 1,565.138 105.495    1,300     1,600     1,850
## grosswage          218 2,652.406 494.384 1,330.000 2,535.000 3,784.760
## bonustotal         218 1,087.269 438.508   30.000    985.340  2,134.760
## age                218  21.844    1.314      20        22        23
## tenure             218  15.486    12.637      4         4        34
## commute            218  33.482    27.988      1        30        75
## gender_num         218   0.000    0.000       0         0         0
## married_num        218   0.000    0.000       0         0         0
## high_educ_num      218   1.000    0.000       1         1         1
## children_num       218   0.000    0.000       0         0         0
## costofcommute      149  11.664    9.548       2         8        25
## promote_switch_num 218   0.000    0.000       0         0         0
## quitjob_num        218   1.000    0.000       1         1         1
## group_index        218  48.000    0.000      48        48        48
## ------------------------------------------------------------------------
```

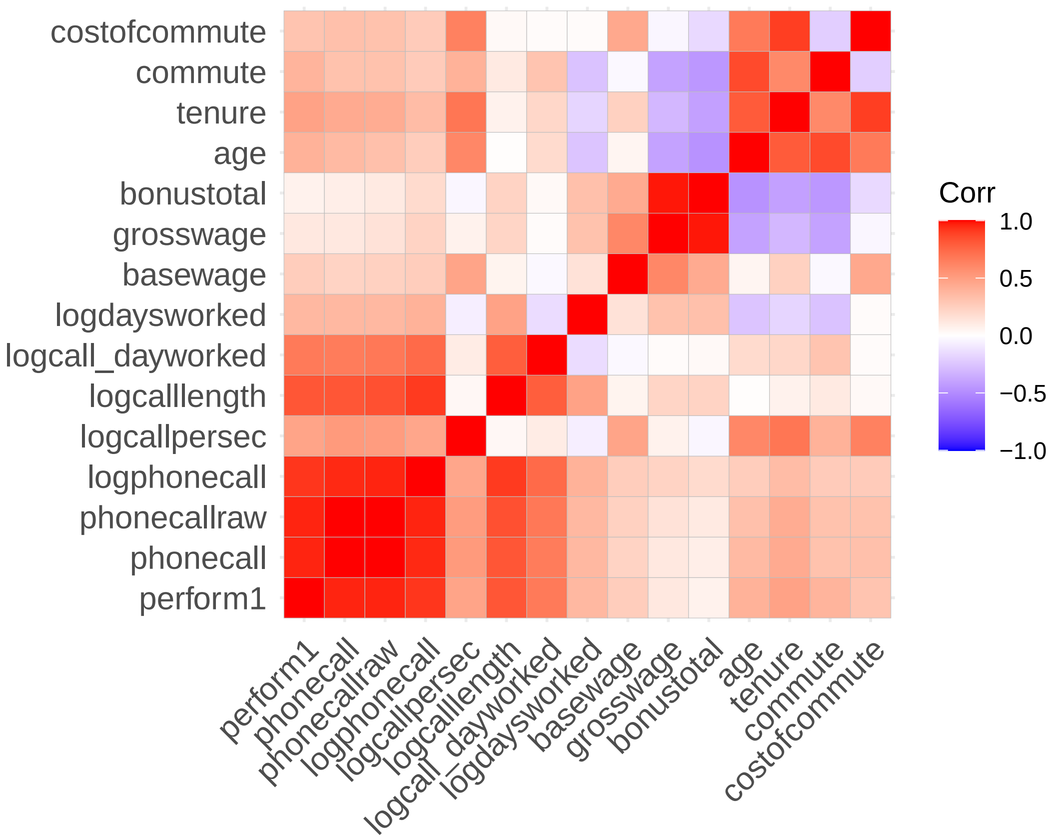
```r
# group definition
#group48 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatwee
# distinct persons
unique(group48 %>%select(personid))
```

```
##    personid
## 1     32804
## 2     24324
## 9     42618
## 48    42634
```

```r
nums <- dplyr::select(group48, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```r
ggcorrplot(cm, lab = FALSE)
```




```r
# radar plot
radar_by_group(df, group_col="group_index", include_group_indices=c(48), title="Group 48 characteristic
```

```
##  [1] "commute"           "costofcommute"     "negative"
##  [4] "positive"          "group_index"       "gender_num"
##  [7] "married_num"       "bedroom"           "children_num"
## [10] "high_educ_num"     "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num"  "perform1"          "grosswage"
## [16] "exhaustion"        "tenure"
## # A tibble: 62 x 17
##    group_index commute costofcommute negative positive gender married bedroom
##          <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
##  1           1   103.           6.45     12.4     29.2      1       0       0
##  2           2   180           18        21.8     20.5      1       0       0
##  3           3   120            9        18.8     23        1       0       0
##  4           4   119.           6.71     22.0     22.4      1       0       1
##  5           5    92.0          5.12     14.5     26.6      1       0       1
##  6           6   131.           7.58     18.2     24.8      1       0       1
##  7           7   118.           4.11    NaN      NaN        1       0       1
##  8           8    20            0       NaN      NaN        1       0       1
##  9           9   112.           3.88     13.8     16.2      1       0       1
## 10          10    28            0       NaN      NaN        1       0       1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
```

```
## [1] 48
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##         <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1          48    33.5          11.7      NaN      NaN      0       0       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "48"
```

```
## pdf
##   2
```

Group 60 202 observations for 4 volunteers

Persona observations:
male
married
number of bedrooms: 1
children
high education
no promotion
worked for 22 months at the company when left (> avg)
27.8 years old (> avg)

Other average weekly observations:
not working from home
performed: -0.22 (< avg)
basewage per month: 1555 (< avg)
bonus per month: 1256 (< avg)
grosswage per month: 2784 (<=avg)
cost of commute per month: 20.5 (» avg)
commute min per week: 101.5 (= avg)
logdaysworked: 1.5 (<= avg)

We can see a strong correlation age, tenure and commute. And a moderate between tenure, bonustotal, grosswage, basewage (steady salary raises(?) But small sample size (less confidence).

Interpretation
Older than avg. male group, with high education and family, which has never worked from home. Underperformance a bit, while working as a little less days as the average. Earn about the same amount less than what they perform less. High cost of commute and average commute minutes per week. Probably left because of this, and maybe the performance also was affect by that. As the salary was steadily improved while working their, this was probably not the main reason.

Recommendation

Because of the small sample size confidence is low. Tough, as the performance is probably worse because of the high commute cost in combination with having a family. A first important action can be the offering of working from home to reduce negative aspects of commuting and increasing performance. If performance actually rises, salary should be increased as well according to that.

```r
# select group data
group60 <- df %>% filter(group_index==60)
stargazer(as.data.frame(group60), type = "text", median = TRUE, header = TRUE)
```

```
##
## ============================================================================
```

```
## Statistic           N     Mean    St. Dev.    Min      Median      Max
## -----------------------------------------------------------------------
## bedroom            202   1.000    0.000       1          1          1
## children           202   1.000    0.000       1          1          1
## high_educ          202   1.000    0.000       1          1          1
## promote_switch     202   0.000    0.000       0          0          0
## quitjob            202   1.000    0.000       1          1          1
## homethatweek       202   0.000    0.000       0          0          0
## perform1           201  -0.216    1.292    -2.996     -0.008      2.164
## phonecall          202  -0.182    1.343    -3.113      0.077      2.729
## phonecallraw       184  435.120  181.523      2         461        823
## logphonecall       184   5.904    0.790     0.693      6.133      6.713
## logcallpersec      184  -5.074    0.196    -5.578     -5.124     -4.533
## logcalllength      183  10.976    0.754     6.271     11.183     11.880
## logcall_dayworked  183   9.416    0.631     5.578      9.599     10.088
## logdaysworked      202   1.532    0.364     0.000      1.609      1.946
## homethatweek_num   202   0.000    0.000       0          0          0
## basewage           195 1,555.128 109.941    1,400      1,550      1,800
## grosswage          195 2,811.191 652.400  1,820.000  2,549.860  4,434.900
## bonustotal         195 1,256.063 592.845   20.000   1,054.000  2,734.900
## age                202  27.847    1.740      25         28         30
## tenure             202  22.064   16.113       8          9         47
## commute            154  101.494   99.220      30         32        240
## gender_num         202   0.000    0.000       0          0          0
## married_num        202   1.000    0.000       1          1          1
## high_educ_num      202   1.000    0.000       1          1          1
## children_num       202   1.000    0.000       1          1          1
## costofcommute      202  20.515   21.521       2          2         55
## promote_switch_num 202   0.000    0.000       0          0          0
## quitjob_num        202   1.000    0.000       1          1          1
## group_index        202  60.000    0.000      60         60         60
## -----------------------------------------------------------------------
```

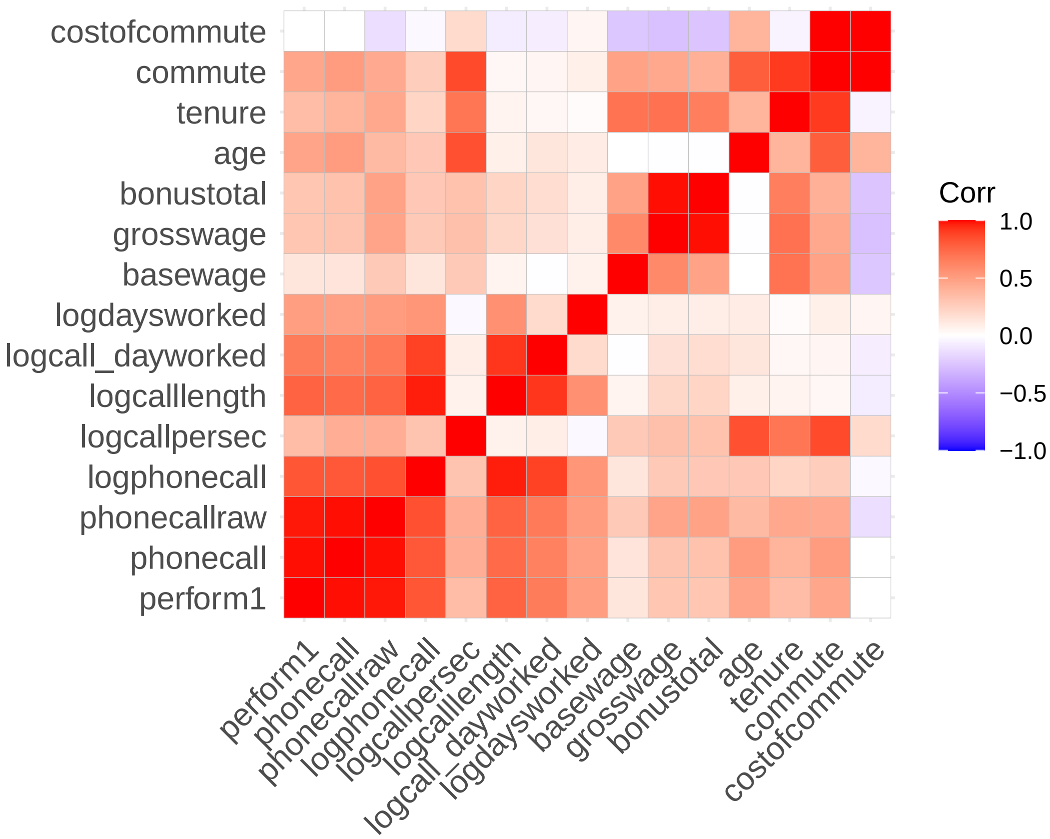
```
# group definition
#group60 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatwee
# distinct persons
unique(group60 %>%select(personid))
```

```
##    personid
## 1     40174
## 2     39942
## 6     16334
## 17    29808
```

```
nums <- dplyr::select(group60, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```
ggcorrplot(cm, lab = FALSE)
```




```r
# radar plot
radar_by_group(df, group_col="group_index", include_group_indices=c(60), title="Group 60 characteristic
```

```
##  [1] "commute"           "costofcommute"     "negative"
##  [4] "positive"          "group_index"       "gender_num"
##  [7] "married_num"       "bedroom"           "children_num"
## [10] "high_educ_num"     "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num"  "perform1"          "grosswage"
## [16] "exhaustion"        "tenure"
## # A tibble: 62 x 17
##    group_index commute costofcommute negative positive gender married bedroom
##          <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
##  1           1   103.           6.45     12.4     29.2      1       0       0
##  2           2   180           18        21.8     20.5      1       0       0
##  3           3   120            9        18.8     23        1       0       0
##  4           4   119.           6.71     22.0     22.4      1       0       1
##  5           5    92.0          5.12     14.5     26.6      1       0       1
##  6           6   131.           7.58     18.2     24.8      1       0       1
##  7           7   118.           4.11    NaN      NaN        1       0       1
##  8           8    20            0       NaN      NaN        1       0       1
##  9           9   112.           3.88     13.8     16.2      1       0       1
## 10          10    28            0       NaN      NaN        1       0       1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
```

```
## [1] 60
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##         <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1          60    101.          20.5      NaN      NaN      0       1       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "60"

## pdf
##   2
```

```r
stargazer( group60, type = "text", omit.stat = c("f", "adj.rsq", "ser"), no.space = TRUE, header = FALSE
```

```
##
## =============================================================
## Statistic           N    Mean    St. Dev.    Min       Max
## -------------------------------------------------------------
## bedroom            202   1.000    0.000       1         1
## children           202   1.000    0.000       1         1
## high_educ          202   1.000    0.000       1         1
## promote_switch     202   0.000    0.000       0         0
## quitjob            202   1.000    0.000       1         1
## homethatweek       202   0.000    0.000       0         0
## perform1           201  -0.216    1.292    -2.996     2.164
## phonecall          202  -0.182    1.343    -3.113     2.729
## phonecallraw       184  435.120  181.523      2        823
## logphonecall       184   5.904    0.790     0.693     6.713
## logcallpersec      184  -5.074    0.196    -5.578    -4.533
## logcalllength      183  10.976    0.754     6.271    11.880
## logcall_dayworked  183   9.416    0.631     5.578    10.088
## logdaysworked      202   1.532    0.364     0.000     1.946
## homethatweek_num   202   0.000    0.000       0         0
## basewage           195 1,555.128 109.941    1,400     1,800
## grosswage          195 2,811.191 652.400  1,820.000 4,434.900
## bonustotal         195 1,256.063 592.845   20.000   2,734.900
## age                202  27.847    1.740      25        30
## tenure             202  22.064   16.113       8        47
## commute            154  101.494   99.220     30        240
## gender_num         202   0.000    0.000       0         0
## married_num        202   1.000    0.000       1         1
## high_educ_num      202   1.000    0.000       1         1
## children_num       202   1.000    0.000       1         1
## costofcommute      202  20.515   21.521       2        55
## promote_switch_num 202   0.000    0.000       0         0
## quitjob_num        202   1.000    0.000       1         1
## group_index        202  60.000    0.000      60        60
## -------------------------------------------------------------
```

Compare the quitter groups

```r
df <- volunteer_endperiod_attitude_performance_wage

# plot characteristic radar plot for quitters (per group and summary)
```

```r
quitter_group_vec<-c(7,16,40,48,60)
radar_by_group(df, group_col="group_index", include_group_indices=quitter_group_vec, title="Quitter pro
```

```
##  [1] "commute"            "costofcommute"      "negative"
##  [4] "positive"           "group_index"        "gender_num"
##  [7] "married_num"        "bedroom"            "children_num"
## [10] "high_educ_num"      "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num"   "perform1"           "grosswage"
## [16] "exhaustion"         "tenure"
## # A tibble: 62 x 17
##    group_index commute costofcommute negative positive gender married bedroom
##          <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
##  1           1   103.           6.45     12.4     29.2      1       0       0
##  2           2   180           18        21.8     20.5      1       0       0
##  3           3   120            9        18.8     23        1       0       0
##  4           4   119.           6.71     22.0     22.4      1       0       1
##  5           5    92.0          5.12     14.5     26.6      1       0       1
##  6           6   131.           7.58     18.2     24.8      1       0       1
##  7           7   118.           4.11    NaN      NaN        1       0       1
##  8           8    20            0       NaN      NaN        1       0       1
##  9           9   112.           3.88     13.8     16.2      1       0       1
## 10          10    28            0       NaN      NaN        1       0       1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
## [1]  7 16 40 48 60
## # A tibble: 5 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##         <int>   <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1           7   118.           4.11      NaN      NaN      1       0       1
## 2          16    52.1          3.57      NaN      NaN      1       0       1
## 3          40   120.           8.28      NaN      NaN      0       0       1
## 4          48    33.5         11.7       NaN      NaN      0       0       1
## 5          60   101.          20.5       NaN      NaN      0       1       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "7"  "16" "40" "48" "60"
```

```
## pdf
##   2
```

```r
radar_by_group(df, group_col="quitjob", include_group_indices=c(TRUE), title="General quitter profile")
```

```
##  [1] "commute"            "costofcommute"      "negative"
##  [4] "positive"           "quitjob"            "gender_num"
##  [7] "married_num"        "bedroom"            "children_num"
## [10] "high_educ_num"      "promote_switch_num" "homethatweek_num"
## [13] "perform1"           "grosswage"          "exhaustion"
## [16] "tenure"
## # A tibble: 2 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
```

```
## 1 FALSE      103.           6.59     16.7     24.2  0.513   0.163   0.966
## 2 TRUE       105.           9.14    NaN      NaN    0.450   0.239   1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "quitjob"
## [1] TRUE
## # A tibble: 1 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1 TRUE       105.          9.14      NaN      NaN  0.450   0.239       1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "TRUE"
```

```
## pdf
##   2
```

```r
# plot characteristic radar plot for non quitters (per group and summary)
# get all available groups
all_groups_vec <- unique(df %>% pull(group_index))
non_quitter_group_vec <- setdiff(all_groups_vec, quitter_group_vec)

radar_by_group(df, group_col="quitjob", include_group_indices=c(FALSE), title="General non-quitter profi
```

```
##  [1] "commute"            "costofcommute"      "negative"
##  [4] "positive"           "quitjob"            "gender_num"
##  [7] "married_num"        "bedroom"            "children_num"
## [10] "high_educ_num"      "promote_switch_num" "homethatweek_num"
## [13] "perform1"           "grosswage"          "exhaustion"
## [16] "tenure"
## # A tibble: 2 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1 FALSE      103.          6.59     16.7     24.2  0.513   0.163   0.966
## 2 TRUE       105.          9.14    NaN      NaN    0.450   0.239   1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "quitjob"
## [1] FALSE
## # A tibble: 1 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1 FALSE      103.          6.59     16.7     24.2  0.513   0.163   0.966
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "FALSE"
```

```
## pdf
##   2
```

```r
#radar_by_group(df)
```

Top performing 10%

Select

```r
df <- volunteer_endperiod_attitude_performance_wage

#df <- na.omit(df$perform1)

# define threshold
threshold <- quantile(df$perform1, 0.9, na.rm = TRUE)

# select data
df <- df %>%
  mutate(isTop10 = if_else(perform1 >= threshold, TRUE, FALSE)) %>% filter(!is.na(isTop10))

top10 <- df %>% filter(isTop10 == TRUE)
bottom90 <- df %>% filter(isTop10 == FALSE)
```

Compare

Here we use the same approach as for the quitters:

```r
# compare general statistics
stargazer(as.data.frame(top10), type = "text", median = TRUE, header = TRUE)
```

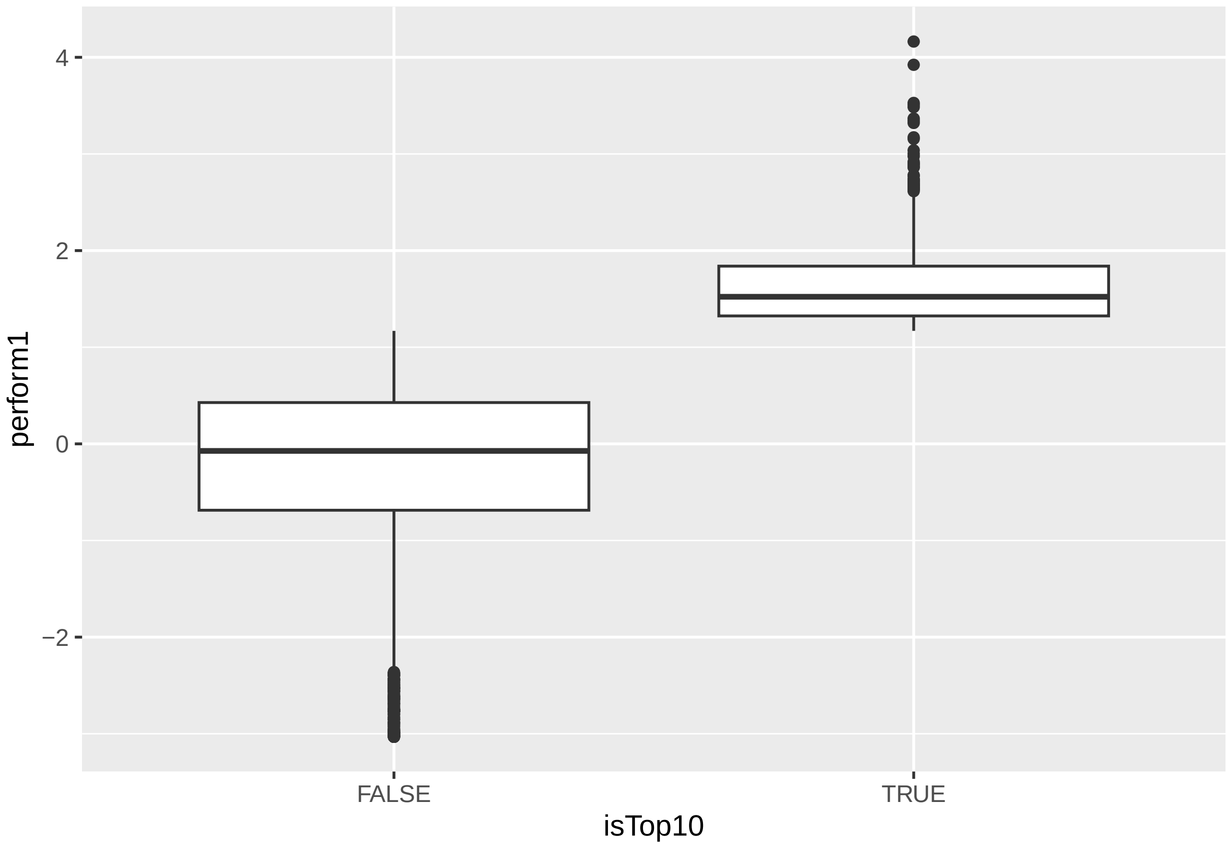
```
##
## ========================================================================
## Statistic           N     Mean    St. Dev.    Min      Median     Max
## ------------------------------------------------------------------------
## bedroom            983   0.961     0.193       0         1         1
## children           983   0.239     0.427       0         0         1
## high_educ          983   0.481     0.500       0         0         1
## promote_switch     983   0.217     0.412       0         0         1
## quitjob            983   0.182     0.386       0         0         1
## homethatweek       983   0.254     0.436       0         0         1
## exhaustion         189   6.804     6.105       0         6        25
## negative           189  14.947     5.966       8        13        40
## positive           189  26.619     5.710       8        28        40
## perform1           983   1.643     0.430     1.169     1.522     4.163
## phonecall          982   1.480     0.488     0.039     1.415     5.828
## phonecallraw       980  665.388   93.648      426      649.5     1,264
## logphonecall       980   6.491     0.134     6.054     6.475     7.142
## logcallpersec      979  -5.125     0.140    -5.458    -5.120    -3.832
## logcalllength      980  11.615     0.160    10.231    11.610    12.116
## logcall_dayworked  981   9.818     0.162     8.285     9.827    10.359
## logdaysworked      982   1.796     0.137     1.386     1.792     1.946
## homethatweek_num   983   0.254     0.436       0         0         1
## basewage           980 1,658.520  183.865    1,300     1,600     2,300
## grosswage          969 3,750.751 1,055.672 1,375.000 3,546.670 8,418.000
## bonustotal         980 2,096.829  953.693   70.000   1,943.205 6,268.000
## age                983  23.834     5.606       1        23        34
## tenure             983  32.765    25.032       2        28        150
## commute            947  107.360   69.337       2        80        300
## gender_num         983   0.752     0.432       0         1         1
## married_num        983   0.308     0.462       0         0         1
```

```
## high_educ_num      983    0.481     0.500      0         0          1
## children_num       983    0.239     0.427      0         0          1
## costofcommute      905    6.407     6.759     0.000     4.000     55.000
## promote_switch_num 983    0.217     0.412      0         0          1
## quitjob_num        983    0.182     0.386      0         0          1
## group_index        983   23.146    17.373      1         16         62
## isTop10            983    1.000     0.000      1         1          1
## ------------------------------------------------------------------------
```

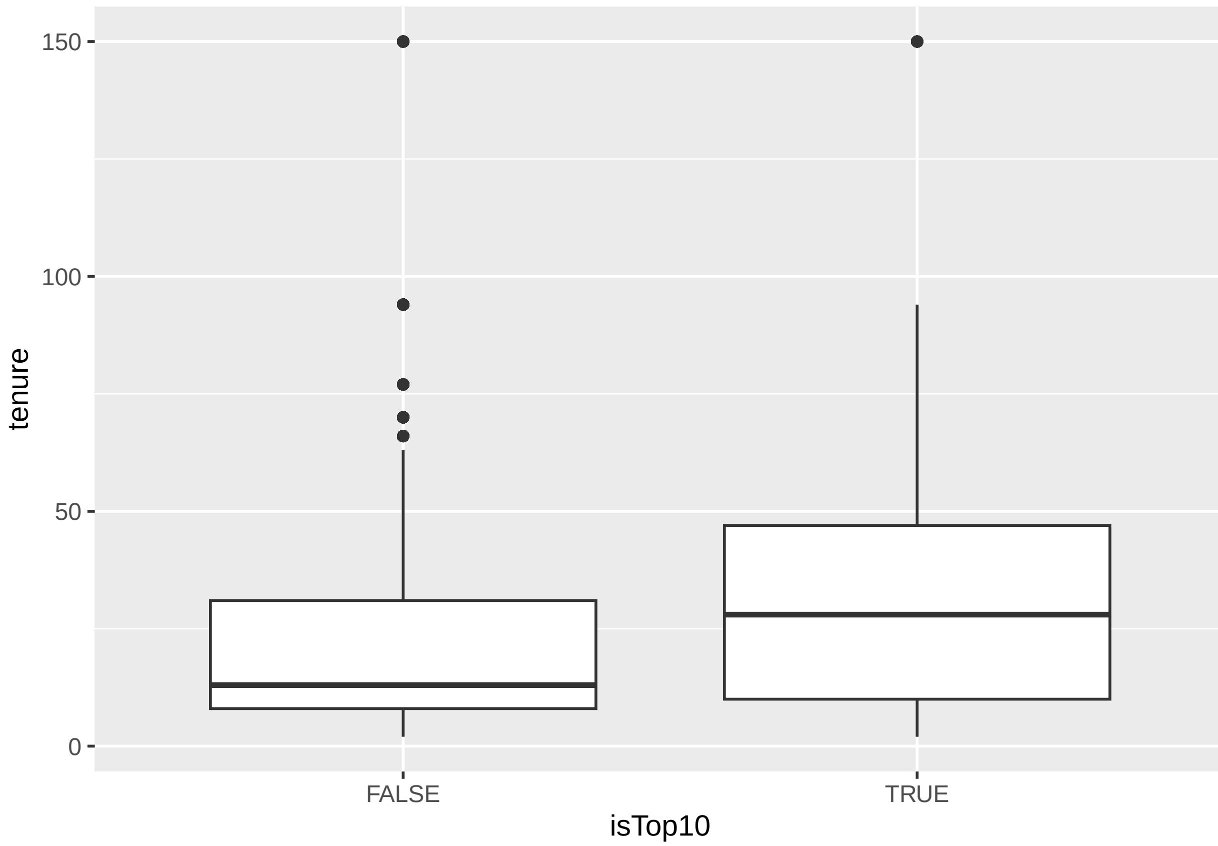
```r
stargazer(as.data.frame(bottom90), type = "text", median = TRUE, header = TRUE)
```

```
##
## ==========================================================================
## Statistic             N     Mean    St. Dev.    Min     Median     Max
## --------------------------------------------------------------------------
## bedroom            8,844   0.975    0.156       0         1         1
## children           8,844   0.132    0.339       0         0         1
## high_educ          8,844   0.392    0.488       0         0         1
## promote_switch     8,844   0.177    0.381       0         0         1
## quitjob            8,844   0.249    0.433       0         0         1
## homethatweek       8,844   0.190    0.392       0         0         1
## exhaustion         1,976   8.742    7.861     0.000     8.000     36.000
## negative           1,976  16.800    6.861     8.000    16.000     40.000
## positive           1,976  23.948    6.699     8.000    24.000     40.000
## perform1           8,844  -0.199    0.851    -3.031    -0.073     1.169
## phonecall          8,714  -0.172    0.854    -3.113    -0.046     3.812
## phonecallraw       8,561  414.722  123.299      1        429       977
## logphonecall       8,574   5.955    0.476     0.000     6.061     6.884
## logcallpersec      8,571  -5.175    0.160    -5.951    -5.173    -1.099
## logcalllength      8,571  11.128    0.511     2.485    11.237    11.883
## logcall_dayworked  8,569   9.438    0.448     2.485     9.514    10.186
## logdaysworked      8,830   1.674    0.288     0.000     1.792     1.946
## homethatweek_num   8,844   0.190    0.392       0         0         1
## basewage           8,794 1,590.322 170.628   527.350  1,550.000 2,450.000
## grosswage          8,702 3,009.238 942.059 1,140.280 2,808.740 14,553.000
## bonustotal         8,794 1,417.370 872.446   0.000   1,208.000 12,853.000
## age                8,844  23.444    4.026       1        23        35
## tenure             8,844  23.011   24.617       2        13        150
## commute            7,936  103.154  67.580       1        80        300
## gender_num         8,844   0.472    0.499       0         0         1
## married_num        8,844   0.170    0.375       0         0         1
## high_educ_num      8,844   0.392    0.488       0         0         1
## children_num       8,844   0.132    0.339       0         0         1
## costofcommute      8,336   7.328    7.314     0.000     6.000     55.000
## promote_switch_num 8,844   0.177    0.381       0         0         1
## quitjob_num        8,844   0.249    0.433       0         0         1
## group_index        8,844  29.117   17.646       1        37         62
## isTop10            8,844   0.000    0.000       0         0         0
## --------------------------------------------------------------------------
```

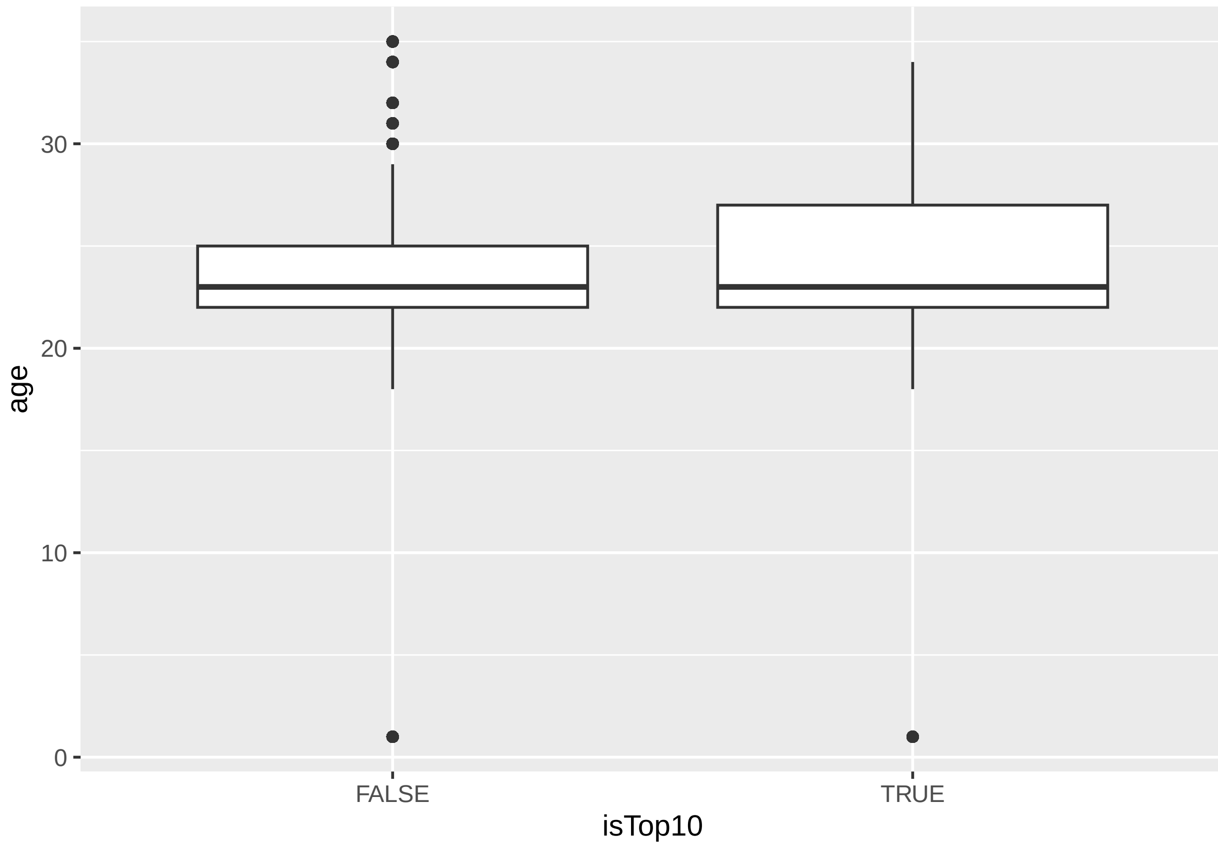
```r
# compare distributions
ggplot(df, aes(x=isTop10, y=perform1)) + geom_boxplot()
```




```r
ggplot(df, aes(x=isTop10, y=tenure)) + geom_boxplot()
```

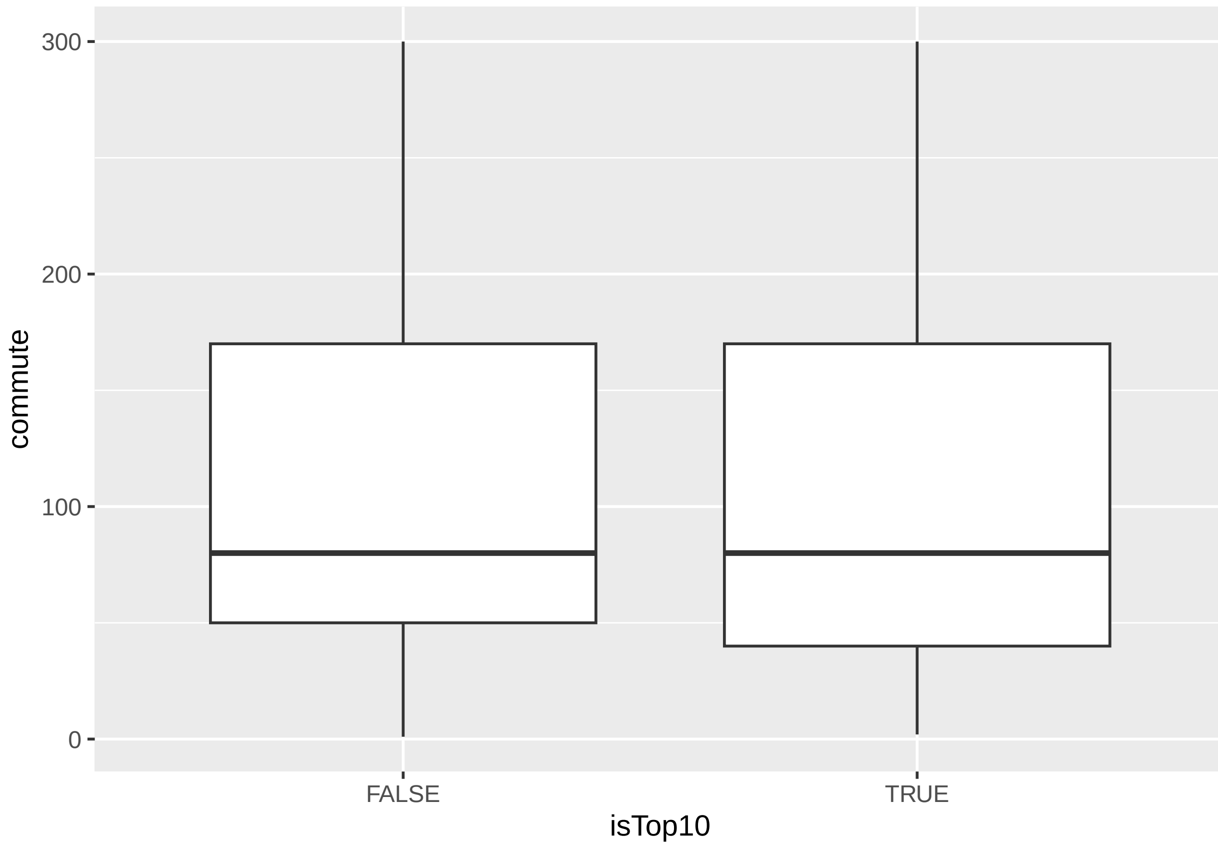



```
ggplot(df, aes(x=isTop10, y=age)) + geom_boxplot()
```



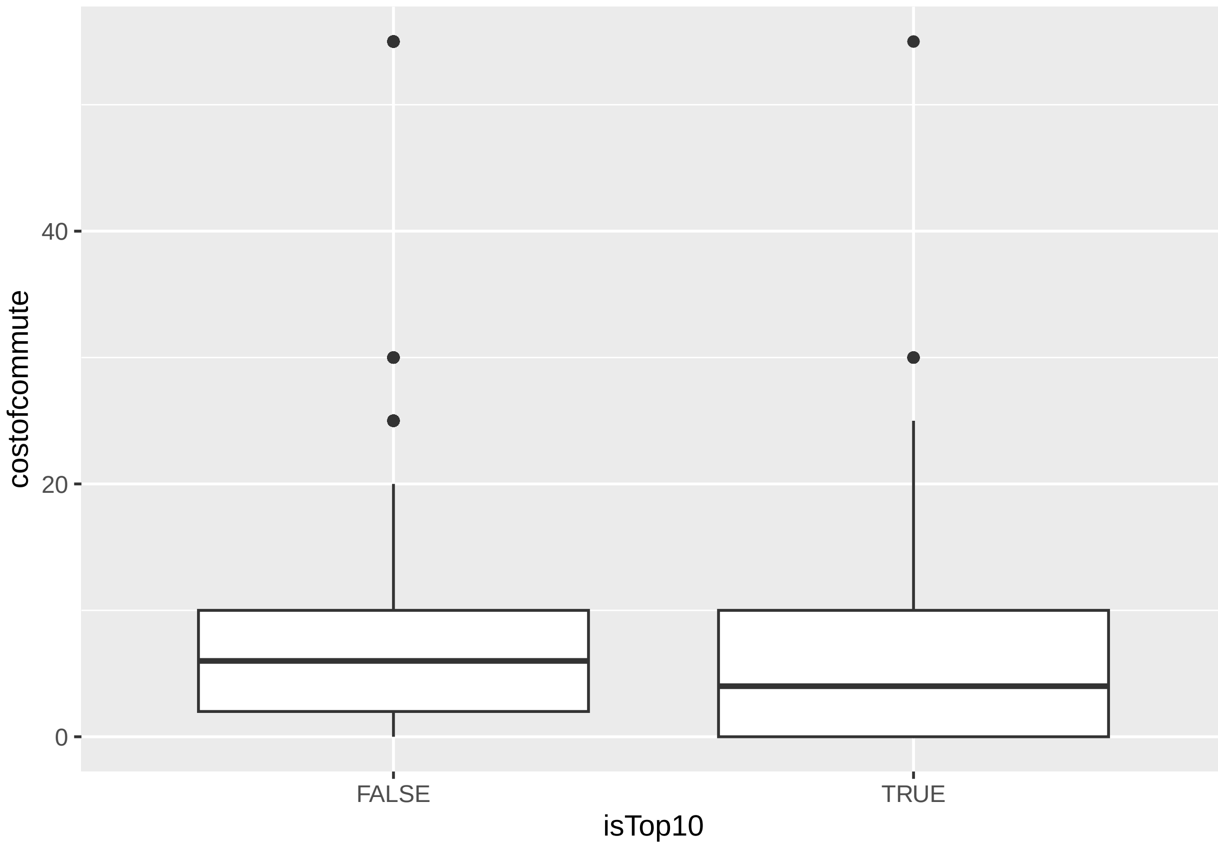

```r
ggplot(df, aes(x=isTop10, y=commute)) + geom_boxplot()
```

```
## Warning: Removed 944 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



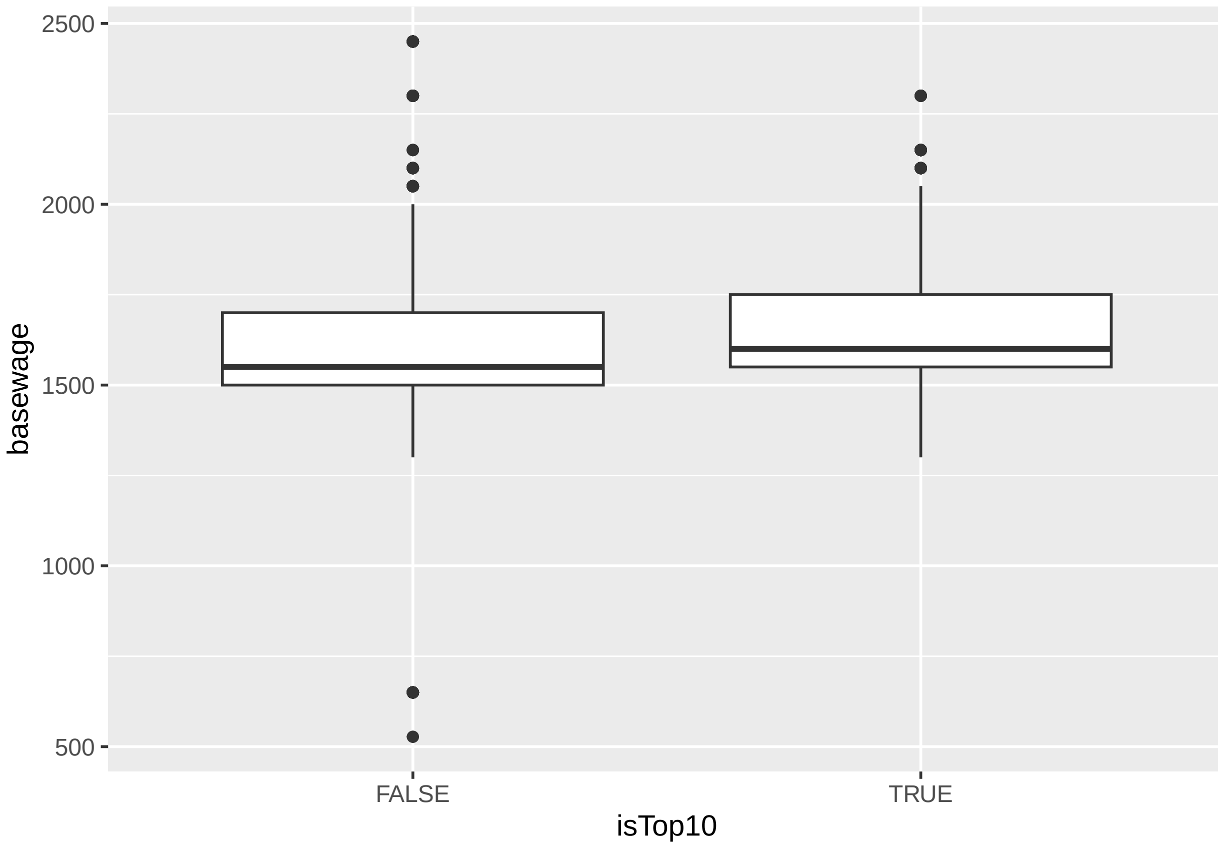

```
ggplot(df, aes(x=isTop10, y=costofcommute)) + geom_boxplot()
```

```
## Warning: Removed 586 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



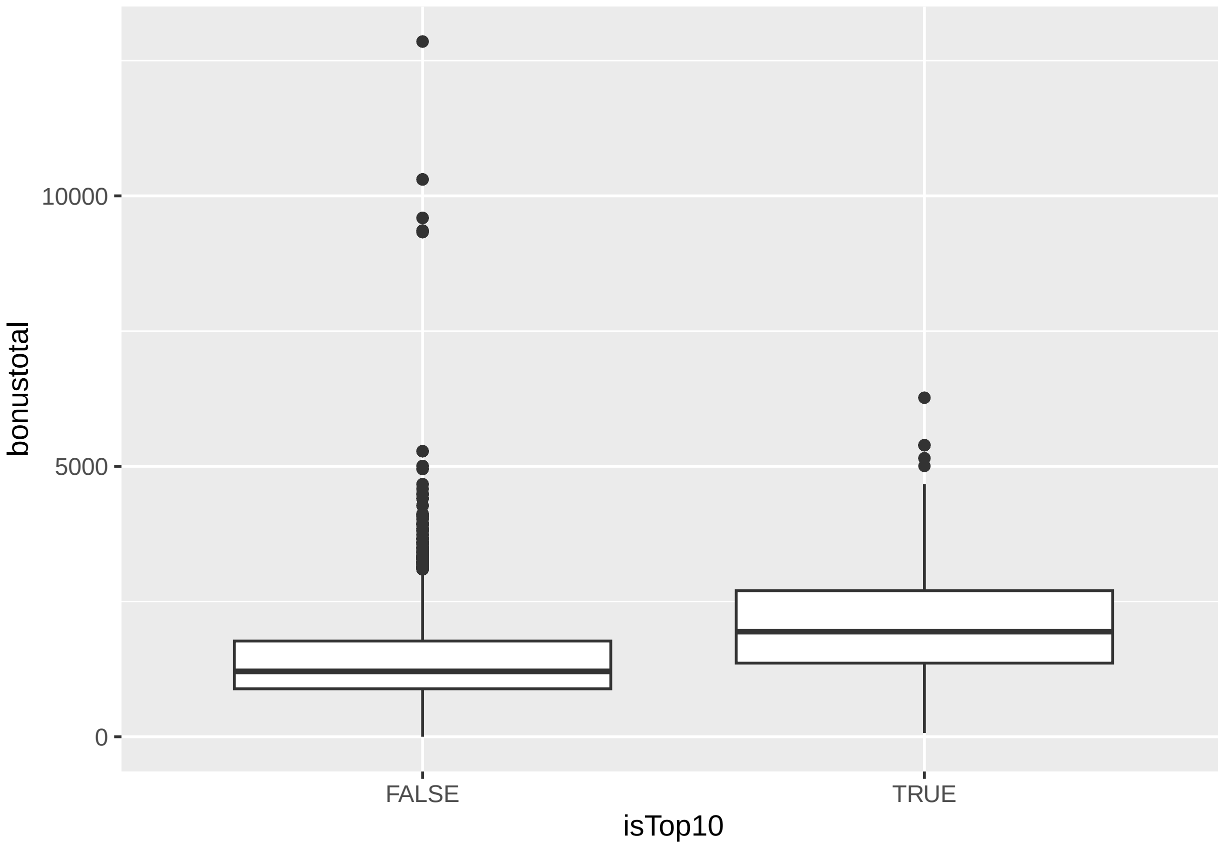

```
ggplot(df, aes(x=isTop10, y=basewage)) + geom_boxplot()
```

```
## Warning: Removed 53 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



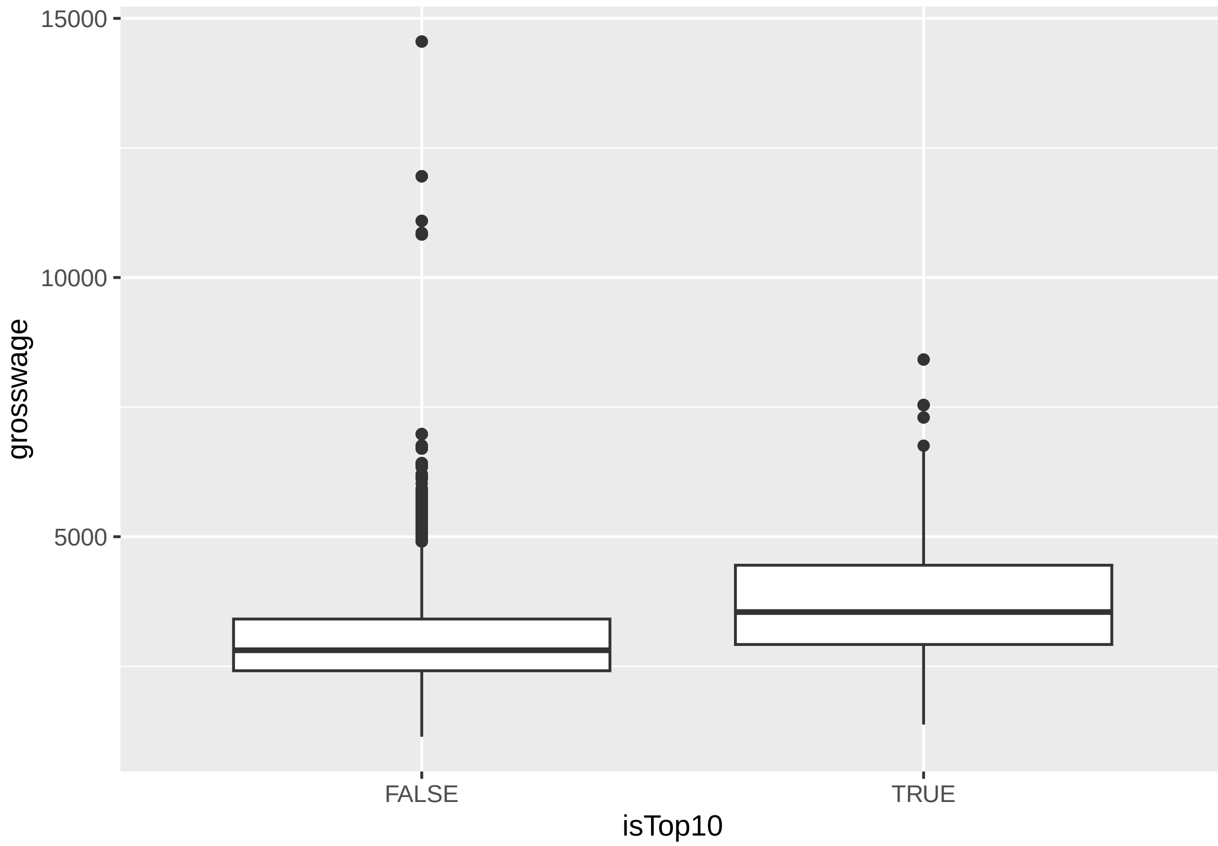

```
ggplot(df, aes(x=isTop10, y=bonustotal)) + geom_boxplot()
```

```
## Warning: Removed 53 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
ggplot(df, aes(x=isTop10, y=grosswage)) + geom_boxplot()
```

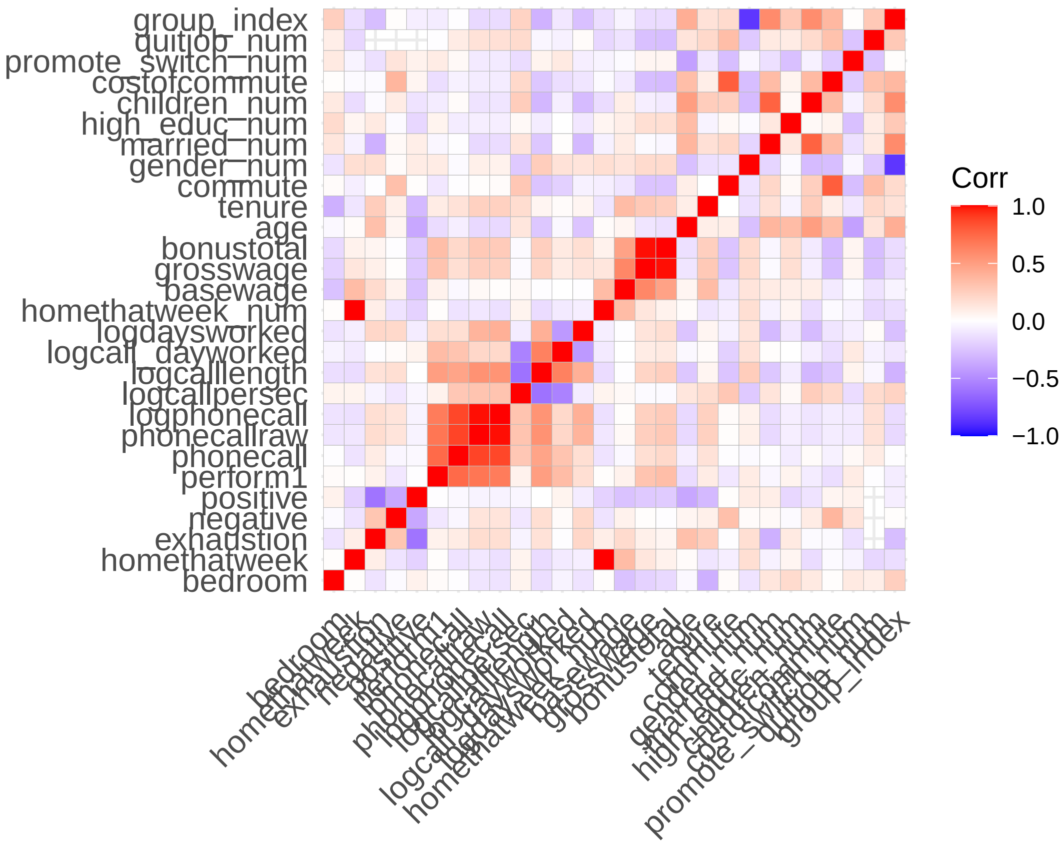
```
## Warning: Removed 156 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```r
# create correlation matrices
nums <- dplyr::select(top10, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

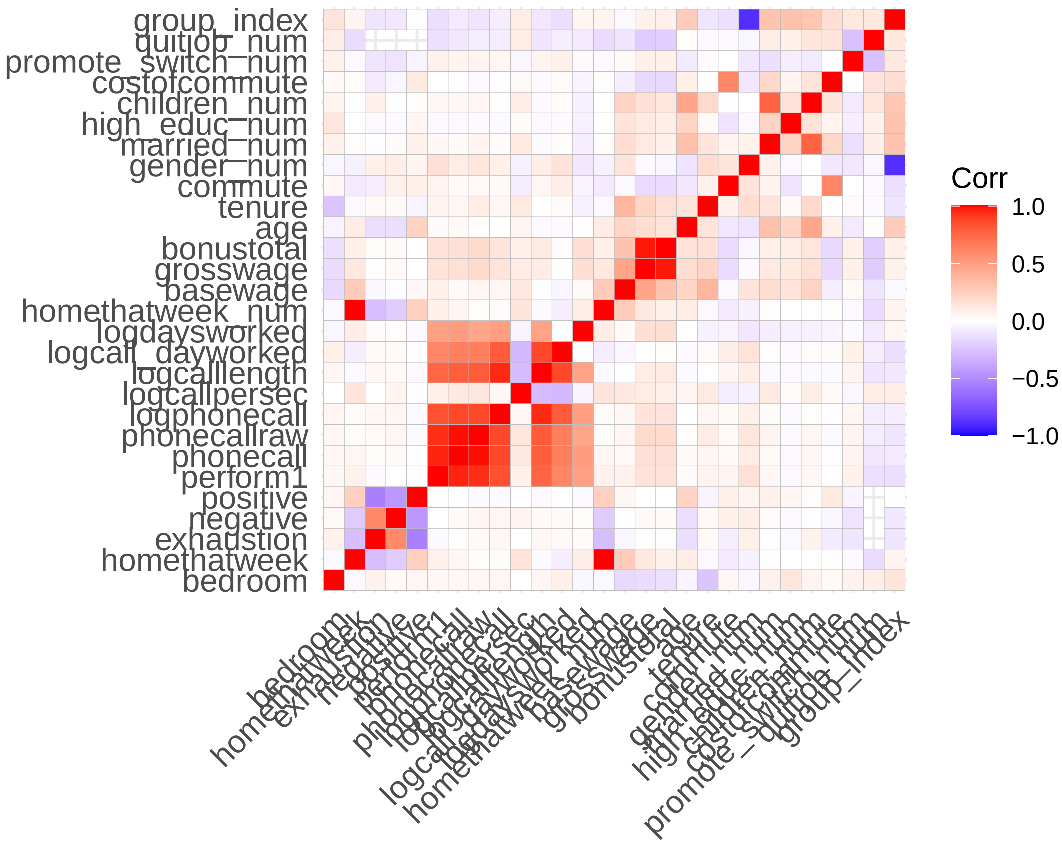
```r
ggcorrplot(cm, lab = FALSE)
```




```r
nums <- dplyr::select(bottom90, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```r
ggcorrplot(cm, lab = FALSE)
```




```
# radar plot
radar_by_group(df, group_col="isTop10", include_group_indices=c(TRUE, FALSE), title="Top vs low performe
```

```
##  [1] "commute"            "costofcommute"      "negative"
##  [4] "positive"           "isTop10"            "gender_num"
##  [7] "married_num"        "bedroom"            "children_num"
## [10] "high_educ_num"      "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num"   "perform1"           "grosswage"
## [16] "exhaustion"         "tenure"
## # A tibble: 2 x 17
##   isTop10 commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1 FALSE      103.          7.33     16.8     23.9  0.472   0.170   0.975
## 2 TRUE       107.          6.41     14.9     26.6  0.752   0.308   0.961
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "isTop10"
## [1]  TRUE FALSE
## # A tibble: 2 x 17
##   isTop10 commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1 FALSE      103.          7.33     16.8     23.9  0.472   0.170   0.975
## 2 TRUE       107.          6.41     14.9     26.6  0.752   0.308   0.961
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
```

```
## [1] "FALSE" "TRUE"
```

```
## pdf
##   2
```

The average employee among the top 10% performing ones, has no children, though it is more likely than for the other 90%, works 25% oft the time from home, is almost 24 years old, works for 32 months at the company and an average commute and average cost of commute. Almost 22% receive a promotion, whereas 17.7% of lower performing people receive one. They earn over the average by bonuses, but have a average base salary. Though some not high performing individuals receive enormous bonuses. Only 18% high performing people quit while 25% quit from the other 90%.

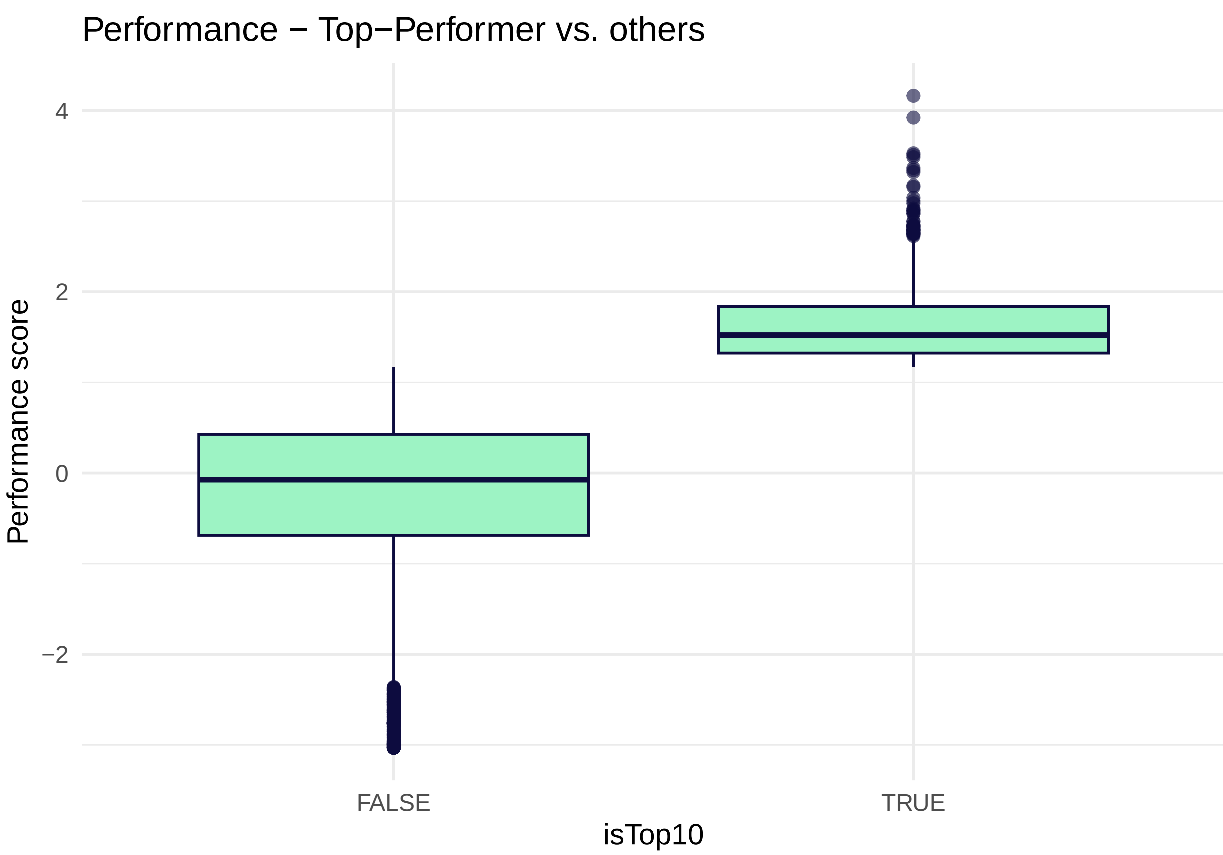
Top performer are less likely to quit when receiving a higher bonus than non performer, but more likely if they have a long commute as this is negatively correlative with their negative attitude. Working from home does not affect performance of both groups. Working from home increases positivity while lowering negativity and exhaustion for the top performer.

Recommendations

- Bonuses combined with goals/milestones to motivate people and increase performance, but only if you reach them
- check existing bonuses for not high performing people and reevaluate if that is fair, this might be a motivation killer
- offer more working from home for high performer, as it does not affect their performance and rather increases positivity

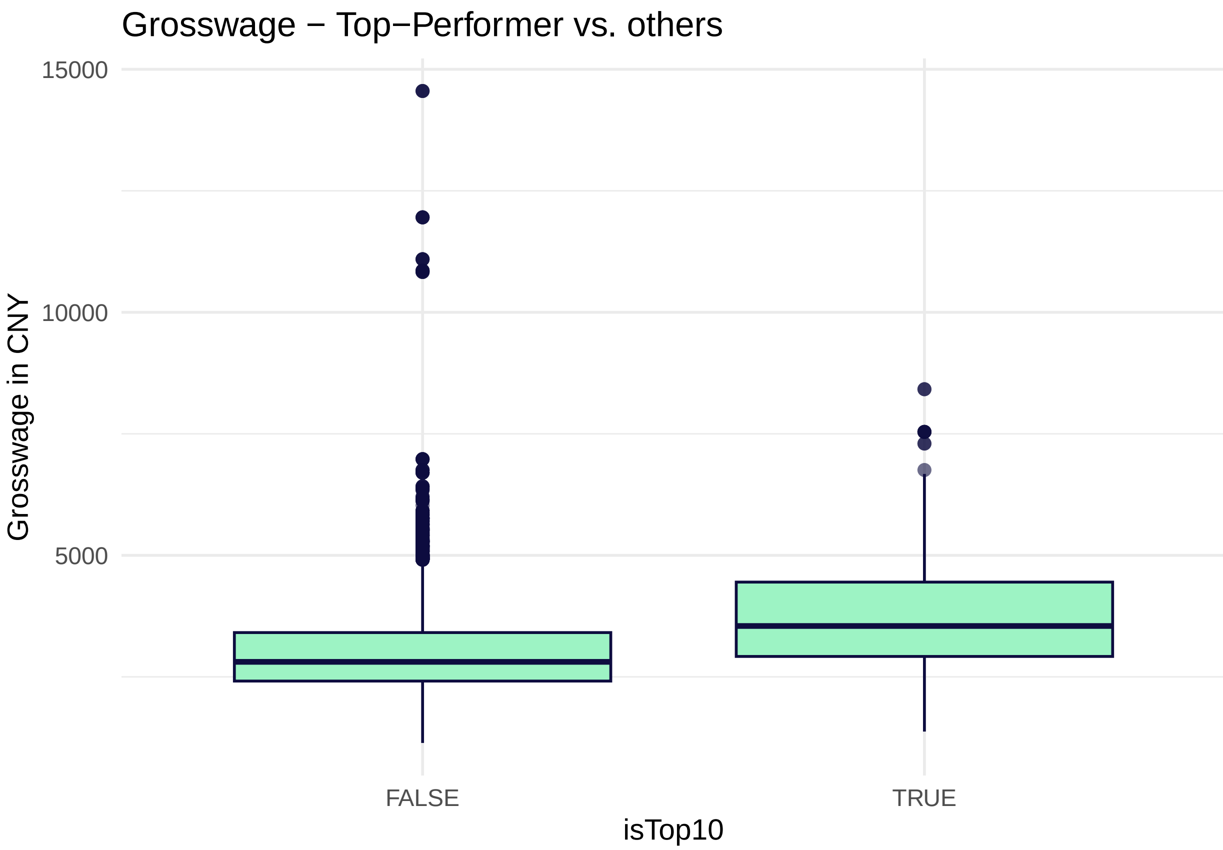
Make perform1 and grosswage poxplots look nice

```r
p <- ggplot(df, aes(y = perform1, x=isTop10)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Performance score", title = "Performance - Top-Performer vs. others") +
  theme_minimal()

# add annotations
p
```




```
ggsave("performance_top_low.png", width = 7, height = 5, dpi = 300)


p <- ggplot(df, aes(y = grosswage, x=isTop10)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Grosswage in CNY", title = "Grosswage - Top-Performer vs. others") +
  theme_minimal()

# add annotations
p
```

```
## Warning: Removed 156 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```r
colnames(df)
```

```
##  [1] "gender"             "married"            "bedroom"
##  [4] "children"           "high_educ"          "promote_switch"
##  [7] "quitjob"            "homethatweek"       "personid"
## [10] "year_month"         "year_week"          "exhaustion"
## [13] "negative"           "positive"           "perform1"
## [16] "phonecall"          "phonecallraw"       "logphonecall"
## [19] "logcallpersec"      "logcalllength"      "logcall_dayworked"
## [22] "logdaysworked"      "date"               "homethatweek_num"
## [25] "week_date"          "basewage"           "grosswage"
## [28] "bonustotal"         "age"                "tenure"
## [31] "commute"            "gender_num"         "married_num"
## [34] "high_educ_num"      "children_num"       "costofcommute"
## [37] "promote_switch_num" "quitjob_num"        "group_index"
## [40] "isTop10"
```

```r
ggsave("grosswage_top_low.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 156 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

Hypothesis testing

Quitters vs non quitters

```r
# H0: There is no difference in earnings between quitters and non-quitters
# H1: There is a significant difference in earnings between quitters and non-quitters
volunteer_endperiod_attitude_performance_wage %>%
  group_by(personid, quitjob_num) %>%
  summarise(mean_grossw = mean(grosswage, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(mean_grossw)) %>%
  t.test(mean_grossw ~ quitjob_num, data = ., alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  mean_grossw by quitjob_num
## t = 5.1008, df = 97.864, p-value = 8.285e-07
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0
## 95 percent confidence interval:
##  367.794     Inf
## sample estimates:
## mean in group 0 mean in group 1
##        3210.301        2664.978
```

```r
# we reject the null hypothesis, as p-value is below 0.05


#H0: There is no difference between perform1 of quitters and non-quitters
#H1: There is a significant difference between perform1 of quitters and non-quitters
volunteer_endperiod_attitude_performance_wage %>%
  group_by(personid, quitjob_num) %>%
  summarise(mean_perform1 = mean(perform1, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(mean_perform1)) %>%
  t.test(mean_perform1 ~ quitjob_num, data = ., alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  mean_perform1 by quitjob_num
## t = 3.0921, df = 76.915, p-value = 0.9986
## alternative hypothesis: true difference in means between group 0 and group 1 is less than 0
## 95 percent confidence interval:
##       -Inf 0.4601867
## sample estimates:
## mean in group 0 mean in group 1
##      0.02464293     -0.27448432
```

```r
# we reject the null hypothesis, as p-value is below 0.05


#H0: There is no difference in tenure between quitters and non-quitters
#H1: There is a significant difference in tenure between quitters and non-quitters
volunteer_endperiod_attitude_performance_wage %>%
  group_by(personid, quitjob_num) %>%
  summarise(mean_tenure = mean(tenure, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(mean_tenure)) %>%
  t.test(mean_tenure ~ quitjob_num, data = ., alternative = "greater")
```

```
##
```

```
##  Welch Two Sample t-test
##
## data:  mean_tenure by quitjob_num
## t = 0.37039, df = 64.932, p-value = 0.3561
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0
## 95 percent confidence interval:
##  -6.401416       Inf
## sample estimates:
## mean in group 0 mean in group 1
##        23.92632        22.10000
```

```r
# we fail to reject the null hypothesis, as p-value is greater than 0.05


#H0: there is no difference in commutecost between quitters and non-quitters
#H1: there is a significant difference in commutecost between quitters and non-quitters
volunteer_endperiod_attitude_performance_wage %>%
  group_by(personid, quitjob_num) %>%
  summarise(mean_costofcommute = mean(costofcommute, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(mean_costofcommute)) %>%
  t.test(mean_costofcommute ~ quitjob_num, data = ., alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  mean_costofcommute by quitjob_num
## t = -1.3774, df = 50.59, p-value = 0.9128
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0
## 95 percent confidence interval:
##  -5.04748      Inf
## sample estimates:
## mean in group 0 mean in group 1
##        6.608471        8.885781
```

```r
# we fail to reject the null hypothesis, as p-value is greater than 0.05


#H0: there is no difference in homeoffice_days between quitters and non-quitters
#H1: there is a significant difference in homeoffice_days between quitters and non-quitters
volunteer_endperiod_attitude_performance_wage %>%
  group_by(personid, quitjob_num) %>%
  summarise(mean_homethatweek = mean(homethatweek_num, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(mean_homethatweek)) %>%
  t.test(mean_homethatweek ~ quitjob_num, data = ., alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  mean_homethatweek by quitjob_num
## t = 4.9726, df = 113.28, p-value = 1.185e-06
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0
## 95 percent confidence interval:
##  0.1012484       Inf
## sample estimates:
## mean in group 0 mean in group 1
```

```
##      0.22867153      0.07675795
```

```r
# we reject the null hypothesis, as p-value is below 0.05


#H0: there is no correlation between age and children_num among quitters
#H1: there is a correlation between age and children_num among quitters
volunteer_endperiod_attitude_performance_wage %>%
  filter(quitjob_num == 1) %>%
  t.test(children_num ~ (age > median(age, na.rm = TRUE)),
         data = ., alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  children_num by age > median(age, na.rm = TRUE)
## t = -27.221, df = 1135.4, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##        -Inf -0.4341011
## sample estimates:
## mean in group FALSE  mean in group TRUE
##          0.03795721          0.50000000
```

```r
# we reject the null hypothesis, as p-value is below 0.05


#H0: there is no correlation between age and married_num among quitters
#H1: there is a correlation between age and married_num among quitters
volunteer_endperiod_attitude_performance_wage %>%
  filter(quitjob_num == 1) %>%
  t.test(married_num ~ (age > median(age, na.rm = TRUE)),
         data = ., alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  married_num by age > median(age, na.rm = TRUE)
## t = -17.35, df = 1467.5, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##        -Inf -0.2842208
## sample estimates:
## mean in group FALSE  mean in group TRUE
##           0.1145618           0.4285714
```

```r
# we reject the null hypothesis, as p-value is below 0.05


#H0: there is no correlation between age and high_educ_num among quitters
#H1: there is a correlation between age and high_educ_num among quitters
volunteer_endperiod_attitude_performance_wage %>%
  filter(quitjob_num == 1) %>%
  t.test(high_educ_num ~ (age > median(age, na.rm = TRUE)),
         data = ., alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  high_educ_num by age > median(age, na.rm = TRUE)
## t = -15.705, df = 2020.9, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##        -Inf -0.2790255
## sample estimates:
## mean in group FALSE  mean in group TRUE
##           0.3374741           0.6491597
```

```
# we reject the null hypothesis, as p-value is below 0.05


#H0: there is no correlation between mean(perform1) and age among quitters
#H1: there is a correlation between mean(perform1) and age among quitters
volunteer_endperiod_attitude_performance_wage %>%
  filter(quitjob_num == 1) %>%
  t.test(perform1 ~ (age > median(age, na.rm = TRUE)),
         data = ., alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  perform1 by age > median(age, na.rm = TRUE)
## t = -6.0454, df = 1700.1, p-value = 9.138e-10
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##        -Inf -0.1953103
## sample estimates:
## mean in group FALSE  mean in group TRUE
##         -0.34598213         -0.07761384
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

Group Hypotheses

Group 7

```
#H0: There is no difference between group 7 mean(quitjob_num) compared to overall average
#H1: Group 7 has higher mean(quitjob_num) compared to overall average
t.test(group7$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "gr
```

```
##
##  Welch Two Sample t-test
##
## data:  group7$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
## t = 179.52, df = 10078, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.7548015       Inf
## sample estimates:
## mean of x mean of y
## 1.0000000 0.2382181
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between group 7 mean(perform1) compared to overall average
#H1: Group 7 has lower mean(perform1) compared to overall average
t.test(group7$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group7$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
## t = -6.218, df = 377.04, p-value = 6.683e-10
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##        -Inf -0.2149481
## sample estimates:
##   mean of x   mean of y
## -0.30751216 -0.01499374
```

```r
# we reject the null hypothesis, as p-value is below 0.05
```

Group 16

```r
#H0: There is no difference between group 16 mean(quitjob_num) compared to overall average
#H1: Group 16 has higher mean(quitjob_num) compared to overall average
t.test(group16$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "g
```

```
##
##  Welch Two Sample t-test
##
## data:  group16$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
## t = 179.52, df = 10078, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.7548015       Inf
## sample estimates:
## mean of x mean of y
## 1.0000000 0.2382181
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between group 16 mean(perform1) compared to overall average
#H1: Group 16 has lower mean(perform1) compared to overall average
t.test(group16$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group16$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
## t = 0.24946, df = 356.69, p-value = 0.5984
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##      -Inf 0.113049
## sample estimates:
##     mean of x     mean of y
## -0.0001399055 -0.0149937416
```

```r
# we fail to reject the null hypothesis, as p-value is above 0.05

#H0: There is no correlation between age and tenure
#H1: There is a correlation between age and tenure
t.test(group16$tenure ~ (group16$age > median(group16$age, na.rm = TRUE)),
       data = group16, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group16$tenure by group16$age > median(group16$age, na.rm = TRUE)
## t = -26.446, df = 230.34, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##       -Inf -9.530795
## sample estimates:
## mean in group FALSE  mean in group TRUE
##            18.58286            28.74847
```

```r
# we reject the null hypothesis, as p-value is below 0.05
```

Group 40

```r
#H0: There is no difference between group 40 mean(quitjob_num) compared to overall average
#H1: Group 40 has higher mean(quitjob_num) compared to overall average
t.test(group40$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "g
```

```
##
##  Welch Two Sample t-test
##
## data:  group40$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
## t = 179.52, df = 10078, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.7548015       Inf
## sample estimates:
## mean of x mean of y
## 1.0000000 0.2382181
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between group 40 mean(perform1) compared to overall average
#H1: Group 40 has lower mean(perform1) compared to overall average
t.test(group40$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group40$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
## t = -12.126, df = 745.84, p-value < 2.2e-16
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##        -Inf -0.3860627
## sample estimates:
##   mean of x   mean of y
```

```
## -0.46173283 -0.01499374
```

```r
# we reject the null hypothesis, as p-value is below 0.05
```

Group 48

```r
#H0: There is no difference between group 48 mean(quitjob_num) compared to overall average
#H1: Group 48 has higher mean(quitjob_num) compared to overall average
t.test(group48$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "g
```

```
##
##  Welch Two Sample t-test
##
## data:  group48$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
## t = 179.52, df = 10078, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.7548015       Inf
## sample estimates:
## mean of x mean of y
## 1.0000000 0.2382181
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between group 48 mean(perform1) compared to overall average
#H1: Group 48 has lower mean(perform1) compared to overall average
t.test(group48$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group48$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
## t = -12.806, df = 237.14, p-value < 2.2e-16
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##        -Inf -0.5333124
## sample estimates:
##   mean of x   mean of y
## -0.62725468 -0.01499374
```

```r
# we reject the null hypothesis, as p-value is below 0.05
```

Group 60

```r
#H0: There is no difference between group 60 mean(quitjob_num) compared to overall average
#H1: Group 60 has higher mean(quitjob_num) compared to overall average
t.test(group60$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "g
```

```
##
##  Welch Two Sample t-test
##
## data:  group60$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
## t = 179.52, df = 10078, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
```

```
##  0.7548015       Inf
## sample estimates:
## mean of x mean of y
## 1.0000000 0.2382181
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between group 60 mean(perform1) compared to overall average
#H1: Group 60 has lower mean(perform1) compared to overall average
t.test(group60$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group60$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
## t = -2.1893, df = 204.81, p-value = 0.01485
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##         -Inf -0.04921906
## sample estimates:
##   mean of x   mean of y
## -0.21566135 -0.01499374
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between age and commute
#H1: Older Members have higher commute
t.test(group60$commute ~ (group60$age > median(group60$age, na.rm = TRUE)),
       data = group60, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group60$commute by group60$age > median(group60$age, na.rm = TRUE)
## t = -2116.3, df = 101, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##       -Inf -208.9536
## sample estimates:
## mean in group FALSE  mean in group TRUE
##            30.88235           240.00000
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between age and tenure
#H1: Older Members have higher tenure
t.test(group60$tenure ~ (group60$age > median(group60$age, na.rm = TRUE)),
       data = group60, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group60$tenure by group60$age > median(group60$age, na.rm = TRUE)
## t = -53.988, df = 149, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
```

```
##       -Inf -32.55052
## sample estimates:
## mean in group FALSE  mean in group TRUE
##               13.42               47.00
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between tenure on grosswage
#H1: Higher tenure leads to higher grosswage
t.test(group60$grosswage ~ (group60$tenure > median(group60$tenure, na.rm = TRUE)),
       data = group60, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group60$grosswage by group60$tenure > median(group60$tenure, na.rm = TRUE)
## t = -16.582, df = 120.89, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##       -Inf -905.1534
## sample estimates:
## mean in group FALSE  mean in group TRUE
##            2321.245            3326.925
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between tenure on basewage
#H1: Higher tenure leads to higher basewage
t.test(group60$basewage ~ (group60$tenure > median(group60$tenure, na.rm = TRUE)),
       data = group60, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group60$basewage by group60$tenure > median(group60$tenure, na.rm = TRUE)
## t = -16.28, df = 188.92, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##       -Inf -149.6183
## sample estimates:
## mean in group FALSE  mean in group TRUE
##            1474.000            1640.526
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between tenure on bonustotal
#H1: Higher tenure leads to higher bonustotal
t.test(group60$bonustotal ~ (group60$tenure > median(group60$tenure, na.rm = TRUE)),
       data = group60, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group60$bonustotal by group60$tenure > median(group60$tenure, na.rm = TRUE)
## t = -13.722, df = 122.17, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
```

```
## 95 percent confidence interval:
##       -Inf -737.7938
## sample estimates:
## mean in group FALSE  mean in group TRUE
##            847.2447           1686.3986
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relationship between commute and perform1
#H1: Higher commute leads to lower perform1
t.test(group60$perform1 ~ (group60$commute > median(group60$commute, na.rm = TRUE)),
       data = group60, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  group60$perform1 by group60$commute > median(group60$commute, na.rm = TRUE)
## t = -6.6431, df = 113.4, p-value = 5.568e-10
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##        -Inf -0.9848502
## sample estimates:
## mean in group FALSE  mean in group TRUE
##          -0.5102064           0.8023030
```

```r
# we reject the null hypothesis, as p-value is below 0.05
```

Top 10 performers

```r
#H0: The variance(basewage) among top performers is equal to the variance(basewage) among the rest
#H1: The variance(basewage) among top performers is ~ 0
var.test(top10$basewage, bottom90$basewage)
```

```
##
##  F test to compare two variances
##
## data:  top10$basewage and bottom90$basewage
## F = 1.1612, num df = 979, denom df = 8793, p-value = 0.00132
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  1.059462 1.277145
## sample estimates:
## ratio of variances
##           1.161167
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: The mean(grosswage) of Top 10 is equal to the mean(grosswage) of the rest
#H1: The mean(grosswage) of Top 10 is greater than the mean(grosswage) of the rest
t.test(top10$grosswage, bottom90$grosswage, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$grosswage and bottom90$grosswage
## t = 20.956, df = 1146.3, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  683.2639      Inf
## sample estimates:
## mean of x mean of y
##  3750.751  3009.238
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0 the mean(children_num) of Top 10 is equal to the mean(children_num) of the rest
#H1: The mean(children_num) of Top 10 is higher than the mean(children_num) of the rest
t.test(top10$children_num, bottom90$children_num, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$children_num and bottom90$children_num
## t = 7.5753, df = 1123.9, p-value = 3.729e-14
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.08347982        Inf
## sample estimates:
## mean of x mean of y
## 0.2390641 0.1324062
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: The mean(homethatweek_num) of Top 10 is equal to the mean(homethatweek_num) of the rest
#H1: The mean(homethatweek_num) of Top 10 is = 0.25
t.test(top10$homethatweek_num, bottom90$homethatweek_num, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$homethatweek_num and bottom90$homethatweek_num
## t = 4.4439, df = 1165.8, p-value = 4.837e-06
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.04059306        Inf
## sample estimates:
## mean of x mean of y
## 0.2543235 0.1898462
```

```r
# we reject the null hypothesis, as p-value is below 0.05
# result for top 10 is ~ 0.25 (25.4)

#H0: The mean(age) of top performes is equal to the mean(age) of the rest
#H1: The mean(age) of top performes = 24
t.test(top10$age, bottom90$age, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$age and bottom90$age
## t = 2.1209, df = 1097.4, p-value = 0.01708
## alternative hypothesis: true difference in means is greater than 0
```

```
## 95 percent confidence interval:
##  0.08725801        Inf
## sample estimates:
## mean of x mean of y
##  23.83418  23.44426
```

```r
# we reject the null hypothesis, as p-value is below 0.05
# result for top 10 is ~ 24 (23.8)

#H0: The mean(tenure) of top performes is equal to the mean(tenure) of the rest
#H1: The mean(tenure) of top performes > mean(tenure) of the rest
t.test(top10$tenure, bottom90$tenure, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$tenure and bottom90$tenure
## t = 11.61, df = 1202.9, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  8.37142     Inf
## sample estimates:
## mean of x mean of y
##  32.76501  23.01052
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: the mean(promote_switch_num) of top performes is equal to the mean(promote_switch_num) of the rest
#H1: The mean(promote_switch_num) of top performes > mean(promote_switch_num) of the rest
t.test(top10$promote_switch_num, bottom90$promote_switch_num, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$promote_switch_num and bottom90$promote_switch_num
## t = 2.9122, df = 1176.6, p-value = 0.001828
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.01741865        Inf
## sample estimates:
## mean of x mean of y
## 0.2166836 0.1766169
```

```r
# we reject the null hypothesis, as p-value is below 0.05

#H0: The mean (quitjob_num) of top performes is equal to the mean(quitjob_num) of the rest
#H1: The mean(quitjob_num) of top performes < mean(quitjob_num) of the rest
t.test(top10$quitjob_num, bottom90$quitjob_num, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$quitjob_num and bottom90$quitjob_num
## t = -5.1051, df = 1272.4, p-value = 1.905e-07
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
```

```
##        -Inf -0.04547338
## sample estimates:
## mean of x mean of y
## 0.1820956 0.2492085
```

```
# we reject the null hypothesis, as p-value is below 0.05

#H0: Among top vs. rest, mean(bonustotal) has no effect on quit probability
#H1: Top performers with higher mean(bonustotal) are less likely to quit than non-performers
t.test(top10$bonustotal ~ (top10$bonustotal > median(top10$bonustotal, na.rm = TRUE)),
       data = top10, alternative = "less")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$bonustotal by top10$bonustotal > median(top10$bonustotal, na.rm = TRUE)
## t = -40.524, df = 710.31, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##       -Inf -1447.914
## sample estimates:
## mean in group FALSE  mean in group TRUE
##            1342.202            2851.456
```

```
# we reject the null hypothesis, as p-value is below 0.05

#H0: mean(positive) does not differ from homethatweek_num
#H1: higher homethatweek_num leads to higher mean(positive)
t.test(top10$positive ~ (top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)),
       data = top10, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$positive by top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)
## t = 3.2093, df = 186.93, p-value = 0.000783
## alternative hypothesis: true difference in means between group FALSE and group TRUE is greater than
## 95 percent confidence interval:
##  1.104165      Inf
## sample estimates:
## mean in group FALSE  mean in group TRUE
##              28.125              25.848
```

```
# we reject the null hypothesis, as p-value is below 0.05

#H0: mean(negative) does not differ from homethatweek_num
#H1: higher homethatweek_num leads to lower mean(negative)
t.test(top10$negative ~ (top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)),
       data = top10, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$negative by top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)
## t = 1.5896, df = 121.93, p-value = 0.05726
## alternative hypothesis: true difference in means between group FALSE and group TRUE is greater than
```

```
## 95 percent confidence interval:
##  -0.0628927        Inf
## sample estimates:
## mean in group FALSE  mean in group TRUE
##            15.92188            14.44800
```

```r
# we fail to reject the null hypothesis, as p-value is above 0.05

#H0: mean(exhaustion) does not differ from homethatweek_num
#H1: higher homethatweek_num leads to lower mean(exhaustion)
t.test(top10$exhaustion ~ (top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)),
       data = top10, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  top10$exhaustion by top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)
## t = -1.1678, df = 126.13, p-value = 0.8775
## alternative hypothesis: true difference in means between group FALSE and group TRUE is greater than 0
## 95 percent confidence interval:
##  -2.655678       Inf
## sample estimates:
## mean in group FALSE  mean in group TRUE
##            6.078125            7.176000
```

```r
# we fail to reject the null hypothesis, as p-value is above 0.05
```